

MDGTB301

Digital Economy

EU Business School · Masters Year 1 · 3 ECTS

Framework Sheet Library

Master Index · 20 Slibraries · 100 Reference Sheets

5	Course Units <i>covering foundations, concepts, markets, adaptation, ethics</i>
20	Slibraries (thematic clusters) <i>4 per unit, colour-coded</i>
100	Framework Reference Sheets <i>5 per Slibrary, print-ready A4</i>
400+	Harvard-referenced sources <i>peer-reviewed and practitioner</i>

Design Philosophy

Each sheet follows a consistent nine-section layout: professional definition, who-developed-it, why-it-was-developed, a custom illustrative figure, how/when/by-whom-used (three columns), innovation versus the predecessor, related frameworks in the course, and Harvard-referenced sources. The structure enables students and practitioners to navigate the library as a reference work — treating the 100 sheets as a mental Slibrary (per Meier, 2025) that scales from single-framework lookup to full strategic reasoning across multiple connected tools.

Unit 1

Digitalization of the Global Economy

4 Slibraries · 20 framework sheets

SLIBRARY 1 · Digital Transformation

5 sheets

- 1 Atoms → Bits
- 2 OECD 3-Layer Model
- 3 S-Curve of Adoption
- 4 Gartner Hype Cycle
- 5 Digitisation · Digitalisation · DT

SLIBRARY 2 · Digital Economy Data

5 sheets

- 1 UNCTAD Digital Trade Flows
- 2 ICT Sector % of GDP
- 3 Global Internet Penetration
- 4 DCO Regional Digital Trends
- 5 The Digital Divide Gap

SLIBRARY 3 · Technology & Ecosystems

5 sheets

- 1 Platform vs. Pipeline
- 2 SMAC Stack
- 3 Iansiti-Levien Ecosystem Patterns
- 4 Network Effects
- 5 Super-App Model

SLIBRARY 4 · Disruption Patterns

5 sheets

- 1 Christensen's Disruption
- 2 Industry Digitization Index
- 3 BCG 10-20-70 Rule
- 4 Before/After: Music · Retail · Finance
- 5 WEF Future of Jobs

Unit 2

Fundamental Concepts of the Digital Economy

4 Libraries · 20 framework sheets

SLIBRARY 5 · E-Business Architecture

5 sheets

- 1 E-Business vs E-Commerce
- 2 Digital Infrastructure Stack
- 3 B2B / B2C / C2C / B2G Models
- 4 API Economy
- 5 Cloud Computing — IaaS, PaaS, SaaS

SLIBRARY 6 · Digital Networks

5 sheets

- 1 Metcalfe's Law
- 2 Direct & Indirect Network Effects
- 3 Two-Sided Markets
- 4 Connection Density (MIT)
- 5 Tipping Point

SLIBRARY 7 · Digital Value Creation

5 sheets

- 1 Value Chain → Value Network
- 2 Freemium Model
- 3 Subscription Models
- 4 Data as Asset
- 5 Long Tail

SLIBRARY 8 · Information Goods

5 sheets

- 1 Zero Marginal Cost
- 2 Experience Goods
- 3 Versioning
- 4 Price Discrimination
- 5 Lock-In & Switching Costs

Unit 3

Digital Markets

4 Libraries · 20 framework sheets

SLIBRARY 9 · Market Structures

5 sheets

- 1 Winner-Takes-All Markets
- 2 Two-Sided Markets — Market Structure View
- 3 Multi-Sided Platforms
- 4 Marketplace vs. Platform
- 5 GAFAM Concentration

SLIBRARY 10 · Digital Verticals

5 sheets

- 1 Fintech
- 2 Legaltech
- 3 Insurtech
- 4 Regtech
- 5 Proptech

SLIBRARY 11 · Platform Economics

5 sheets

- 1 Chicken-and-Egg Problem
- 2 Subsidised Side Pricing
- 3 Platform Governance
- 4 Switching Costs in Platforms
- 5 Data Economies of Scale

SLIBRARY 12 · Market Failures & Regulation

5 sheets

- 1 DMA — Digital Markets Act
- 2 DSA — Digital Services Act
- 3 Antitrust in Digital Markets
- 4 Algorithmic Collusion
- 5 Essential Facilities Doctrine

Unit 4

Adapting to the Digital Economy

4 Libraries · 20 framework sheets

SLIBRARY 13 · Digital Transformation Strategy

5 sheets

- 1 BCG 10-20-70 Rule
- 2 McKinsey 7S — Digital Edition
- 3 Digital Maturity Model
- 4 Transformation Roadmap
- 5 Change Resistance

SLIBRARY 14 · The Future of Work

5 sheets

- 1 Reskilling vs. Upskilling
- 2 Task Automation Potential
- 3 Step-Function AI Impact (MIT)
- 4 Gig Economy Platforms
- 5 Skills Matrix 2025-2030

SLIBRARY 15 · AI in Business

5 sheets

- 1 AI Spectrum
- 2 Generative AI Economics
- 3 AI ROI — The \$252B Question (MIT)
- 4 Human-AI Pairing (MIT)
- 5 Cognitive Trade-Off (MIT)

SLIBRARY 16 · The New Consumer

5 sheets

- 1 5A Framework (Kotler)
- 2 ZMOT — Zero Moment of Truth
- 3 Hyper-Personalisation
- 4 Customer Journey Loop
- 5 Experience Economy

Unit 5

Ethical & Legal Issues in the Digital Economy

4 Libraries · 20 framework sheets

SLIBRARY 17 · Data Rights & Privacy

5 sheets

- 1 GDPR Principles
- 2 Personal Data as Property
- 3 Consent & Legitimate Interest
- 4 Data Minimisation
- 5 Right to be Forgotten

SLIBRARY 18 · Intellectual Property

5 sheets

- 1 Copyright in the Digital Age
- 2 Creative Commons
- 3 Patent Trolls
- 4 AI & Copyright
- 5 Trade Secrets vs Patents

SLIBRARY 19 · Surveillance Capitalism

5 sheets

- 1 Zuboff's Definition
- 2 Behavioural Futures Markets
- 3 Data Extraction Imperative
- 4 Instrumentarian Power
- 5 Privacy Equation (MIT)

SLIBRARY 20 · AI Ethics & Governance

5 sheets

- 1 EU AI Act
- 2 Algorithmic Bias
- 3 Responsibility Gap
- 4 AI Risk & Mitigation Matrix (MIT)
- 5 Human-Centered AI Blueprint (MIT)

How to Use This Library

For students

Each Slibrary is designed as a mental reference — a 'Thumbnail Thinking' collection (Meier, 2025). Use individual sheets for framework lookup, entire Slibraries for thematic deep-dives, and the full library as a reference catalogue. The Related-Frameworks pills on every sheet are navigation links to the most relevant neighbouring ideas in the course.

For practising executives

The library functions as a decision-support reference. When facing a digital strategy, market entry, or governance decision, the relevant framework(s) can be located by unit, keyword, or cross-reference. Sheet structure is optimised for quick comprehension: definition, visual, how/when/who used. Most sheets can be absorbed in under two minutes.

For MDGTB301 assessments

The midterm (40%, Units 1-3) and final (60%, all units) briefs expect explicit application of named frameworks to the assigned case studies (Spotify and AI in financial services). Students should cite the specific sheet (e.g. 'Applying the Long Tail from Slibrary 7 Sheet 5...') to demonstrate framework mastery. The resit brief similarly requires framework citation.

Reading order

Linear reading (Slibrary 1 through 20) follows the course's weekly sequence and is recommended for first-time readers. Topic-specific reading (e.g. ethics → Slibraries 17–20) is recommended for practitioners with focused questions. Cross-reference navigation via the Related-Frameworks pills produces exploration paths that reveal deeper connections.

Design conventions

Colour coding signals theme: blue/slate/indigo/violet = digitalization; teal/cyan/sky/emerald = fundamental concepts; orange/amber/red/magenta = digital markets; greens = adapting; rose/purple/plum/navy = ethics & legal. Each sheet's left-edge accent bar uses its Slibrary's signature colour. Harvard author-date referencing is used throughout, with full citations at the bottom of each sheet.

Atoms → Bits

SLIBRARY 01 · SHEET 1 / 5

Tapscott's foundational shift — physical things become information

PROFESSIONAL DEFINITION

The Atoms-to-Bits framing describes the macro-economic transition in which value, products, and transactions migrate from physical (atomic) carriers to digital (bit) carriers. Once a good or process is rendered as bits, it inherits the economic properties of information — non-rivalry, near-zero marginal cost of reproduction, and frictionless distribution (Negroponte, 1995; Tapscott, 1996).

WHO DEVELOPED IT

The phrase was popularised by Nicholas Negroponte (1995) in 'Being Digital' and by Don Tapscott (1996) in 'The Digital Economy' — the book that gave the field its name. Brynjolfsson and Kahin (2000) extended the framing into formal economic analysis of how bit-based goods rewrite cost structures.

WHY IT WAS DEVELOPED

By the mid-1990s the personal computer, the public Internet, and digital media were converging. Managers needed a single, memorable mental model to grasp why the rules of business were changing simultaneously across so many industries. Atoms-vs-bits gave them that one-line model — and a way to predict which sectors would be hit next.

FIGURE 1 · TWO ECONOMIES

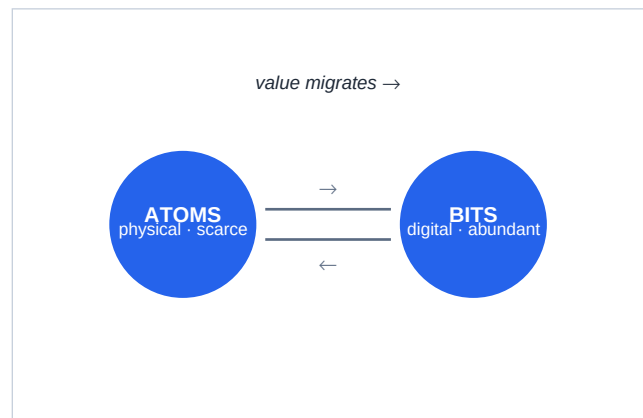


Figure 1. The atoms-to-bits transition. Physical goods carry atoms — costly to produce, ship, and store. Digital goods carry bits — copied at zero cost and delivered instantly across borders (Negroponte, 1995; Tapscott, 1996).

HOW IT IS USED

Strategists use the framing to map a company's product portfolio along the atoms-bits axis and to identify which atomic activities can be substituted, augmented, or replaced by bit-based alternatives.

WHEN IT IS USED

At the start of any digital strategy review, in MBA case discussions of industry transformation, and whenever a sector first encounters a digital substitute.

BY WHOM IT IS USED

Strategy consultants, board-level executives during digital strategy formulation, policymakers framing the digital economy, and educators introducing the field.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with earlier IT-productivity discussions that treated computing as a support function (Solow, 1987; Brynjolfsson, 1993), the atoms-to-bits framing elevated digitalisation to a core economic transition. Compared with sector-specific analyses, it works as a single, cross-sector heuristic that students can apply on a whiteboard in 30 seconds.

RELATED FRAMEWORKS IN THIS COURSE

[OECD 3-Layer Model · Slibrary 1](#)
[S-Curve of Adoption · Slibrary 1](#)
[Zero Marginal Cost · Slibrary 8](#)
[Christensen Disruption · Slibrary 4](#)

SOURCES (HARVARD REFERENCING STYLE)

Brynjolfsson, E. and Kahin, B. (eds.) (2000) Understanding the Digital Economy. Cambridge, MA: MIT Press.
 Negroponte, N. (1995) Being Digital. New York, NY: Knopf.
 Solow, R.M. (1987) 'We'd better watch out', New York Times Book Review, 12 July, p. 36.
 Tapscott, D. (1996) The Digital Economy: Promise and Peril in the Age of Networked Intelligence. New York, NY: McGraw-Hill.

OECD 3-Layer Model

SLIBRARY 01 · SHEET 2 / 5

Core · Narrow · Broad — the three scopes for measuring the digital economy

PROFESSIONAL DEFINITION
The OECD 3-Layer Model is a measurement framework that distinguishes three nested scopes of the digital economy: a Core layer (the ICT sector itself — hardware, software, telecommunications), a Narrow layer (digital services and platforms), and a Broad layer (any economic activity intensively enabled by digital technology). The layers are designed to support comparable national statistics (OECD, 2020; Bukht and Heeks, 2017).

WHO DEVELOPED IT
Developed by the OECD's Working Party on Measurement and Analysis of the Digital Economy. The conceptual foundations were articulated by Bukht and Heeks (2017) at the University of Manchester, and adopted in OECD's 'Roadmap toward a Common Framework for Measuring the Digital Economy' (OECD, 2020).

WHY IT WAS DEVELOPED
National statistical agencies disagreed on what counted as 'the digital economy', producing GDP-share estimates ranging from 4% to 30%. The 3-layer model gave ministries a shared vocabulary so that international comparisons became meaningful.

FIGURE 1 · THE THREE NESTED SCOPES

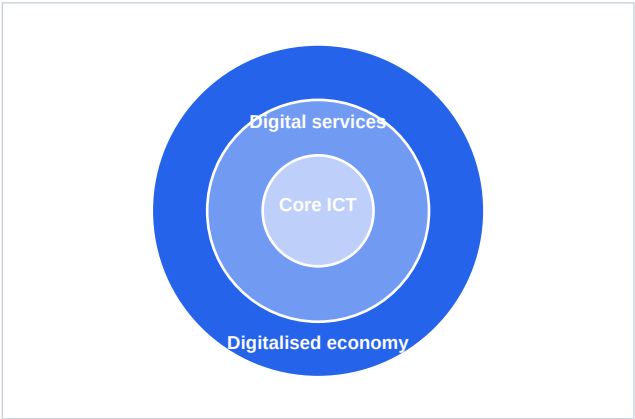


Figure 1. The Core scope captures the ICT sector itself; the Narrow scope adds digital platforms and services; the Broad scope captures any sector intensively enabled by digital technology (OECD, 2020).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Statisticians attribute economic activity to one of the three layers, then sum value added by layer to estimate the digital economy's GDP contribution. Strategists use it to position competitors and to forecast where digital intensity will spread	During national statistical reporting, in industry studies of digital intensity, in policy briefings, and whenever an executive must justify which slice of the economy to include in a 'digital' market sizing.	OECD member-state statistical agencies, central banks, ministries of digital affairs, industry associations, and consultancies producing market sizing reports.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL
Compared with single-number estimates (e.g., 'ICT is X% of GDP'), the 3-layer model captures the gradient between pure ICT and digitalisation of legacy sectors. Compared with country-specific definitions, it offers an internationally comparable standard endorsed by the OECD's 38 member states.

RELATED FRAMEWORKS IN THIS COURSE

- Atoms → Bits · Slibrary 1
- Digitisation · Digitalisation · DT · Slibrary 1
- UNCTAD Digital Trade Flows · Slibrary 2
- ICT Sector % of GDP · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

Bukht, R. and Heeks, R. (2017) 'Defining, conceptualising and measuring the digital economy', Development Informatics Working Paper No. 68. Manchester: GDI.
OECD (2020) Roadmap toward a Common Framework for Measuring the Digital Economy. Paris: OECD Publishing.
OECD (2024) OECD Digital Economy Outlook 2024 (Volume 1). Paris: OECD Publishing.

S-Curve of Adoption

SLIBRARY 01 · SHEET 3 / 5

Innovations diffuse slowly, then explosively, then saturate

PROFESSIONAL DEFINITION

The S-Curve describes the cumulative adoption of an innovation over time as a logistic function: a slow start while early adopters experiment, a steep middle as the early and late majority join, and a flat plateau as the laggards complete the diffusion. The shape was formalised by Rogers (1962) and remains the canonical diffusion model.

WHO DEVELOPED IT

Drawn from Everett Rogers's 'Diffusion of Innovations' (1962). The mathematical form predates Rogers — the logistic function was studied by Verhulst in 1838 — but Rogers gave it the social-science framing that connects adopter categories to the curve's segments.

WHY IT WAS DEVELOPED

Practitioners needed a way to forecast the timing and shape of technology adoption without forecasting individual decisions. The S-curve gave them a single curve that summarises hundreds of empirical studies — from hybrid corn to smartphones.

FIGURE 1 · CUMULATIVE ADOPTION

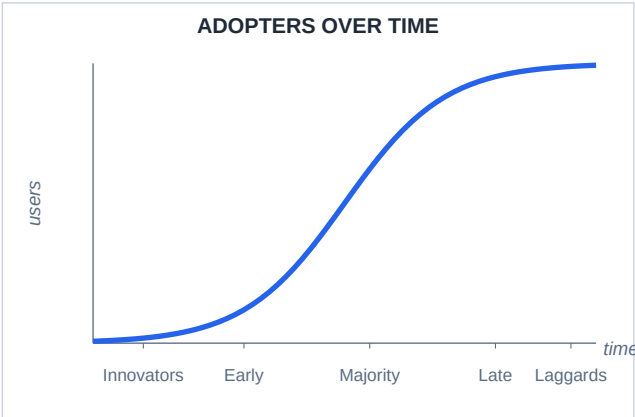


Figure 1. The classic S-curve of adoption. Slow take-off, steep middle, saturation plateau — driven by sequential entry of five adopter segments (Rogers, 1962).

HOW IT IS USED

Strategists plot a technology's cumulative adoption against time, identify the current point on the curve, and adjust go-to-market and investment intensity to match the segment about to enter.

WHEN IT IS USED

During market entry decisions, technology forecasting exercises, capacity planning for scaling firms, and competitive analysis of when to fast-follow vs. pioneer.

BY WHOM IT IS USED

Marketing strategists, product managers, venture capitalists, technology forecasters, and policymakers planning infrastructure roll-out.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with linear forecasting, the S-curve explicitly models non-linear take-off and saturation. Compared with descriptive case histories, it provides a parametric form (logistic) that can be fitted to early data and used for prediction. The curve's main weakness — assuming all adoption is voluntary and irreversible — has been addressed by extensions like the Bass model (Bass, 1969).

RELATED FRAMEWORKS IN THIS COURSE

- Atoms → Bits · Slibrary 1
- Gartner Hype Cycle · Slibrary 1
- Christensen Disruption · Slibrary 4
- Industry Digitization Index · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

Bass, F.M. (1969) 'A new product growth model for consumer durables', Management Science, 15(5), pp. 215–227.
Rogers, E.M. (1962) Diffusion of Innovations. New York, NY: Free Press.
Rogers, E.M. (2003) Diffusion of Innovations. 5th edn. New York, NY: Free Press.

Gartner Hype Cycle

SLIBRARY 01 · SHEET 4 / 5

Trigger · Peak · Trough · Slope · Plateau — the emotional arc of new tech

PROFESSIONAL DEFINITION

The Gartner Hype Cycle is a graphical representation of the maturity of an emerging technology as it moves through five named phases: Innovation Trigger, Peak of Inflated Expectations, Trough of Disillusionment, Slope of Enlightenment, and Plateau of Productivity. Each annual edition positions ~25 technologies along the curve (Gartner, 2024).

WHO DEVELOPED IT

Created by Jackie Fenn at Gartner in 1995. Gartner has published the cycle annually since 1995, and now produces specialised hype-cycles for AI, cloud, marketing technology, and 30+ other domains (Fenn and Raskino, 2008).

WHY IT WAS DEVELOPED

Gartner's clients struggled to distinguish genuine technology shifts from short-lived fads. The hype cycle gave them a vocabulary for the emotional cycle that reliably follows every new technology — first overhyped, then dismissed, then quietly mature.

FIGURE 1 · THE FIVE PHASES

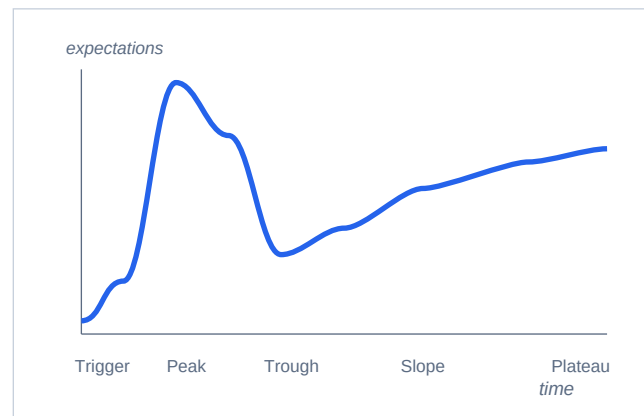


Figure 1. Gartner's five-phase curve plots expectations against time. The peak and trough are emotional, not technical — the underlying technology improves monotonically (Gartner, 2024).

HOW IT IS USED

Executives locate a technology on the current cycle, then map the technology's phase to a procurement posture: experiment in the trigger phase, resist sales pitches at the peak, accelerate adoption on the slope, and commit at the plateau.

WHEN IT IS USED

During annual technology strategy reviews, IT investment committee meetings, and vendor evaluations — particularly when distinguishing 'shiny object' announcements from production-grade technology.

BY WHOM IT IS USED

CIOs, CTOs, IT investment committees, technology analysts, venture capitalists, and vendor product managers benchmarking their own roadmaps.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the S-curve, the hype cycle adds an explicit psychological dimension — expectations rise faster than performance, then over-correct downward. Compared with academic technology forecasting, it is qualitative, opinionated, and updated annually, which suits practitioner timeframes. Critics note the cycle is descriptive rather than predictive (Aranda and Easterbrook, 2014).

RELATED FRAMEWORKS IN THIS COURSE

[S-Curve of Adoption · Slibrary 1](#)
[AI Spectrum · Slibrary 15](#)
[Generative AI Economics · Slibrary 15](#)
[Christensen Disruption · Slibrary 4](#)

SOURCES (HARVARD REFERENCING STYLE)

Aranda, J. and Easterbrook, S. (2014) 'Anchoring and adjustment in software estimation', *Empirical Software Engineering*, 19(4), pp. 866–906.
 Fenn, J. and Raskino, M. (2008) *Mastering the Hype Cycle*. Boston, MA: Harvard Business Press.
 Gartner (2024) *Hype Cycle for Emerging Technologies, 2024*. Stamford, CT: Gartner.

Digitisation · Digitalisation · DT

SLIBRARY 01 · SHEET 5 / 5

Three precise concepts students must not conflate

PROFESSIONAL DEFINITION

Three distinct but commonly confused concepts: Digitisation is the technical conversion of analogue information into digital form (e.g., scanning a paper file). Digitalisation is the use of digital technologies to change a business process. Digital Transformation (DT) is the holistic, organisation-wide reinvention of strategy, operating model and value proposition enabled by digital technology (Gartner, 2023; Verhoef et al., 2021).

WHO DEVELOPED IT

The taxonomy crystallised in the practitioner literature around 2015-2018, with academic codification by Verhoef et al. (2021) at Groningen and the Gartner glossary (Gartner, 2023). German-speaking academia distinguishes 'Digitalisierung' from 'Digitale Transformation' along similar lines (Schallmo and Williams, 2018).

WHY IT WAS DEVELOPED

Executives, consultants and academics were using the three terms interchangeably, muddying strategic conversations. The three-tier vocabulary disciplines discussions of scope: a scanning project is not a transformation, and a transformation is not merely a scanning project.

FIGURE 1 · THE THREE TIERS

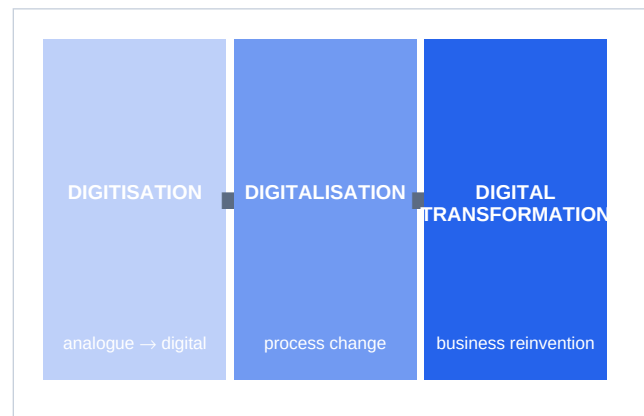


Figure 1. The three concepts ordered by scope of organisational change. Each tier subsumes and depends on the previous one (Verhoef et al., 2021; Gartner, 2023).

HOW IT IS USED

Project leaders classify each digital initiative into one of the three tiers, then tailor governance, budget, and KPIs to the appropriate scope — avoiding the common error of asking transformational metrics from a scanning project.

WHEN IT IS USED

At project intake, in board-level strategy discussions, in academic writing where precision matters, and whenever a vendor pitches 'digital transformation' for what is actually a process digitisation.

BY WHOM IT IS USED

Project managers, transformation leads, board members evaluating proposals, academic researchers, and management consultants.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-term usage of 'digital', the three-tier vocabulary forces explicit scope decisions. Compared with consulting frameworks that conflate the tiers, it allows executives to identify when consultants are over-promising scope (or under-delivering it). The taxonomy is now standard in MBA curricula globally.

RELATED FRAMEWORKS IN THIS COURSE

[OECD 3-Layer Model · Slibrary 1](#)
[BCG 10-20-70 Rule · Slibrary 4](#)
[Digital Maturity Model · Slibrary 13](#)
[Transformation Roadmap · Slibrary 13](#)

SOURCES (HARVARD REFERENCING STYLE)

Gartner (2023) Gartner Glossary: Digital, Digitization, Digitalization. Stamford, CT: Gartner.
 Schallmo, D. and Williams, C.A. (2018) Digital Transformation Now! Cham: Springer.
 Verhoef, P.C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Dong, J.Q., Fabian, N. and Haenlein, M. (2021) 'Digital transformation: A multidisciplinary reflection and research agenda', Journal of Business Research, 122, pp. 889–901.

UNCTAD Digital Trade Flows

SLIBRARY 02 · SHEET 1 / 5

Digitally-deliverable services exports — the world's fastest-growing trade

PROFESSIONAL DEFINITION

UNCTAD's Digital Trade indicator measures the global value of digitally-deliverable services exports — services that can in principle be supplied across borders via ICT networks (financial, computing, telecommunications, business, royalties). In 2024, this trade reached approximately USD 4.5 trillion, accounting for over half of all services exports (UNCTAD, 2024).

WHO DEVELOPED IT

The indicator is published annually in UNCTAD's Digital Economy Report and harmonised with the IMF's Balance of Payments framework (BPM6). Its conceptual foundations rest on Manyika et al. (2016) at McKinsey Global Institute and the WTO/UNCTAD/OECD joint Handbook on Measuring Digital Trade (2023).

WHY IT WAS DEVELOPED

Conventional trade statistics, designed for shipping containers, were systematically missing the boom in cross-border digital services. UNCTAD's indicator was created to make the digital trade boom visible to ministries that were still focused on goods trade.

FIGURE 1 · DIGITAL VS NON-DIGITAL SERVICES (USD TN, 2024)

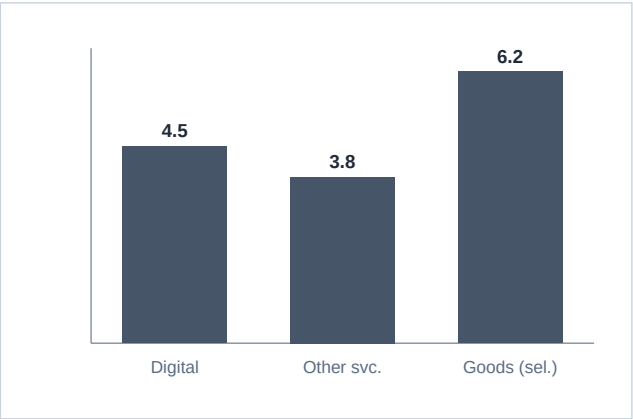


Figure 1. Digitally-deliverable services exports reached USD 4.5 trillion in 2024 — over half of total services exports and growing roughly twice as fast as goods trade (UNCTAD, 2024).

HOW IT IS USED

Analysts use the indicator to track the share of services that are deliverable online, to benchmark countries' digital export capacity, and to identify which sectors lead in cross-border digital services.

WHEN IT IS USED

In country-level competitiveness studies, in negotiations on digital trade chapters of trade agreements (e.g., DEPA), and in market sizing for international expansion of digital firms.

BY WHOM IT IS USED

Trade ministries, central banks, OECD/WTO/UNCTAD analysts, consultancies advising on cross-border digital strategy, and academics studying globalisation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional balance-of-payments services categories, the digital indicator isolates the cross-border-deliverable subset. Compared with industry estimates of 'the digital economy', it is internationally comparable, methodologically transparent, and produced annually by an intergovernmental body.

RELATED FRAMEWORKS IN THIS COURSE

OECD 3-Layer Model · Slibrary 1

ICT Sector % of GDP · Slibrary 2

DCO Regional Digital Trends · Slibrary 2

The Digital Divide Gap · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

Manyika, J., Lund, S., Bughin, J., Woetzel, J., Stamenov, K. and Dhingra, D. (2016) Digital Globalization: The New Era of Global Flows. New York, NY: McKinsey Global Institute.
UNCTAD (2024) Digital Economy Report 2024. Geneva: United Nations Conference on Trade and Development.
WTO/UNCTAD/OECD (2023) Handbook on Measuring Digital Trade. 2nd edn. Geneva and Paris: WTO/UNCTAD/OECD.

ICT Sector % of GDP

SLIBRARY 02 · SHEET 2 / 5

Core ICT 4-6% · digitalised economy 30%+ in advanced markets

PROFESSIONAL DEFINITION

The ICT Sector share of GDP is the value added by the Information and Communication Technology sector — manufacturers, telecoms, software publishers, IT services — expressed as a percentage of total GDP. In OECD economies the Core ICT sector typically contributes 4-6% of GDP, while the broader 'digitalised economy' (see OECD 3-Layer Model) reaches 30% or more in the most advanced markets (OECD, 2024).

WHO DEVELOPED IT

Methodology developed by the OECD Working Party on Indicators for the Information Society. National figures are produced by statistical offices (Eurostat, BEA, ONS, Destatis) using the ISIC Rev. 4 classification. Mesenbourg's (2001) US Census Bureau framework provided the conceptual scaffolding.

WHY IT WAS DEVELOPED

Policymakers needed a comparable, standardised measure of the size of the ICT sector to benchmark national digital competitiveness. Without it, every country claimed its own definition and headline number.

FIGURE 1 · CORE ICT % OF GDP (SELECTED 2024)

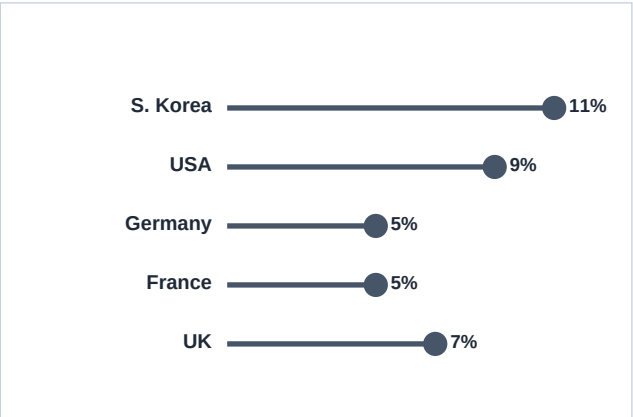


Figure 1. Core ICT sector share of GDP varies from ~4-5% in some EU economies to 9-11% in technology-export-led economies. Total digitalised economy reaches 30%+ in advanced markets (OECD, 2024).

HOW IT IS USED

Ministries use the figure as a headline competitiveness indicator, benchmarked against peers. Researchers decompose the figure to identify whether ICT growth is driven by hardware, software, or services.

WHEN IT IS USED

In annual digital competitiveness reports, in industrial policy discussions, in FDI promotion materials, and whenever a minister of digital affairs is asked 'how big is our digital economy?'

BY WHOM IT IS USED

Ministries of digital affairs, ICT industry associations, OECD/Eurostat analysts, and consultancies producing country competitiveness reports.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with employment counts, the value-added measure aligns with GDP and permits sectoral decomposition. Compared with total-economy 'digital intensity' indicators, the Core ICT figure is narrower but internationally comparable. Critics note that it understates digitalisation in non-ICT sectors — which the OECD 3-Layer Model explicitly addresses.

RELATED FRAMEWORKS IN THIS COURSE

- OECD 3-Layer Model · Slibrary 1
- UNCTAD Digital Trade Flows · Slibrary 2
- Global Internet Penetration · Slibrary 2
- DCO Regional Digital Trends · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

Mesenbourg, T.L. (2001) Measuring the Digital Economy. Suitland, MD: US Bureau of the Census.
OECD (2024) OECD Digital Economy Outlook 2024 (Volume 1). Paris: OECD Publishing.
Eurostat (2024) Digital Economy and Society Statistics. Luxembourg: Publications Office.

Global Internet Penetration

SLIBRARY 02 · SHEET 3 / 5

~68% online in 2024 — but quality and use divides persist

PROFESSIONAL DEFINITION

Internet Penetration is the percentage of a population that uses the internet (typically defined as use within the past three months, ITU standard). In 2024, approximately 68% of the world's population was online — equivalent to ~5.5 billion people, leaving 2.6 billion still offline. The headline figure conceals vast divides in connectivity quality, affordability, and meaningful use (ITU, 2024).

WHO DEVELOPED IT

The official global figure is published by the International Telecommunication Union (ITU), the UN agency for ICT, in its annual 'Facts and Figures' report. National regulators (Ofcom, BNetzA, FCC) report country-level data feeding into the ITU aggregate.

WHY IT WAS DEVELOPED

Internet penetration is the foundational metric for any digital-economy analysis: without users, no digital market exists. The ITU figure was created to give policymakers a Sustainable Development Goal (SDG 9.c) target and a way to track the closing of the digital divide.

FIGURE 1 · PENETRATION BY REGION (% 2024)

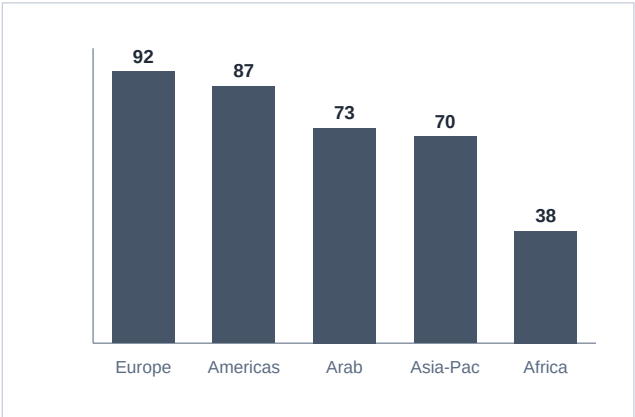


Figure 1. Internet penetration by region, 2024. Europe and the Americas have neared saturation; Africa remains the largest unconnected region. Global mean: ~68% (ITU, 2024).

HOW IT IS USED

Analysts use the indicator as a denominator for digital market sizing, as a policy target for universal-access programmes, and as input to digital-divide indices that combine penetration with affordability and skills.

WHEN IT IS USED

In market-entry studies for digital firms, in donor and World Bank funding decisions for connectivity infrastructure, and in annual SDG monitoring reports.

BY WHOM IT IS USED

Telecom regulators, donor agencies (World Bank, AfDB), ITU and UN agencies, digital firms sizing emerging markets, and academic researchers in development studies.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with subscription counts, internet-use measures actual access by individuals. Compared with averaged figures, regional and gender-disaggregated breakdowns reveal the structure of the digital divide. Critics note the binary 'use/don't use' definition obscures gradient differences in connectivity quality (van Deursen and Helsper, 2015).

RELATED FRAMEWORKS IN THIS COURSE

UNCTAD Digital Trade Flows · Slibrary 2

The Digital Divide Gap · Slibrary 2

DCO Regional Digital Trends · Slibrary 2

Connection Density (MIT) · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

ITU (2024) Measuring Digital Development: Facts and Figures 2024. Geneva: International Telecommunication Union.
van Deursen, A.J.A.M. and Helsper, E.J. (2015) 'The third-level digital divide: Who benefits most from being online?', in Robinson et al. (eds.) Communication and Information Technologies Annual. Bingley: Emerald, pp. 29–52.
World Bank (2024) Digital Progress and Trends Report 2024. Washington, DC: World Bank.

DCO Regional Digital Trends

SLIBRARY 02 · SHEET 4 / 5

MENA and GCC digital GDP rising sharply — a counterweight narrative

PROFESSIONAL DEFINITION

The Digital Cooperation Organization (DCO) Digital Economy Trends report tracks the digital GDP contribution of MENA, GCC, and partner states — a region underweighted in OECD-centric data. The 2024 edition reports digital economy contributions of 12-18% of GDP in leading GCC states and rapid growth in mobile-first ecosystems (DCO, 2024).

WHO DEVELOPED IT

The Digital Cooperation Organization is an intergovernmental body founded in 2020 and headquartered in Riyadh; its 16 member states span MENA, Africa and Asia. The flagship Digital Economy Trends report is produced annually by DCO Insights and reviewed by member-state ministries (DCO, 2025).

WHY IT WAS DEVELOPED

Conventional digital-economy reports (OECD, EU) underweight the GCC and broader MENA region's transformation. DCO created the report to give member states regionally-relevant benchmarks and to make their digital growth visible to international investors.

HOW IT IS USED

DCO members use the indicator to benchmark progress on national digital strategies (Saudi Vision 2030, UAE D33), to attract investment, and to coordinate cross-border digital regulation (e.g., e-residency, data flows).

WHEN IT IS USED

In annual DCO ministerial summits, in sovereign wealth fund investment reviews, in MENA-focused consulting reports, and in academic studies of non-Western digital transformation.

BY WHOM IT IS USED

GCC ministries of digital affairs, sovereign wealth fund analysts, regional consultancies (Strategy&, Roland Berger MENA), and academics specialising in Gulf or MENA economic transformation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with OECD-centric reporting, DCO offers a regionally-grounded view that captures mobile-first growth, sovereign-cloud strategies, and Islamic-finance fintech. Compared with national reports alone, it provides cross-country comparability within the DCO bloc.

RELATED FRAMEWORKS IN THIS COURSE

UNCTAD Digital Trade Flows · Slibrary 2

Global Internet Penetration · Slibrary 2

ICT Sector % of GDP · Slibrary 2

OECD 3-Layer Model · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

DCO (2024) Digital Economy Trends 2024. Riyadh: Digital Cooperation Organization.
DCO (2025) Digital Economy Trends 2025. Riyadh: Digital Cooperation Organization.
Saudi Government (2016) Saudi Vision 2030. Riyadh: Council of Economic and Development Affairs.

FIGURE 1 · DIGITAL GDP SHARE — DCO MEMBERS (%)

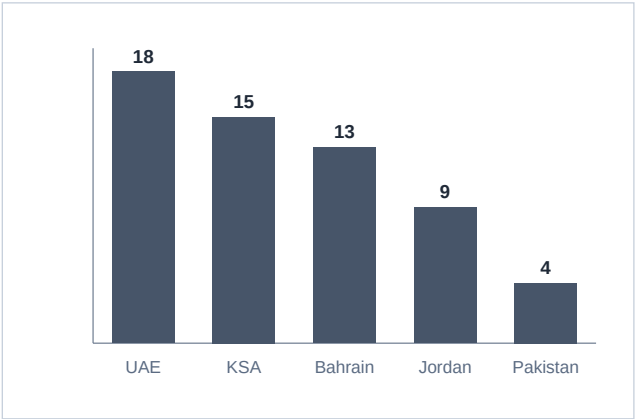


Figure 1. Digital economy share of GDP in selected DCO member states, 2024. GCC states lead, driven by Vision 2030 programmes and aggressive digital infrastructure investment (DCO, 2024).

The Digital Divide Gap

SLIBRARY 02 · SHEET 5 / 5

2.6 billion offline — and the gap is widening on quality, not access

PROFESSIONAL DEFINITION

The Digital Divide is the gap between those who have effective access to digital technologies and those who do not. Modern usage distinguishes three levels: first-level (physical access), second-level (skills and use), and third-level (outcomes and benefits). The 2.6 billion still offline are concentrated in rural areas, low-income economies, and among older women (van Deursen and Helsper, 2015; ITU, 2024).

WHO DEVELOPED IT

The first-level concept entered policy discourse in the late 1990s (NTIA, 1999). The three-level taxonomy was developed by van Dijk (2005) and extended by van Deursen and Helsper (2015) at Twente and the LSE. ITU and the World Bank produce the most-cited measurements.

WHY IT WAS DEVELOPED

Early framings — 'who has internet access' — masked the more enduring divides in skills and outcomes. As physical access neared saturation in advanced economies, researchers needed a richer model to explain why digital inequality persisted even when connections were universal.

FIGURE 1 · THREE LEVELS OF THE DIVIDE



Figure 1. The three-level digital divide. As Level-1 closes, Levels 2 and 3 become more salient — and the gap is widening on outcomes (van Deursen and Helsper, 2015).

HOW IT IS USED

Policymakers diagnose which level dominates in their context and target the intervention accordingly: subsidised devices for Level 1, digital-skills programmes for Level 2, inclusive design and outcome targeting for Level 3.

WHEN IT IS USED

In national digital-inclusion strategy formulation, in donor programme design, in ESG and DEI corporate reporting, and in academic work on inequality.

BY WHOM IT IS USED

Ministries of digital affairs, World Bank and donor agencies, NGOs (Web Foundation, ITU Partner2Connect), academic researchers, and CSR teams in digital firms.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the original binary (online vs. offline), the three-level model captures the persistent inequalities in skills and outcomes that survive even universal access. Compared with single-country studies, the ITU framework provides internationally comparable indicators across all three levels.

RELATED FRAMEWORKS IN THIS COURSE

Global Internet Penetration · Slibrary 2

UNCTAD Digital Trade Flows · Slibrary 2

DCO Regional Digital Trends · Slibrary 2

ICT Sector % of GDP · Slibrary 2

SOURCES (HARVARD REFERENCING STYLE)

NTIA (1999) Falling Through the Net: Defining the Digital Divide. Washington, DC: US Department of Commerce.
van Deursen, A.J.A.M. and Helsper, E.J. (2015) 'The third-level digital divide: Who benefits most from being online?', in Robinson et al. (eds.) Communication and Information Technologies Annual. Bingley: Emerald, pp. 29–52.
van Dijk, J.A.G.M. (2005) The Deepening Divide: Inequality in the Information Society. London: SAGE.

Platform vs. Pipeline

SLIBRARY 03 · SHEET 1 / 5

Linear value chains vs. multi-sided value creation

PROFESSIONAL DEFINITION

A Pipeline business creates value through a linear sequence — design, produce, sell, deliver — and captures margin at each step (Porter, 1985). A Platform business creates value by enabling direct interactions between two or more user groups, capturing margin from the matching itself (Parker, Van Alstyne and Choudary, 2016). The distinction is the most consequential business-model shift of the digital era.

WHO DEVELOPED IT

The platform-business framing was developed by Geoffrey Parker (Tuck), Marshall Van Alstyne (BU) and Sangeet Paul Choudary in 'Platform Revolution' (2016), synthesising work by Eisenmann, Parker and Van Alstyne (2006) and Hagiu and Wright (2015). The pipeline framing originates with Porter (1985).

WHY IT WAS DEVELOPED

Conventional strategy frameworks (Porter's Five Forces, Value Chain) were calibrated to pipeline firms. They systematically misdiagnosed the strategic problems of Airbnb, Uber, eBay and YouTube — firms that own no inventory and produce no content but capture value by orchestrating others.

FIGURE 1 · TWO ARCHITECTURES

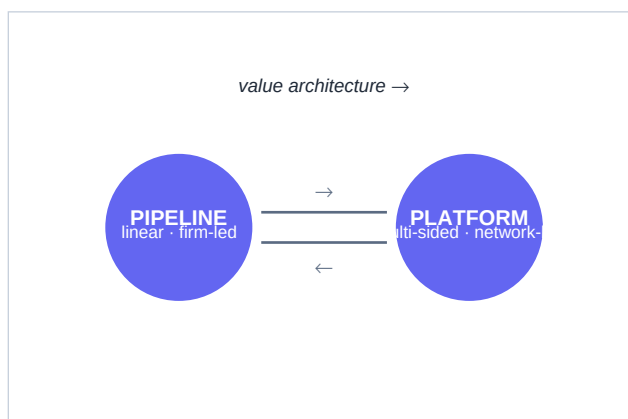


Figure 1. The pipeline produces sequentially; the platform orchestrates. Strategic logic, KPIs, and competitive moats differ across the two architectures (Parker, Van Alstyne and Choudary, 2016).

HOW IT IS USED

Strategists classify a business as pipeline or platform, then apply the appropriate playbook: pipelines optimise for cost, quality and supply-chain control; platforms optimise for network density, governance and the chicken-and-egg

WHEN IT IS USED

At the start of any business model design, in M&A due diligence (acquiring a platform requires different valuation logic), and when an incumbent considers platformisation.

BY WHOM IT IS USED

Founders and product leaders designing new businesses, M&A and PE analysts, antitrust regulators, and strategy consultants.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with Porter's value chain, the platform framing acknowledges that the firm need not own production. Compared with earlier 'two-sided market' economics (Rochet and Tirole, 2003), Parker et al. add the strategic playbook — chicken-and-egg tactics, governance design, openness decisions — that turns theory into a build manual.

RELATED FRAMEWORKS IN THIS COURSE

SMAC Stack · Slibrary 3

Network Effects · Slibrary 3

Two-Sided Markets · Slibrary 6

Multi-Sided Platforms · Slibrary 9

SOURCES (HARVARD REFERENCING STYLE)

Eisenmann, T., Parker, G. and Van Alstyne, M. (2006) 'Strategies for two-sided markets', Harvard Business Review, 84(10), pp. 92–101.
 Hagiu, A. and Wright, J. (2015) 'Multi-sided platforms', International Journal of Industrial Organization, 43, pp. 162–174.
 Parker, G.G., Van Alstyne, M.W. and Choudary, S.P. (2016) Platform Revolution. New York, NY: Norton.
 Porter, M.E. (1985) Competitive Advantage. New York, NY: Free Press.

SMAC Stack

SLIBRARY 03 · SHEET 2 / 5

Social · Mobile · Analytics · Cloud — the substrate of digital business

PROFESSIONAL DEFINITION

The SMAC Stack is a four-pillar mental model of the foundational technologies underpinning all digital businesses: Social (networked communication and media), Mobile (smartphone-first interaction), Analytics (data-driven decision support), and Cloud (elastic, on-demand computing). Together they form the 'third platform' of computing, after mainframe and client-server (Cognizant, 2013; IDC, 2014).

WHO DEVELOPED IT

Coined by Cognizant analysts in 2013 and popularised by IDC's 'Third Platform' research programme. The conceptual ancestors are Brynjolfsson and McAfee's (2014) 'second machine age' technologies and Westerman, Bonnet and McAfee's (2014) 'Leading Digital'.

WHY IT WAS DEVELOPED

Executives needed a memorable shorthand for what specifically had changed in the technology stack between ~2008 and 2014. SMAC compressed dozens of technology trends into a single four-letter mnemonic that survived board-level conversations.

FIGURE 1 · FOUR PILLARS

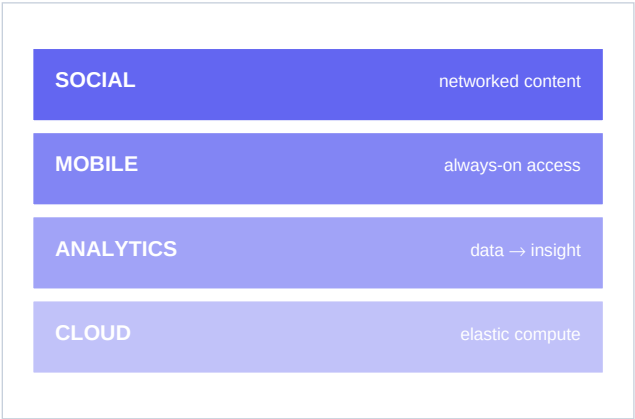


Figure 1. The SMAC stack. Each layer is independently powerful but their combination is what produced the 2010s digital boom (Cognizant, 2013).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
CIOs assess their organisation's maturity on each of the four pillars, identify the weakest link, and prioritise investment to bring all four to a consistent standard before pursuing higher-order digital initiatives (AI, IoT).	In digital strategy formulation, in IT-budget prioritisation, in vendor evaluation (does this vendor cover all four?), and in executive-education programmes.	CIOs and CTOs, technology consultants, IT-budget owners, and vendors positioning their offering against the four pillars.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with vendor-by-vendor technology lists, SMAC offers a vendor-neutral architecture. Compared with later expansions (SMAC + IoT + AI = SMACIT or SMACIB), the original four-pillar version is more memorable and was the right level of abstraction for 2014. The framing has aged but remains pedagogically useful.

RELATED FRAMEWORKS IN THIS COURSE

Platform vs. Pipeline · Slibrary 3

E-Business Infrastructure Stack · Slibrary 5

Cloud: IaaS · PaaS · SaaS · Slibrary 5

API Economy · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

Brynjolfsson, E. and McAfee, A. (2014) The Second Machine Age. New York, NY: Norton.
Cognizant (2013) The SMAC Code: Embracing New Technologies for Future Business. Teaneck, NJ: Cognizant.
IDC (2014) The Third Platform: A Powerful Framework. Framingham, MA: IDC.
Westerman, G., Bonnet, D. and McAfee, A. (2014) Leading Digital. Boston, MA: HBR Press.

Iansiti-Levien Ecosystem Patterns

SLIBRARY 03 · SHEET 3 / 5

Keystone · Niche · Dominator · Hub Landlord — the four ecosystem roles

PROFESSIONAL DEFINITION

The Iansiti-Levien framework classifies firms by the role they play in a digital business ecosystem. keystones (Apple, Microsoft) create platforms that increase value for all participants; Niche players specialise in narrow capabilities; Dominators absorb the ecosystem's value into themselves; Hub Landlords extract rents without creating value (Iansiti and Levien, 2004).

WHO DEVELOPED IT

Marco Iansiti and Roy Levien at Harvard Business School, in 'The Keystone Advantage' (2004). Their work synthesised earlier ecosystem thinking by Moore (1996) and was extended by Adner (2017) on ecosystem strategy.

WHY IT WAS DEVELOPED

Strategy frameworks treated firms as autonomous units competing on Porter's Five Forces. But by the early 2000s, Apple, Microsoft, and major retailers were clearly succeeding by orchestrating ecosystems — a phenomenon outside the existing vocabulary.

FIGURE 1 · FOUR ECOSYSTEM ROLES



Figure 1. Four roles based on what the firm contributes versus extracts. Keystones create value for the whole ecosystem; Hub Landlords extract without creating (Iansiti and Levien, 2004).

HOW IT IS USED

Executives diagnose their firm's current role and target a healthier role for the long term. Investors use the framing to evaluate ecosystem health: dominators and hub landlords erode their own foundations over time.

WHEN IT IS USED

In ecosystem strategy reviews, in antitrust analyses (regulators distinguish platform stewardship from rent extraction), and in M&A integration decisions.

BY WHOM IT IS USED

Strategy executives at platform firms, antitrust regulators, ecosystem partners (developers, app makers) evaluating which platform to invest skills in, and academics.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with Porter's bilateral five-forces logic, the ecosystem framing is explicitly multilateral and includes the firm's effect on partners. Compared with Moore's (1996) ecosystem metaphor, Iansiti-Levien adds typology and prescriptive advice. The framework anticipated by 20 years many themes of contemporary platform regulation (DMA, gatekeeper rules).

RELATED FRAMEWORKS IN THIS COURSE

Platform vs. Pipeline · Slibrary 3

Network Effects · Slibrary 3

Multi-Sided Platforms · Slibrary 9

Platform Governance · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Adner, R. (2017) 'Ecosystem as structure: An actionable construct for strategy', *Journal of Management*, 43(1), pp. 39–58.
 Iansiti, M. and Levien, R. (2004) *The Keystone Advantage*. Boston, MA: HBR Press.
 Moore, J.F. (1996) *The Death of Competition*. New York, NY: HarperBusiness.

Network Effects

SLIBRARY 03 · SHEET 4 / 5

The product becomes more valuable as more people use it

PROFESSIONAL DEFINITION

A Network Effect exists when a product's value to each user depends on the number of other users. Direct network effects arise when users benefit from same-side users (telephones, messaging). Indirect (cross-side) network effects arise when users on one side benefit from more users on another side (operating systems and developers, marketplaces and merchants) (Katz and Shapiro, 1985; Eisenmann et al., 2006).

WHO DEVELOPED IT

Formalised by Michael Katz and Carl Shapiro (1985) and extended into multi-sided settings by Rochet and Tirole (2003). The strategic implications were popularised by Shapiro and Varian (1999) in 'Information Rules', and by Parker, Van Alstyne and Choudary (2016) in 'Platform Revolution'.

WHY IT WAS DEVELOPED

Conventional micro-economics assumed that demand for a good was independent of the number of other consumers. Network effects break that assumption — and explain why digital markets reliably produce winner-takes-most outcomes.

FIGURE 1 · VALUE GROWS WITH USERS

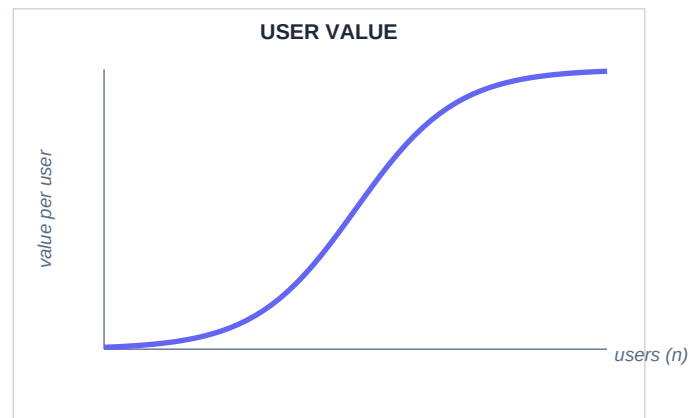


Figure 1. As user count grows, per-user value rises non-linearly. Once past critical mass, the dynamic becomes self-reinforcing (Katz and Shapiro, 1985).

HOW IT IS USED

Strategists test whether a product has network effects by asking whether its value depends on user count. If yes, growth strategy emphasises rapid adoption (subsidies, free tiers, viral loops) over short-term margin.

WHEN IT IS USED

When evaluating a digital business model, when a startup faces the chicken-and-egg problem, and when antitrust regulators assess concentration.

BY WHOM IT IS USED

Founders, product managers, growth marketers, venture capitalists, and antitrust regulators.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional supply-and-demand framing, network effects predict monopolistic outcomes from competitive dynamics alone. Compared with simple 'first-mover advantage' framing, network effects identify the specific mechanism (positive feedback) and conditions (critical mass) under which first-mover advantage actually materialises.

RELATED FRAMEWORKS IN THIS COURSE

Platform vs. Pipeline · Slibrary 3

Metcalfe's Law · Slibrary 6

Direct vs. Indirect Network Effects · Slibrary 6

Tipping Point · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

Eisenmann, T., Parker, G. and Van Alstyne, M. (2006) 'Strategies for two-sided markets', Harvard Business Review, 84(10), pp. 92–101.
 Katz, M.L. and Shapiro, C. (1985) 'Network externalities, competition, and compatibility', American Economic Review, 75(3), pp. 424–440.
 Rochet, J.-C. and Tirole, J. (2003) 'Platform competition in two-sided markets', Journal of the European Economic Association, 1(4), pp. 990–1029.
 Shapiro, C. and Varian, H.R. (1999) Information Rules. Boston, MA: HBR Press.

Super-App Model

SLIBRARY 03 · SHEET 5 / 5

WeChat · Grab · Gojek — bundled ecosystems inside one app

PROFESSIONAL DEFINITION

A Super-App is a single mobile application that bundles many distinct services — messaging, payments, shopping, transport, food delivery, financial services — behind one identity and one wallet. Originating in Asia, the model exploits cross-service network effects and switching-cost lock-in to capture user time and transaction share (Tencent, 2024; Grab, 2024).

WHO DEVELOPED IT

Pioneered by WeChat (Tencent, China) from 2013 with the launch of the Mini Programs platform. Replicated by Grab (Southeast Asia), Gojek (Indonesia), Paytm (India), and Kakao (South Korea). Western firms (Meta, X) have repeatedly attempted super-app strategies with limited success.

WHY IT WAS DEVELOPED

In emerging Asian markets, leapfrogging desktop computing meant users entered the internet via mobile only. Bundling services into one app reduced data costs, simplified onboarding for new internet users, and exploited the smartphone's limited home-screen real estate.

FIGURE 1 · BUNDLE OF SERVICES

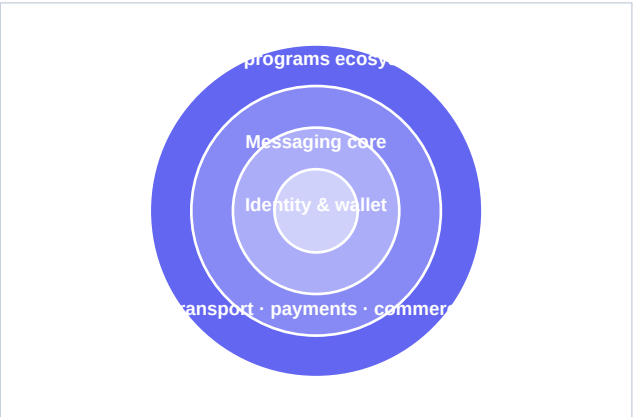


Figure 1. The super-app architecture. Identity and wallet sit at the core; messaging is the engagement engine; commerce, transport and finance sit on top; mini-programs extend the ecosystem to third-party developers (Tencent, 2024).

HOW IT IS USED

Strategists evaluate the super-app pattern as a market-entry option in mobile-first, low-trust, cash-light economies. Conditions for success: high mobile penetration, low desktop penetration, weak existing payment rails, and regulatory tolerance for

WHEN IT IS USED

In market-entry decisions for emerging markets, in fintech-adjacency strategy, and in competitive-response analysis when a super-app threatens incumbent single-purpose apps.

BY WHOM IT IS USED

CEOs of Asian conglomerates, Western firms attempting Asian expansion, antitrust and financial regulators (super-apps blur banking and tech), and consultancies advising emerging-market entry.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with Western single-purpose app strategies, the super-app captures more user time and transaction share by bundling. Compared with native operating systems, the super-app is OS-independent and bypasses Apple/Google app stores. The model is now a target of antitrust attention precisely because it concentrates many markets behind one operator (Cusumano et al., 2019).

RELATED FRAMEWORKS IN THIS COURSE

- Platform vs. Pipeline · Slibrary 3
- Network Effects · Slibrary 3
- Multi-Sided Platforms · Slibrary 9
- Platform Governance · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Cusumano, M.A., Gawer, A. and Yoffie, D.B. (2019) The Business of Platforms. New York, NY: Harper Business.
Grab (2024) Annual Report 2024. Singapore: Grab Holdings.
Steinberg, M. (2020) 'LINE as super app: Platformization in East Asia', Social Media + Society, 6(2).
Tencent (2024) WeChat Open Class 2024. Shenzhen: Tencent.

Christensen's Disruption

SLIBRARY 04 · SHEET 1 / 5

Low-end attackers with 'good-enough' products unseat incumbents

PROFESSIONAL DEFINITION

Christensen's theory of disruptive innovation describes how incumbents lose to new entrants that initially serve low-end or unserved customers with simpler, cheaper, 'good-enough' products. As the entrant improves, it climbs upmarket and displaces the incumbent — who, rationally optimising for high-margin customers, leaves the low end uncontested (Christensen, 1997; Christensen, Raynor and McDonald, 2015).

WHO DEVELOPED IT

Clayton Christensen at Harvard Business School, in 'The Innovator's Dilemma' (1997). The theory was refined in 'The Innovator's Solution' (Christensen and Raynor, 2003) and clarified in a 2015 HBR article addressing widespread misuse of the term.

WHY IT WAS DEVELOPED

Incumbent firms with strong management, attentive customers, and disciplined investment processes were nonetheless losing to upstarts. Christensen showed that the same management practices that produce success in mature markets predictably blind firms to disruptive entrants.

FIGURE 1 · DISRUPTIVE TRAJECTORY

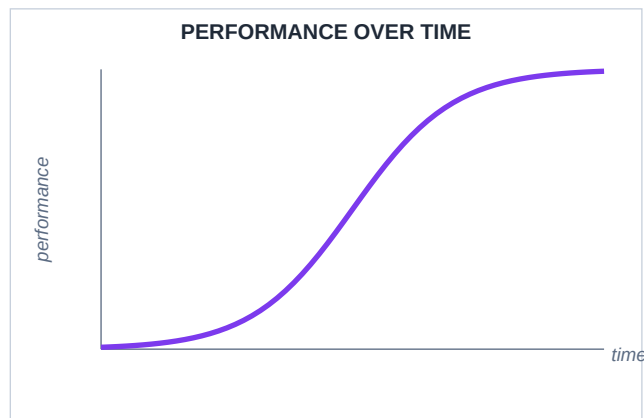


Figure 1. The disruptive entrant starts below incumbents on traditional metrics but improves faster, eventually overtaking the incumbent's offering (Christensen, 1997).

HOW IT IS USED

Incumbents use the theory to monitor for low-end attackers, set up separate business units to compete with disruptive offerings, and evaluate whether their discipline of 'listening to customers' is producing strategic blindness.

WHEN IT IS USED

In annual strategy reviews, in investment committee discussions of seemingly-marginal new entrants, and in corporate-venturing decisions about whether to build, buy, or partner with potential disruptors.

BY WHOM IT IS USED

Strategy leaders at incumbent firms, corporate venture capital teams, board members evaluating market threats, and academic strategy researchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with generic 'innovation' framings, Christensen's theory identifies a specific mechanism (asymmetric incentives at the low end) and predicts a specific trajectory (low-end → upmarket). Compared with Porter's competitive analysis, it explains why incumbents lose despite being well-managed — a paradox Porter's framework cannot resolve.

RELATED FRAMEWORKS IN THIS COURSE

S-Curve of Adoption · Slibrary 1

Industry Digitization Index · Slibrary 4

BCG 10-20-70 Rule · Slibrary 4

Before/After Music Retail Finance · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

Christensen, C.M. (1997) *The Innovator's Dilemma*. Boston, MA: HBR Press.
 Christensen, C.M. and Raynor, M. (2003) *The Innovator's Solution*. Boston, MA: HBR Press.
 Christensen, C.M., Raynor, M.E. and McDonald, R. (2015) 'What is disruptive innovation?', *Harvard Business Review*, 93(12), pp. 44–53.

Industry Digitization Index

SLIBRARY 04 · SHEET 2 / 5

McKinsey/MGI ranks sectors by digital intensity and pace

PROFESSIONAL DEFINITION

The MGI Industry Digitization Index is a composite indicator that ranks economic sectors on three dimensions of digital intensity: digital assets (hardware, software, data), digital usage (transactions, business processes, customer interactions), and digital workers (skills, work routines, remuneration premia). The Index reveals a wide and persistent gap between leaders (ICT, finance, media) and laggards (mining, agriculture, construction) (MGI, 2015; MGI, 2024).

WHO DEVELOPED IT

Developed by the McKinsey Global Institute (MGI) under James Manyika and colleagues. First published in 2015 for the US economy and subsequently extended to Europe and Asia. Inputs are drawn from BEA, BLS, and MGI proprietary surveys.

WHY IT WAS DEVELOPED

Cross-sector digital comparisons were dominated by anecdote and headline consultancy claims. MGI created the Index to give boards a defensible benchmark — where am I, relative to my sector, and where is my sector relative to others?

FIGURE 1 · DIGITIZATION INDEX (US, 2024)

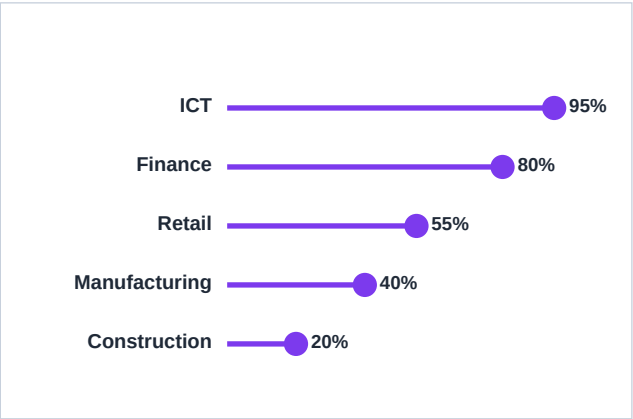


Figure 1. Sector rankings on the digitization index. ICT and finance lead; construction and agriculture lag. The gap is widening, not closing (MGI, 2024).

HOW IT IS USED

Boards use the Index to benchmark their firm against sector peers and against leading sectors. Investors use it to identify under-digitised sectors with high transformation upside. Policymakers use it to target reskilling and infrastructure.

WHEN IT IS USED

In annual strategy reviews, in PE/VC sector-thesis development, in industrial policy formulation, and in academic productivity analyses.

BY WHOM IT IS USED

Boards and CEOs, McKinsey clients, PE/VC investors, ministries of industry, and academic researchers studying sectoral productivity differences.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-metric rankings (e.g., 'IT spend / revenue'), the composite index captures three dimensions of digital intensity. Compared with subjective consultancy maturity scores, it is grounded in publicly verifiable data inputs. The Index has become the de-facto benchmark cited in every major McKinsey digital publication since 2015.

RELATED FRAMEWORKS IN THIS COURSE

- Christensen Disruption · Slibrary 4
- BCG 10-20-70 Rule · Slibrary 4
- Digital Maturity Model · Slibrary 13
- WEF Future of Jobs · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

MGI (2015) Digital America: A Tale of the Haves and Have-Mores. Washington, DC: McKinsey Global Institute.
MGI (2024) The State of Digital America 2024. New York, NY: McKinsey Global Institute.
Manyika, J., Ramaswamy, S., Khanna, S., Sarrazin, H., Pinkus, G., Sethupathy, G. and Yaffe, A. (2015) 'Digital America: A tale of the haves and have-mores', McKinsey Quarterly, December.

BCG 10-20-70 Rule

SLIBRARY 04 · SHEET 3 / 5

10% algorithms · 20% tech · 70% people & process

PROFESSIONAL DEFINITION

The BCG 10-20-70 Rule is a budgeting heuristic for AI and digital transformation investment: roughly 10% of effort and budget should go to algorithm and model development, 20% to underlying technology and data infrastructure, and 70% to people, process change and organisational integration. The rule emerged from BCG's analysis of why so many AI investments fail to capture value (BCG, 2023; Sukharevsky and Westra, 2024).

WHO DEVELOPED IT

Articulated by BCG's AI practice in studies of AI value capture, building on earlier research with MIT Sloan Management Review (Ransbotham et al., 2017-2024). Variants of the rule appear under different names in McKinsey, Accenture, and Deloitte transformation literature.

WHY IT WAS DEVELOPED

Executive teams habitually over-invested in glamorous algorithm development and under-invested in the unglamorous work of change management, process redesign, and skills development — producing the well-documented gap between AI 'pilots' and AI value at scale.

FIGURE 1 · THE 10-20-70 SPLIT



Figure 1. The 10-20-70 split of AI/transformation investment. The largest share goes to organisational change — the most-cited cause of AI value-capture failure (BCG, 2023).

HOW IT IS USED

Programme leaders re-baseline their planned investment against the 10-20-70 split, identify under-invested categories (almost always: people and process), and rebalance budget and headcount accordingly.

WHEN IT IS USED

In AI investment proposal reviews, in transformation programme design, in post-mortem analyses of underperforming AI initiatives, and in board-level AI strategy discussions.

BY WHOM IT IS USED

CIOs, Chief AI Officers, transformation programme directors, board AI committees, and consultancies advising on AI scale-up.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with technology-centric AI investment templates, 10-20-70 elevates organisational change to the dominant budget line. Compared with vague calls for 'change management', it gives a quantitative target that boards can monitor. Critics note that the exact split varies by sector and project type — the rule is a directional heuristic, not a precise budget formula.

RELATED FRAMEWORKS IN THIS COURSE

- Christensen Disruption · Slibrary 4
- Industry Digitization Index · Slibrary 4
- McKinsey 7S — Digital · Slibrary 13
- Transformation Roadmap · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

BCG (2023) Where's the Value in AI? Boston, MA: Boston Consulting Group.
Ransbotham, S., Khodabandeh, S., LaFountain, B., Kiron, D. and Sukharevsky, A. (2024) Expanding AI's Impact with Organizational Learning. Cambridge, MA: MIT Sloan Management Review and BCG.
Sukharevsky, A. and Westra, B. (2024) Generative AI: From Hype to Real Value. Boston, MA: BCG.

Before/After: Music · Retail · Finance

SLIBRARY 04 · SHEET 4 / 5

Three canonical sector transformations in one diagram

PROFESSIONAL DEFINITION

The Before/After framing is a pedagogical exhibit comparing three industries — Music, Retail, and Finance — across a 25-year window (~1995-2025). In each, the value chain was vertically integrated and physical-asset-heavy in the 'before'; in the 'after' it has been unbundled, dematerialised, and reorganised around platforms (IFPI, 2024; eMarketer, 2024; BIS, 2024).

WHO DEVELOPED IT

The Before/After exhibit is a compilation rather than a single author's work. Source data: IFPI Global Music Report (music), eMarketer / Digital Commerce 360 (retail), Bank for International Settlements (finance). Pedagogical synthesis is common in MBA strategy and digital-economy textbooks (Gans, 2016; Brynjolfsson and McAfee, 2014).

WHY IT WAS DEVELOPED

Students and executives needed a memorable, evidence-based way to grasp the magnitude of digital transformation. Three contrasting sectors — each with different digitisation drivers — together demonstrate that the pattern is general, not industry-specific.

FIGURE 1 · BEFORE → AFTER

Sector	1995 BEFORE	2025 AFTER
Music	CDs · stores	streaming · subs.
Retail	high street	e-comm · platform
Finance	branch · cash	mobile · neobank

Figure 1. Three sectors compared across 30 years of digitisation. Each shifted from physical-vertical integration to digital-platform-mediated delivery (IFPI, 2024; eMarketer, 2024; BIS, 2024).

HOW IT IS USED

Educators use the exhibit to anchor the digital-transformation conversation. Strategists use the analogy to argue that their own sector — even if it appears stable — is on the same trajectory.

WHEN IT IS USED

In MBA introductions to the digital economy, in board-level briefings on industry trajectory, and in due-diligence reports for late-digitising sectors.

BY WHOM IT IS USED

MBA faculty, McKinsey/BCG/Bain partners, due-diligence teams, board strategy advisors, and journalists writing on digital transformation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-sector case studies, the three-sector exhibit shows the generality of the pattern. Compared with abstract technology forecasting, it anchors the transformation in concrete, well-known industries that students already know — making the abstract immediately legible.

RELATED FRAMEWORKS IN THIS COURSE

- Christensen Disruption · Slibrary 4
- Industry Digitization Index · Slibrary 4
- BCG 10-20-70 Rule · Slibrary 4
- Atoms → Bits · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

BIS (2024) Annual Economic Report 2024. Basel: Bank for International Settlements.
Brynjolfsson, E. and McAfee, A. (2014) The Second Machine Age. New York, NY: Norton.
eMarketer (2024) Global E-commerce Forecast 2024. New York, NY: Insider Intelligence.
IFPI (2024) Global Music Report 2024. London: International Federation of the Phonographic Industry.

WEF Future of Jobs

SLIBRARY 04 · SHEET 5 / 5

By 2030: 170M jobs created · 92M displaced · net +78M

PROFESSIONAL DEFINITION

The World Economic Forum's Future of Jobs Report is a biennial survey of 1,000+ global employers projecting labour-market shifts driven by technology, the green transition, demographic change, and economic uncertainty. The 2025 edition forecasts 170 million new jobs and 92 million displacements between 2025 and 2030 — a net gain of 78 million jobs but with severe sectoral and skills churn (WEF, 2025).

WHO DEVELOPED IT

Produced by the World Economic Forum's Centre for the New Economy and Society, with methodological lead by Saadia Zahidi. The report is grounded in a survey of Chief Economists and HR leaders representing some 14 million workers across 22 industries and 55 economies.

WHY IT WAS DEVELOPED

Generic 'AI will destroy jobs' headlines obscured the more nuanced reality of creation, destruction, and skill-shift happening in parallel. The WEF report was designed to give employers, governments and educators a single, regularly-updated evidence base for workforce planning.

FIGURE 1 · NET JOB CHANGE BY 2030 (M)

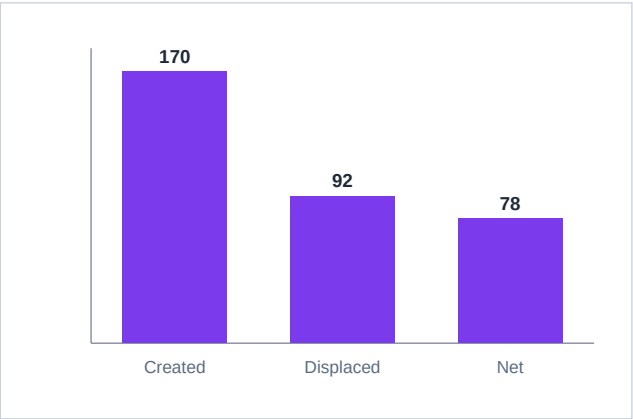


Figure 1. The headline numbers from the WEF 2025 report. Net positive — but the creation and destruction occur in different occupations, requiring large-scale reskilling (WEF, 2025).

HOW IT IS USED

Workforce planners use the sectoral and skills breakdowns to forecast hiring and reskilling needs. Educators align curricula to growing skills clusters. Governments target reskilling subsidies at workers in shrinking categories.

WHEN IT IS USED

In biennial workforce strategy reviews, in national skills-policy discussions, and as evidence base for collective-bargaining and pension-system stress tests.

BY WHOM IT IS USED

Chief Human Resource Officers, ministries of labour, OECD/ILO analysts, education-policy makers, and trade unions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with macroeconomic projections of total employment, the WEF report decomposes change by occupation, skill cluster and reason. Compared with consultancy reports on AI specifically, it includes the green transition, demographics and geo-economic fragmentation as competing forces — producing a richer picture of labour-market futures.

RELATED FRAMEWORKS IN THIS COURSE

Industry Digitization Index · Slibrary 4

BCG 10-20-70 Rule · Slibrary 4

Reskilling vs. Upskilling · Slibrary 14

Skills Matrix 2025-2030 · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

OECD (2024) OECD Employment Outlook 2024. Paris: OECD Publishing.
WEF (2023) The Future of Jobs Report 2023. Geneva: World Economic Forum.
WEF (2025) The Future of Jobs Report 2025. Geneva: World Economic Forum.

E-Business vs E-Commerce

SLIBRARY 05 · SHEET 1 / 5

The architecture of digital firms beyond online sales

PROFESSIONAL DEFINITION

E-Commerce denotes the buying and selling of goods and services over electronic networks. E-Business is the broader concept: the digitalisation of all business processes — internal (HR, finance, operations), inter-organisational (supply chain), and customer-facing (sales, service). Every e-commerce site is part of an e-business; not every e-business sells online (IBM, 1997; Laudon and Traver, 2022).

WHO DEVELOPED IT

The e-business term was coined by IBM in a 1997 marketing campaign and formalised by Weill and Vitale (2001) in 'Place to Space'. Laudon and Traver (2022) provide the canonical textbook treatment used in business schools worldwide.

WHY IT WAS DEVELOPED

Confusing e-commerce with e-business causes managers to under-invest. They build a transactional website but neglect the back-office systems, data architecture, and process re-engineering that turn online sales into competitive advantage.

FIGURE 1 · NESTED SCOPE

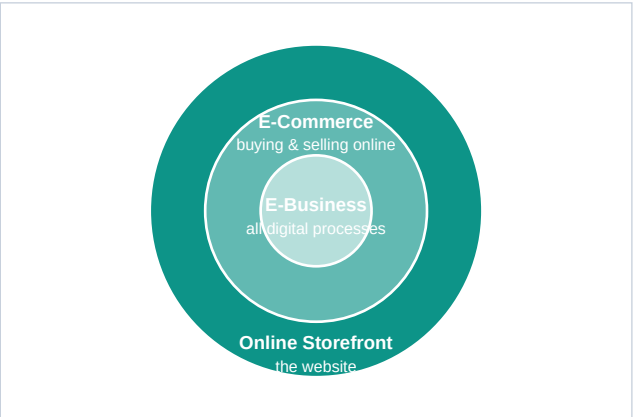


Figure 1. E-commerce is a subset of e-business. The storefront is a subset of e-commerce. Architecting the outer rings — supply chain, ERP, CRM, analytics — is what creates durable digital advantage (Laudon and Traver, 2022).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Map a firm's processes onto the three rings. For each, ask whether it is digitised, digitalised (process redesigned around digital), or paper-based. The gap between rings reveals the next investment priority.	During digital strategy reviews, M&A integration planning, and ERP/CRM business cases.	Chief Information Officers, enterprise architects, digital transformation consultants, and business school instructors introducing digital business.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the narrower e-commerce framing prevalent in the 1990s, the e-business concept directs management attention to the entire enterprise architecture. Compared with later 'digital business' or 'digital-native' framings, it remains the most precise vocabulary for distinguishing transactional websites from digitised firms.

RELATED FRAMEWORKS IN THIS COURSE

Digital Infrastructure Stack · Slibrary 5

Cloud Computing IaaS/PaaS/SaaS · Slibrary 5

Value Chain → Network · Slibrary 7

API Economy · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

IBM (1997) e-business: Solutions for the New Economy. Armonk, NY: IBM Corporation.
Laudon, K.C. and Traver, C.G. (2022) E-Commerce 2022: Business, Technology, Society, 17th edn. Harlow: Pearson.
Weill, P. and Vitale, M.R. (2001) Place to Space: Migrating to eBusiness Models. Boston, MA: Harvard Business School Press.

Digital Infrastructure Stack

SLIBRARY 05 · SHEET 2 / 5

From physical fibre to user-facing applications — the seven layers of digital business

PROFESSIONAL DEFINITION

The Digital Infrastructure Stack is a layered model decomposing the technical foundations of any digital service into stacked dependencies: physical (fibre, data centres), network (Internet protocols), platform (cloud), data, applications, interfaces, and user devices. Each layer supplies services to those above and depends on those below (OECD, 2020; WEF, 2023).

WHO DEVELOPED IT

Layered models of digital infrastructure draw on telecommunications stack thinking (OSI/TCP-IP) and were generalised for the digital economy by the OECD (2020) 'Going Digital' framework and the WEF Digital Infrastructure series (WEF, 2023). Brynjolfsson and McAfee (2014) popularised the idea of 'general-purpose technology stacks'.

WHY IT WAS DEVELOPED

Decision-makers — public and private — increasingly need to invest across multiple layers simultaneously. Without a shared stack vocabulary, debates about 'digital sovereignty', 'cloud independence' or 'AI compute' descend into confusion.

FIGURE 2 · INFRASTRUCTURE LAYERS

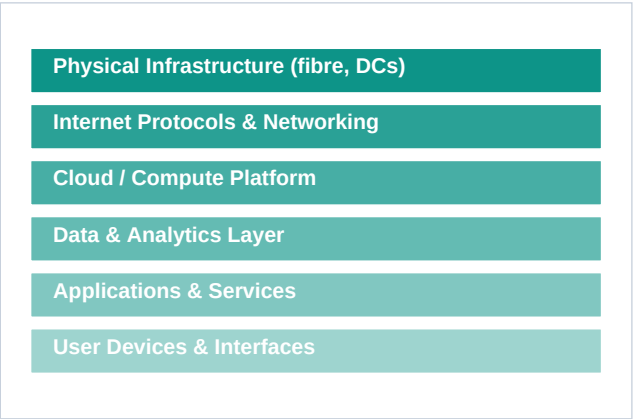


Figure 2. The digital infrastructure stack. Each layer is a market with its own competitive dynamics and policy issues. Vertical integration across layers is a defining feature of the largest digital firms (OECD, 2020).

HOW IT IS USED

Use the stack to scope strategy questions ('which layer are we competing on?'), investment decisions ('which layer are we under-served by?') and policy analysis ('which layer is the regulatory bottleneck?').

WHEN IT IS USED

Sovereignty and resilience reviews, infrastructure investment decisions, vendor selection, and digital industrial-policy formulation.

BY WHOM IT IS USED

OECD economists, Chief Technology Officers, digital-policy ministries, infrastructure investors, and vendor-strategy teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the OSI seven-layer networking model, the digital infrastructure stack is explicitly economic: it labels each layer with its market structure and competitive dynamics. Compared with abstract 'digital economy' diagrams, it grounds the analysis in physical assets and dependencies.

RELATED FRAMEWORKS IN THIS COURSE

Cloud Computing IaaS/PaaS/SaaS · Slibrary 5

API Economy · Slibrary 5

Technology Ecosystems · Slibrary 3

OECD 3-Layer Model · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Brynjolfsson, E. and McAfee, A. (2014) The Second Machine Age. New York, NY: W.W. Norton.
OECD (2020) Going Digital integrated policy framework. Paris: OECD Publishing.
WEF (2023) Digital Infrastructure: Global Public Goods. Geneva: World Economic Forum.

B2B / B2C / C2C / B2G Models

SLIBRARY 05 · SHEET 3 / 5

The four canonical e-business archetypes, by counterparty

PROFESSIONAL DEFINITION

E-business models are classified by who transacts with whom: Business-to-Business (B2B), Business-to-Consumer (B2C), Consumer-to-Consumer (C2C), and Business-to-Government (B2G). Each archetype has distinct economics, sales cycles, regulatory exposures, and growth patterns (Laudon and Traver, 2022; Turban et al., 2018).

WHO DEVELOPED IT

The taxonomy emerged in the late 1990s in industry analyses (Forrester, Gartner) and was codified for academic use by Laudon and Traver (2022) and Turban et al. (2018). The OECD adopts the same classification in its e-commerce statistics.

WHY IT WAS DEVELOPED

Classifying a digital business by counterparty type immediately tells you its sales-cycle length (B2B long, B2C short), customer-acquisition costs, regulatory burden (B2G heavy), and the relevant benchmarks. It is the first cut of any digital-business analysis.

FIGURE 3 · COUNTERPARTY MATRIX

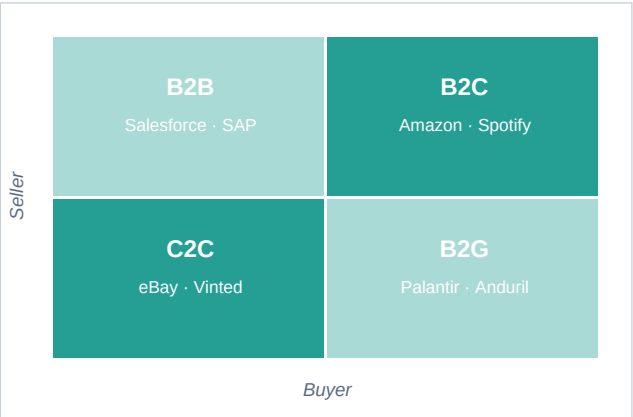


Figure 3. The four canonical e-business archetypes. Most large digital firms operate in more than one cell — Amazon is B2C (retail) and B2B (AWS); Microsoft is B2B (software) and B2G (defence cloud) (Laudon and Traver, 2022).

HOW IT IS USED

When analysing a digital firm, identify the dominant cell and any secondary cells. Apply the relevant benchmark set: B2B firms benchmark against software-as-a-service metrics; B2C against retail and engagement metrics; B2G against procurement-cycle KPIs.

WHEN IT IS USED

Initial competitive scans, investor pitches, market-sizing exercises and any case-study analysis in business-school courses.

BY WHOM IT IS USED

Investment analysts, market-research firms, business-school students, and product managers selecting their go-to-market motion.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with industry-based classifications (retail vs. software vs. media), the counterparty matrix is structural rather than sectoral, which makes it more durable across waves of platform innovation. Compared with a single 'digital business' label, it forces the analyst to separate transactional logics that would otherwise be conflated.

RELATED FRAMEWORKS IN THIS COURSE

E-Business vs E-Commerce · Slibrary 5

Multi-Sided Platforms · Slibrary 11

Two-Sided Markets · Slibrary 6

Platform vs Marketplace · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Laudon, K.C. and Traver, C.G. (2022) E-Commerce 2022: Business, Technology, Society. 17th edn. Harlow: Pearson.
OECD (2019) Unpacking E-commerce: Business Models, Trends and Policies. Paris: OECD Publishing.
Turban, E., Outland, J., King, D., Lee, J.K., Liang, T.-P. and Turban, D.C. (2018) Electronic Commerce 2018. 9th edn. Cham: Springer.

API Economy

SLIBRARY 05 · SHEET 4 / 5

When integration becomes a market — software functionality as a tradable service

PROFESSIONAL DEFINITION

The API Economy refers to the commercial ecosystem that emerges when firms expose their internal capabilities as Application Programming Interfaces (APIs) that other firms can consume — typically priced per call or per tier. APIs convert software functions into services that can be combined, resold, and traded (Iyer and Subramaniam, 2015; Capgemini, 2023).

WHO DEVELOPED IT

The term was popularised by Iyer and Subramaniam (2015) in Harvard Business Review. It draws on earlier service-oriented-architecture literature (Erl, 2007). Capgemini's annual API Economy reports (2020-2024) provide the most-cited industry estimates.

WHY IT WAS DEVELOPED

Once core functions (payments, identity, mapping, language understanding) are available as APIs, new firms can launch in days instead of years — but they also become structurally dependent on the API providers. Understanding the API economy is central to understanding modern competitive dynamics.

FIGURE 4 · API VALUE CHAIN

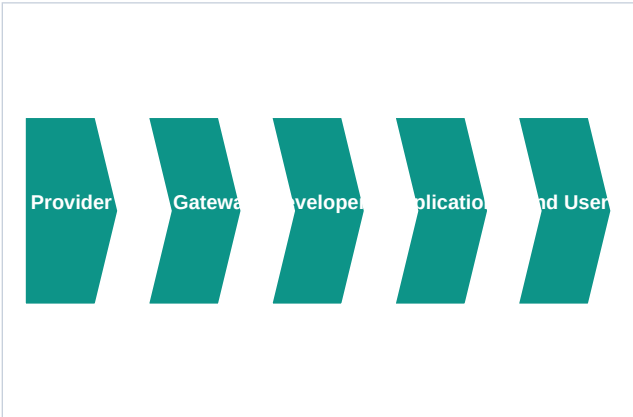


Figure 4. The API value chain. Stripe, Twilio, Plaid and OpenAI are 'API-first' firms whose entire business is selling functionality through APIs to other developers (Iyer and Subramaniam, 2015).

HOW IT IS USED

Firms decide which capabilities to keep internal, which to expose as APIs (creating potential platform leverage), and which to consume from external APIs (creating speed but also dependency).

WHEN IT IS USED

When designing platform strategy, build-vs-buy decisions, vendor concentration analysis, and digital-resilience reviews.

BY WHOM IT IS USED

Chief Technology Officers, platform architects, venture-capital investors evaluating API-first start-ups, and competition authorities analysing platform power.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional software licensing — where firms bought a CD and integrated it themselves — the API economy treats integration as the product. Compared with the broader 'platform economy', the API frame makes the unit of analysis the technical interface, which is what most digital integration disputes turn on (Apple App Store, GDPR data-portability).

RELATED FRAMEWORKS IN THIS COURSE

- Cloud Computing IaaS/PaaS/SaaS · Slibrary 5
- Multi-Sided Platforms · Slibrary 11
- Digital Infrastructure Stack · Slibrary 5
- Platform Governance · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Capgemini (2023) The API Economy: Connected Business at Scale. Paris: Capgemini Research Institute.
Erl, T. (2007) SOA: Principles of Service Design. Upper Saddle River, NJ: Prentice Hall.
Iyer, B. and Subramaniam, M. (2015) 'The strategic value of APIs', Harvard Business Review, 7 January.

Cloud Computing — IaaS, PaaS, SaaS

SLIBRARY 05 · SHEET 5 / 5

The three layers of utility computing and the firms that own them

PROFESSIONAL DEFINITION

Cloud Computing delivers compute, storage and software as on-demand utility services. Three canonical layers exist: Infrastructure-as-a-Service (IaaS, raw compute and storage), Platform-as-a-Service (PaaS, runtime and developer tools), and Software-as-a-Service (SaaS, finished applications). Each is a distinct market with distinct economics (NIST, 2011; Gartner, 2024).

WHO DEVELOPED IT

The IaaS/PaaS/SaaS taxonomy was formalised by the US National Institute of Standards and Technology (Mell and Grance, 2011). The dominant providers are AWS, Microsoft Azure and Google Cloud Platform; Gartner's 'Magic Quadrant' is the standard industry reference.

WHY IT WAS DEVELOPED

Cloud is the single largest infrastructure shift since the Internet itself. Knowing which layer of cloud a firm consumes determines its cost structure, scalability, lock-in exposure, and resilience profile.

FIGURE 5 · CLOUD STACK

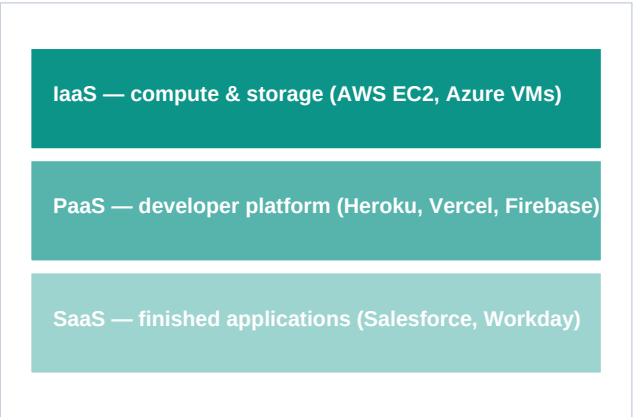


Figure 5. The cloud computing stack. Each layer abstracts more away from the customer — and creates more dependence on the provider. The largest providers operate in all three layers, raising competition concerns (Mell and Grance, 2011; Gartner, 2024).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Map a firm's IT estate onto the three layers. The deeper the abstraction (SaaS), the lower the operational burden but the higher the switching cost. Use this map to negotiate contracts, plan exits, and stress-test resilience.	IT-strategy reviews, vendor-concentration risk assessments, sovereignty audits, and competition-policy analyses (e.g. UK CMA cloud market investigation 2023-2024).	Chief Information Officers, procurement leaders, enterprise architects, competition regulators, and business-school instructors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with on-premise IT — where firms owned and operated their own servers — cloud transforms a fixed cost into a variable cost and a capital expenditure into an operating expenditure, fundamentally changing how digital businesses scale. Compared with simpler 'cloud vs. on-prem' framings, the IaaS/PaaS/SaaS distinction reveals which layer is the actual locus of competition or lock-in.

RELATED FRAMEWORKS IN THIS COURSE

Digital Infrastructure Stack · Slibrary 5

API Economy · Slibrary 5

Lock-In & Switching Costs · Slibrary 8

Subscription Models · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

Gartner (2024) Magic Quadrant for Strategic Cloud Platform Services. Stamford, CT: Gartner.
Mell, P. and Grance, T. (2011) The NIST Definition of Cloud Computing. Special Publication 800-145. Gaithersburg, MD: NIST.
UK Competition and Markets Authority (2024) Cloud Services Market Investigation: Provisional Decision Report. London: CMA.

Metcalfe's Law

SLIBRARY 06 · SHEET 1 / 5

The value of a network grows with the square of its users

PROFESSIONAL DEFINITION

Metcalfe's Law states that the value of a communications network is proportional to the square of the number of connected users ($V \propto n^2$). Originally formulated by Robert Metcalfe for Ethernet, it captures the combinatorial growth of possible connections as the user base expands (Metcalfe, 2013; Shapiro and Varian, 1999).

WHO DEVELOPED IT

Robert Metcalfe (co-inventor of Ethernet) formulated the law in the 1980s; it was popularised by George Gilder in 1993 and formalised by Metcalfe (2013) using empirical Facebook data. Critics including Briscoe, Odlyzko and Tilly (2006) argued for $n \cdot \log(n)$ as a more accurate growth function.

WHY IT WAS DEVELOPED

Almost every digital business with a social or transactional component depends on network effects for its competitive moat. Metcalfe's Law is the simplest and most-cited quantification of why.

FIGURE 6 · QUADRATIC VALUE

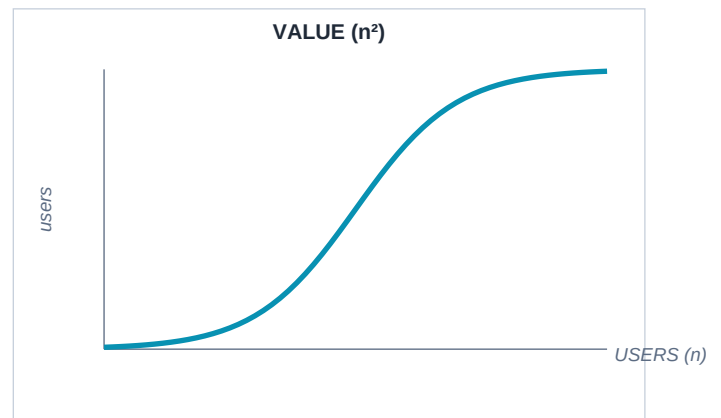


Figure 6. Network value scales with n^2 : doubling users quadruples possible connections. The shape implies large positive feedback once a network reaches critical mass (Metcalfe, 2013).

HOW IT IS USED

Estimate a network's potential value, identify where it sits on the curve, and benchmark user-acquisition spending against the implied marginal value of each additional user.

WHEN IT IS USED

Valuing social networks and marketplaces, investor diligence on network-effect businesses, and academic introductions to network economics.

BY WHOM IT IS USED

Venture capitalists, technology economists, marketing strategists, and competition regulators evaluating network-effect dominance.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with linear growth assumptions, Metcalfe's Law captures the explosive value creation typical of digital networks. Compared with later refinements (Briscoe et al.'s $n \cdot \log(n)$), it remains the dominant heuristic because of its simplicity. The framework predates and underlies virtually all modern platform-economics theory.

RELATED FRAMEWORKS IN THIS COURSE

Direct & Indirect Network Effects · Slibrary 6

Two-Sided Markets · Slibrary 6

Tipping Point · Slibrary 6

Connection Density · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

Briscoe, B., Odlyzko, A. and Tilly, B. (2006) 'Metcalfe's Law is wrong', IEEE Spectrum, 43(7), pp. 34-39.
 Metcalfe, B. (2013) 'Metcalfe's Law after 40 years of Ethernet', IEEE Computer, 46(12), pp. 26-31.
 Shapiro, C. and Varian, H.R. (1999) Information Rules: A Strategic Guide to the Network Economy. Boston, MA: Harvard Business School Press.

Direct & Indirect Network Effects

SLIBRARY 06 · SHEET 2 / 5

When users attract users, and when one user group attracts another

PROFESSIONAL DEFINITION

Direct (same-side) network effects occur when additional users of a service make the service more valuable to other users of the same type — telephone, WhatsApp, Facebook. Indirect (cross-side) network effects occur when more users of one type attract users of another type — riders attract drivers on Uber; merchants attract shoppers on Amazon (Rochet and Tirole, 2003; Parker, Van Alstyne and Choudary, 2016).

WHO DEVELOPED IT

The distinction was formalised in two-sided market theory by Rochet and Tirole (2003). Eisenmann, Parker and Van Alstyne (2006) translated it for managers in Harvard Business Review. Parker, Van Alstyne and Choudary (2016) made it the cornerstone of platform-strategy writing.

WHY IT WAS DEVELOPED

Same-side and cross-side effects require different launch strategies. Same-side network effects need viral growth; cross-side effects need careful subsidy of one side to bootstrap the other.

FIGURE 7 · TWO TYPES OF NE

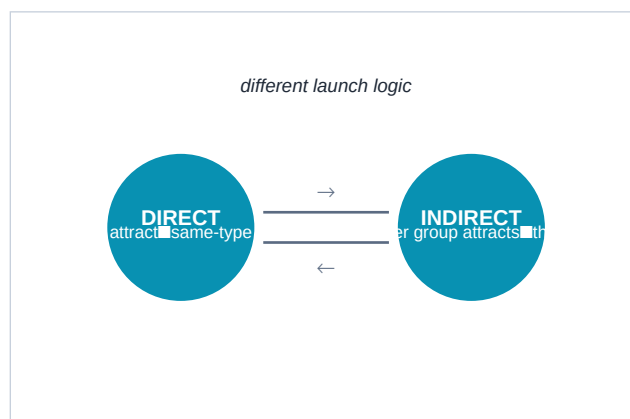


Figure 7. Direct vs indirect network effects. WhatsApp grew through direct effects (more users → more contacts). Uber grew through indirect effects (more drivers → shorter wait times → more riders → more drivers) (Parker et al., 2016).

HOW IT IS USED

Decompose a platform's network effects: which sides exist, which side attracts which, where are the strongest cross-side links. Then design subsidies, pricing and onboarding to ignite the strongest link first.

WHEN IT IS USED

Platform-launch strategy, two-sided pricing decisions, and competition analysis of network-effect-driven dominance.

BY WHOM IT IS USED

Platform-business founders and CMOs, venture-capital investors, antitrust economists, and platform-strategy consultants.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the undifferentiated 'network effects' of 1990s writing, the direct/indirect split allows precise diagnosis of why one platform succeeds and another fails. It remains the foundation of modern platform economics.

RELATED FRAMEWORKS IN THIS COURSE

Metcalfe's Law · Slibrary 6

Two-Sided Markets · Slibrary 6

Chicken-and-Egg Problem · Slibrary 11

Multi-Sided Platforms · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Eisenmann, T., Parker, G. and Van Alstyne, M. (2006) 'Strategies for two-sided markets', Harvard Business Review, 84(10), pp. 92-101.
 Parker, G.G., Van Alstyne, M.W. and Choudary, S.P. (2016) Platform Revolution. New York, NY: W.W. Norton.
 Rochet, J.-C. and Tirole, J. (2003) 'Platform competition in two-sided markets', Journal of the European Economic Association, 1(4), pp. 990-1029.

Two-Sided Markets

SLIBRARY 06 · SHEET 3 / 5

Rochet & Tirole's Nobel-prize-winning framework for platform economics

PROFESSIONAL DEFINITION

A Two-Sided Market is one in which a platform serves two distinct user groups whose value depends on each other — and where the platform's pricing on each side determines participation on the other. The platform's optimal price on either side may be below cost, even free or subsidised, when cross-side externalities are large (Rochet and Tirole, 2003; 2006).

WHO DEVELOPED IT

Jean-Charles Rochet and Jean Tirole established the formal theory in the early 2000s. Tirole's 2014 Nobel Prize in Economics cited this work. David Evans and Richard Schmalensee (2016) developed the managerial implications in 'Matchmakers'.

WHY IT WAS DEVELOPED

Two-sided market theory explains the most counter-intuitive feature of digital platforms: why one side often pays nothing or is even paid (subsidised) while the other pays a premium. Without it, free services like Google Search look economically irrational.

FIGURE 8 · TWO SIDES

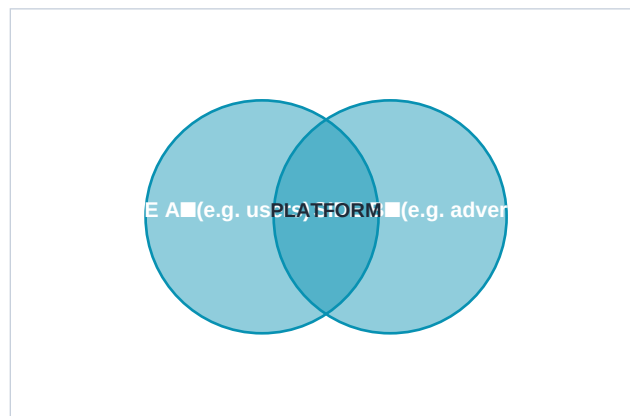


Figure 8. The platform mediates between two interdependent groups. Pricing on each side is set strategically with reference to cross-side externalities — not to side-specific marginal cost (Rochet and Tirole, 2003).

HOW IT IS USED

Identify the two sides; estimate the cross-side externality each direction; set price to the more elastic side below cost where externalities justify it; recoup margin on the less price-sensitive side.

WHEN IT IS USED

Pricing strategy for any platform business, M&A valuation of two-sided firms, and antitrust review of zero-priced services.

BY WHOM IT IS USED

Platform CFOs, antitrust economists, regulators (DMA designation logic), and graduate-school industrial-organisation courses.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-sided market theory, the two-sided framework explains why ad-funded platforms can sustainably price one side at zero. It is the conceptual foundation for much modern competition policy, including the EU Digital Markets Act.

RELATED FRAMEWORKS IN THIS COURSE

Direct & Indirect Network Effects · Slibrary 6

Multi-Sided Platforms · Slibrary 11

Subsidised Side Pricing · Slibrary 11

Platform Governance · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Evans, D.S. and Schmalensee, R. (2016) Matchmakers: The New Economics of Multisided Platforms. Boston, MA: Harvard Business Review Press.
 Rochet, J.-C. and Tirole, J. (2003) 'Platform competition in two-sided markets', Journal of the European Economic Association, 1(4), pp. 990-1029.
 Rochet, J.-C. and Tirole, J. (2006) 'Two-sided markets: a progress report', RAND Journal of Economics, 37(3), pp. 645-667.

Connection Density (MIT)

SLIBRARY 06 · SHEET 4 / 5

Why dense, frequently-used networks are more valuable than large dormant ones

PROFESSIONAL DEFINITION

Connection Density measures the rate at which a network's potential connections actually produce active interactions. A network with one billion users who never message each other is worth less than one with one hundred million who do. The MIT Initiative on the Digital Economy emphasises density as the missing variable in classical network-effect models (Aral, 2021; MIT IDE, 2025).

WHO DEVELOPED IT

Sinan Aral (MIT Sloan) developed connection-density analysis in 'The Hype Machine' (2021), drawing on his research at the MIT Initiative on the Digital Economy. The MIT IDE 2025 Annual Report formalises density metrics for AI-augmented networks.

WHY IT WAS DEVELOPED

Vanity metrics (registered users, total downloads) routinely overstate network value. Density metrics — daily active users, messages per user per day, transactions per node — are what predict revenue and durability.

FIGURE 9 · DENSITY DECOMPOSED

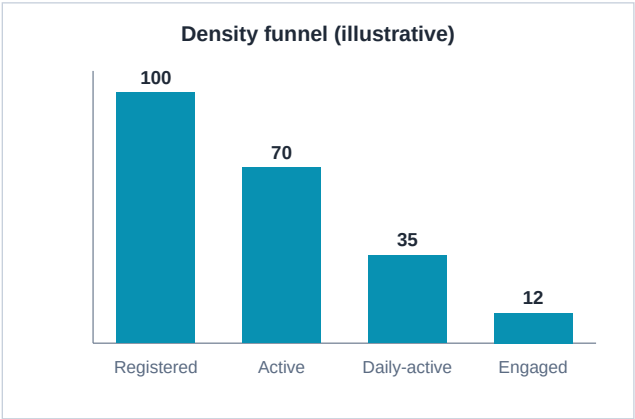


Figure 9. The density funnel. Registered users are necessary but not sufficient. Engagement density — the share of users who interact daily — is what determines advertiser value, retention and durability (Aral, 2021).

HOW IT IS USED

Replace headcount metrics with density metrics in dashboards. Decompose user growth by activity bucket. Diagnose declining engagement before declining membership (the leading indicator).

WHEN IT IS USED

Investor reporting, platform-health monitoring, social-network research, and the DMA's gatekeeper designation analysis (which now considers active rather than registered users).

BY WHOM IT IS USED

Growth and analytics teams at platform firms, digital-marketing analysts, academic network scientists, and competition regulators.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with Metcalfe's Law (which assumes all users are equally connected), density metrics expose hollow networks. Compared with engagement metrics in isolation, the density framework integrates engagement into network theory.

RELATED FRAMEWORKS IN THIS COURSE

- Metcalfe's Law · Slibrary 6
- Direct & Indirect Network Effects · Slibrary 6
- Tipping Point · Slibrary 6
- Hyper-Personalisation · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Aral, S. (2021) The Hype Machine: How Social Media Disrupts Our Elections, Our Economy, and Our Health. New York, NY: Currency.
MIT Initiative on the Digital Economy (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Sloan School of Management.
Watts, D.J. and Strogatz, S.H. (1998) 'Collective dynamics of small-world networks', Nature, 393, pp. 440-442.

Tipping Point

SLIBRARY 06 · SHEET 5 / 5

The critical mass at which a network becomes self-sustaining — and competitors collapse

PROFESSIONAL DEFINITION

The Tipping Point is the threshold of users (or transactions, or content density) above which a network's growth becomes self-sustaining and below which it decays. In digital markets, tipping is sharp because feedback loops are strong: once a network passes the threshold, smaller competitors typically collapse rapidly (Schelling, 1971; Gladwell, 2000; Eisenmann et al., 2006).

WHO DEVELOPED IT

The economic concept dates to Thomas Schelling's segregation models (1971) and Mark Granovetter's threshold models (1978). Malcolm Gladwell (2000) popularised it for general audiences. Eisenmann, Parker and Van Alstyne (2006) applied it to platform markets in Harvard Business Review.

WHY IT WAS DEVELOPED

Knowing where the tipping point sits — and how far a competitor is from it — is the central tactical question in platform launches and platform-vs-platform competition.

FIGURE 10 · TIPPING DYNAMICS

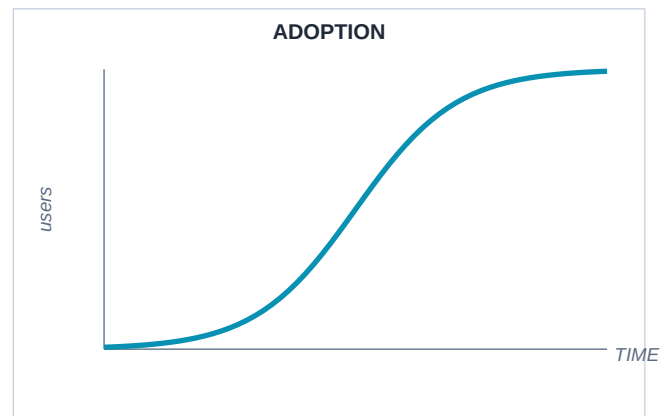


Figure 10. Once a network passes the tipping point, growth accelerates non-linearly. Below the threshold, the same investment yields decay. Identifying the threshold is the difference between scale and shutdown (Eisenmann et al., 2006).

HOW IT IS USED

Estimate the threshold based on per-segment activity densities, design pre-tipping investments (free trials, founder events, content seeding) and concentrate them on the segments closest to the threshold.

WHEN IT IS USED

Platform launch strategy, market-entry decisions in network-effect markets, and the diagnosis of stalled platforms.

BY WHOM IT IS USED

Platform founders, growth-marketing leaders, venture capitalists assessing market winnability, and antitrust economists analysing tipped markets.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with continuous-growth models, the tipping-point framing makes salient the binary 'win-or-die' nature of many digital markets. Compared with the colloquial 'critical mass', it has a formal threshold that can be estimated from data.

RELATED FRAMEWORKS IN THIS COURSE

Direct & Indirect Network Effects · Slibrary 6

Two-Sided Markets · Slibrary 6

Winner-Takes-All Markets · Slibrary 9

Connection Density · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

Eisenmann, T., Parker, G. and Van Alstyne, M. (2006) 'Strategies for two-sided markets', Harvard Business Review, 84(10), pp. 92-101.
 Granovetter, M. (1978) 'Threshold models of collective behavior', American Journal of Sociology, 83(6), pp. 1420-1443.
 Schelling, T.C. (1971) 'Dynamic models of segregation', Journal of Mathematical Sociology, 1(2), pp. 143-186.

Value Chain → Value Network

SLIBRARY 07 · SHEET 1 / 5

From Porter's linear chain to interconnected value creation in digital business

PROFESSIONAL DEFINITION

Porter's Value Chain (1985) decomposes a firm into a sequential pipeline of activities. The Value Network — developed by Stabell and Fjeldstad (1998) for service and platform businesses — replaces the chain with a web of mutually reinforcing relationships. Digital firms typically create value through networks rather than chains, fundamentally altering where margin is captured (Porter, 1985; Stabell and Fjeldstad, 1998).

WHO DEVELOPED IT

Michael Porter introduced the value chain in 'Competitive Advantage' (1985). Stabell and Fjeldstad (1998) generalised it into three configurations: chain, shop, and network. Tapscott (1996) and Parker, Van Alstyne and Choudary (2016) extended the network logic to digital platforms.

WHY IT WAS DEVELOPED

Applying a chain template to a network business misallocates investment: the analyst looks for sequential bottlenecks when the real value driver is connection density. The conceptual upgrade is essential before any digital strategy work begins.

FIGURE 11 · CHAIN VS NETWORK

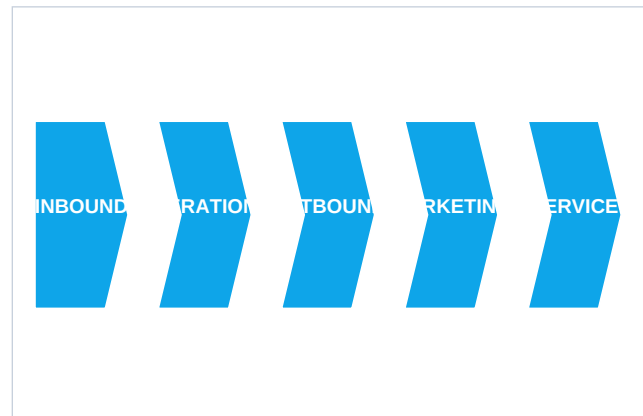


Figure 11. Porter's value chain (illustrated) is the foundational mental model for manufacturing-era firms. In digital businesses, value emerges from interactions between many parties — better captured by a network diagram (Stabell and Fjeldstad, 1998).

HOW IT IS USED

Diagnose the firm's value-creation configuration. If chains, optimise sequentially. If networks, identify the dominant interaction patterns and invest in interaction infrastructure rather than sequential efficiency.

WHEN IT IS USED

Foundational digital-strategy reviews, business-school strategy courses, and the transition from a product-firm mindset to a platform-firm mindset.

BY WHOM IT IS USED

Strategy consultants, board-level transformation programmes, business-school instructors, and corporate-development teams during pivots to platform business models.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with Porter's linear chain, the value-network configuration recognises that in digital businesses, the firm itself is a node in a multi-actor system rather than a production line. Compared with the more recent platform-business literature, Stabell and Fjeldstad's framework provides the bridge from classical strategy to platform thinking.

RELATED FRAMEWORKS IN THIS COURSE

[Atoms → Bits · Slibrary 1](#)
[Multi-Sided Platforms · Slibrary 11](#)
[Two-Sided Markets · Slibrary 6](#)
[Platform vs Marketplace · Slibrary 11](#)

SOURCES (HARVARD REFERENCING STYLE)

Parker, G.G., Van Alstyne, M.W. and Choudary, S.P. (2016) Platform Revolution. New York, NY: W.W. Norton.
 Porter, M.E. (1985) Competitive Advantage: Creating and Sustaining Superior Performance. New York, NY: Free Press.
 Stabell, C.B. and Fjeldstad, Ø.D. (1998) 'Configuring value for competitive advantage: on chains, shops, and networks', Strategic Management Journal, 19(5), pp. 413-437.

Freemium Model

SLIBRARY 07 · SHEET 2 / 5

Free-to-acquire, premium-to-monetise — the canonical digital pricing structure

PROFESSIONAL DEFINITION

The Freemium Model offers a basic version of a product for free and charges for a premium tier with additional features, capacity, or removed restrictions. It exploits the near-zero marginal cost of digital goods to acquire users at scale, then monetises the small fraction with high willingness to pay (Anderson, 2009; Kumar, 2014).

WHO DEVELOPED IT

Jarid Lukin coined the term 'freemium' in 2006. Chris Anderson (2009) elevated it in 'Free: The Future of a Radical Price'. V. Kumar (2014) provided the most-cited Harvard Business Review treatment of conversion economics.

WHY IT WAS DEVELOPED

Freemium dominates digital consumer pricing — Spotify, Dropbox, LinkedIn, Slack all use it. Mastering its conversion economics (typical 2-5% paid conversion) is essential to evaluating any consumer digital business.

FIGURE 12 · CONVERSION FUNNEL

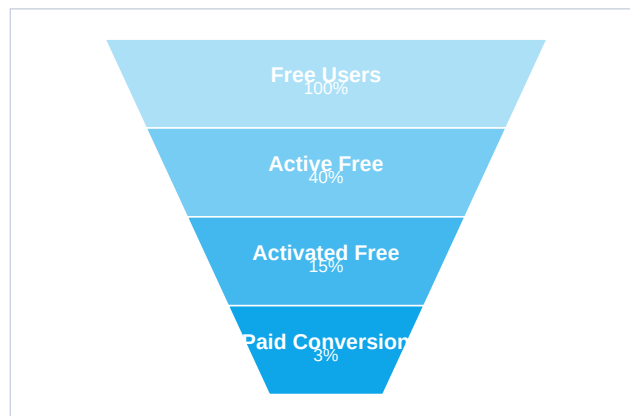


Figure 12. The freemium funnel. Conversion rates vary 1-10% by category. Pricing power lies less in the headline price than in the design of the free/paid feature wall (Kumar, 2014).

HOW IT IS USED

Set the feature wall: which features hook users (must be free) vs. which create high willingness to pay (must be paid). Track activation, not just acquisition. Iterate the wall as cohort data accumulate.

WHEN IT IS USED

Pricing design for any digital consumer or SaaS product, investor diligence on subscription firms, and product-led-growth strategy formulation.

BY WHOM IT IS USED

Product managers, growth-marketing teams, SaaS CFOs, and venture-capital analysts.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with classical pricing — where price discrimination required visible market segmentation — freemium uses self-selection: users choose their tier. Compared with pure free or pure paid models, it captures both the network effects of free and the revenue of paid.

RELATED FRAMEWORKS IN THIS COURSE

[Subscription Models · Slibrary 7](#)
[Versioning · Slibrary 8](#)
[Price Discrimination · Slibrary 8](#)
[Zero Marginal Cost · Slibrary 8](#)

SOURCES (HARVARD REFERENCING STYLE)

Anderson, C. (2009) Free: The Future of a Radical Price. New York, NY: Hyperion.
 Kumar, V. (2014) 'Making freemium work', Harvard Business Review, 92(5), pp. 27-29.
 Pujol, J.M. (2010) 'Freemium business model statistics', European Journal of Information Systems, 19(3), pp. 274-282.

Subscription Models

SLIBRARY 07 · SHEET 3 / 5

Recurring revenue and the predictable economics of digital businesses

PROFESSIONAL DEFINITION

A Subscription Model charges customers a recurring fee for ongoing access to a service. It converts a one-time transaction into a continuing relationship, transforming customer lifetime value, working-capital dynamics, and the firm's incentive to invest in retention (Tzuo and Weisert, 2018; Warrillow, 2015).

WHO DEVELOPED IT

Tien Tzuo (founder of Zuora) popularised the framing in 'Subscribed' (2018). John Warrillow (2015) provided the SME-oriented playbook. Academic treatment goes back to Anderson and Sullivan (1993) on customer-retention economics.

WHY IT WAS DEVELOPED

Subscription economics now dominate software, media, and an increasing share of consumer goods (Dollar Shave Club, HelloFresh). The shift in unit economics (from gross margin to lifetime value vs. customer-acquisition cost) is the key conceptual upgrade managers need.

FIGURE 13 · LTV VS CAC

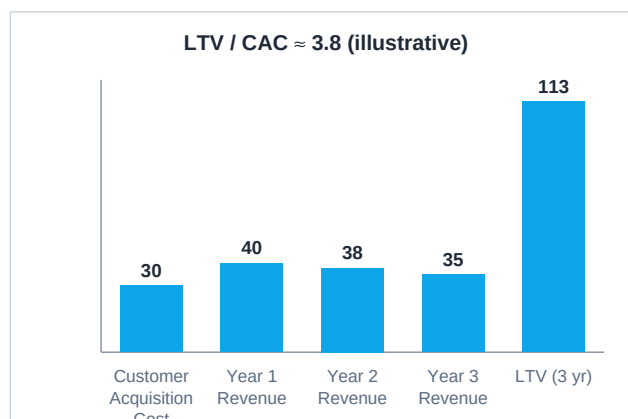


Figure 13. The subscription unit-economics view. A profitable subscription business needs LTV to exceed CAC by a healthy multiple (typically $\geq 3\times$). Churn and retention are now the central operating metrics (Tzuo and Weisert, 2018).

HOW IT IS USED

Move financial reporting from accounting profit to ARR, gross retention, net retention, and CAC. Restructure incentives and product investment around retention.

WHEN IT IS USED

Pivots from licence to subscription, board-level transitions to recurring revenue, and investor-relations updates for SaaS firms.

BY WHOM IT IS USED

SaaS CEOs and CFOs, growth-marketing leads, equity-research analysts, and B-school subscription-economics modules.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with one-off transaction accounting, subscription metrics make the time dimension explicit. Compared with the broader customer-relationship-management literature, the subscription frame ties retention directly to enterprise valuation through deferred-revenue and ARR multiples.

RELATED FRAMEWORKS IN THIS COURSE

Freemium Model · Slibrary 7

Cloud Computing IaaS/PaaS/SaaS · Slibrary 5

Lock-In & Switching Costs · Slibrary 8

Data as Asset · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

Anderson, E.W. and Sullivan, M.W. (1993) 'The antecedents and consequences of customer satisfaction for firms', *Marketing Science*, 12(2), pp. 125-143.
 Tzuo, T. and Weisert, G. (2018) *Subscribed: Why the Subscription Model Will Be Your Company's Future*. New York, NY: Portfolio.
 Warrillow, J. (2015) *The Automatic Customer: Creating a Subscription Business in Any Industry*. New York, NY: Portfolio.

Data as Asset

SLIBRARY 07 · SHEET 4 / 5

When information becomes balance-sheet material — the data-monetisation logic

PROFESSIONAL DEFINITION

The Data-as-Asset framing treats accumulated user, operational and transactional data as a strategic asset that produces compounding economic returns through better products, more accurate models, and increased switching costs. Unlike physical assets, data exhibits increasing returns to scale and can be re-used at near-zero marginal cost (Mayer-Schönberger and Cukier, 2013; Varian, 2018).

WHO DEVELOPED IT

Mayer-Schönberger and Cukier (2013) popularised the asset framing in 'Big Data'. Hal Varian (Google Chief Economist) developed the formal economics in his 2018 NBER paper 'Artificial Intelligence, Economics, and Industrial Organization'. The Economist's (2017) cover 'The world's most valuable resource is no longer oil, but data' carried the framing into general management vocabulary.

WHY IT WAS DEVELOPED

Most firms now hold vast data assets but treat them as cost (storage). The asset framing reorients management toward extraction of value, with implications for organisational design, governance, and competitive strategy.

FIGURE 14 · DATA FLYWHEEL

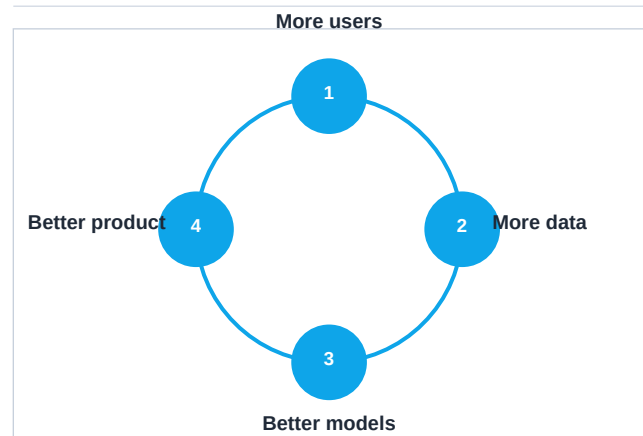


Figure 14. The data flywheel. Each turn improves the product, attracting more users, generating more data — the dynamic at the heart of platform durability and AI-firm moats (Varian, 2018).

HOW IT IS USED

Inventory data assets, classify by re-use potential, identify the dominant flywheel, design governance to enable safe re-use, and monitor the data multiplier explicitly.

WHEN IT IS USED

Digital strategy reviews, AI strategy formulation, M&A diligence (data assets often dominate intangibles), and DMA-style competition reviews.

BY WHOM IT IS USED

Chief Data Officers, AI strategy teams, intangible-asset valuation analysts, and platform-competition regulators.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the IT-cost view of data, the asset framing connects data to enterprise value. Compared with general intangibles literature, it makes explicit the unique compounding logic of data-driven flywheels.

RELATED FRAMEWORKS IN THIS COURSE

AI in Business — General Spectrum · Slibrary 15

Data Economies of Scale · Slibrary 11

Hyper-Personalisation · Slibrary 16

Surveillance Capitalism · Slibrary 19

SOURCES (HARVARD REFERENCING STYLE)

Mayer-Schönberger, V. and Cukier, K. (2013) Big Data: A Revolution That Will Transform How We Live, Work, and Think. London: John Murray.
 The Economist (2017) 'The world's most valuable resource is no longer oil, but data'. The Economist, 6 May.
 Varian, H.R. (2018) Artificial intelligence, economics, and industrial organization. NBER Working Paper No. 24839. Cambridge, MA: National Bureau of Economic Research.

Long Tail

SLIBRARY 07 · SHEET 5 / 5

Anderson's discovery — niche markets aggregating to majority demand online

PROFESSIONAL DEFINITION

The Long Tail describes the demand distribution of digital marketplaces: while a small set of 'hits' dominates physical-shelf retail, the aggregate demand for the long tail of niche items can rival or exceed the hits — but only when distribution is digital and shelf space is unconstrained (Anderson, 2006; Brynjolfsson, Hu and Smith, 2010).

WHO DEVELOPED IT

Chris Anderson (then editor of Wired) coined the term in 2004 and developed it in his 2006 book 'The Long Tail'. Brynjolfsson, Hu and Smith (2010) provided the empirical validation for Amazon's product mix.

WHY IT WAS DEVELOPED

The long tail explains why digital marketplaces beat physical retailers — not on best-sellers, but on niche selection. It is the structural argument behind Amazon's product-range moat and Spotify's catalogue strategy.

FIGURE 15 · LONG-TAIL DISTRIBUTION

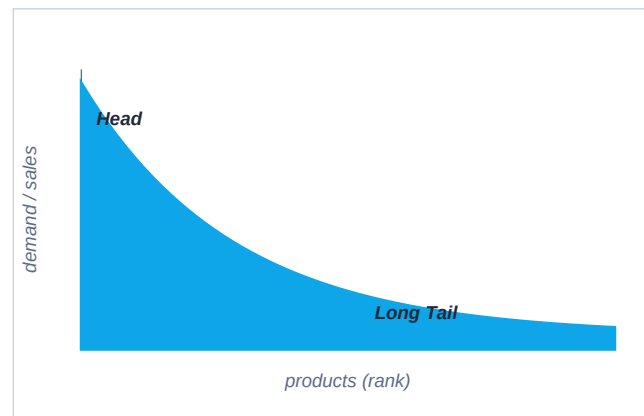


Figure 15. The long tail of demand. The shaded region — long-tail revenue — is inaccessible to physical retailers (limited shelves) but available to digital marketplaces (unlimited catalogue) (Anderson, 2006; Brynjolfsson et al., 2010).

HOW IT IS USED

For incumbents: assess whether digital entrants are eating the tail (often invisible from headline market-share data). For digital firms: design recommender systems and search to monetise the tail.

WHEN IT IS USED

Competitive analysis of digital vs. physical retail, recommender-system strategy, and IP-portfolio-monetisation in publishing and music.

BY WHOM IT IS USED

Marketplace product managers, recommender-system engineers, retail-strategy consultants, and content-IP licensing teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with 80/20 thinking, the long-tail framework recognises that the 80% can exceed the 20% under digital distribution. Compared with niche-marketing theory, it highlights aggregation as the value mechanism — the marketplace, not the niche supplier, captures most of the surplus.

RELATED FRAMEWORKS IN THIS COURSE

[Atoms → Bits · Slibrary 1](#)
[Zero Marginal Cost · Slibrary 8](#)
[Hyper-Personalisation · Slibrary 16](#)
[Multi-Sided Platforms · Slibrary 11](#)

SOURCES (HARVARD REFERENCING STYLE)

Anderson, C. (2006) *The Long Tail: Why the Future of Business is Selling Less of More*. New York, NY: Hyperion.
 Brynjolfsson, E., Hu, Y. and Smith, M.D. (2010) 'Long tails vs. superstars: the effect of information technology on product variety and sales concentration patterns', *Information Systems Research*, 21(4), pp. 736-747.
 Brynjolfsson, E., Hu, Y. and Simester, D. (2011) 'Goodbye Pareto Principle, hello long tail', *Management Science*, 57(8), pp. 1373-1386.

Zero Marginal Cost

SLIBRARY 08 · SHEET 1 / 5

The defining cost structure of information goods — and its strategic consequences

PROFESSIONAL DEFINITION

Information Goods exhibit high fixed costs of production but near-zero marginal cost of reproduction and distribution. A film costs hundreds of millions to make but pennies to stream; a software product takes years to develop but copies cost nothing. This structure drives the strategic logic of digital business — pricing, scale economics, and competitive dynamics (Shapiro and Varian, 1999; Rifkin, 2014).

WHO DEVELOPED IT

Carl Shapiro and Hal Varian formalised the analysis in 'Information Rules' (1999). Jeremy Rifkin (2014) extended the framing to predict a 'zero marginal cost society'. Brynjolfsson and McAfee (2014) connected it to 'the second machine age'.

WHY IT WAS DEVELOPED

Almost every digital strategy decision is shaped by zero marginal cost: it explains why scale is decisive, why pricing tends toward differentiation rather than cost-plus, and why free is so often viable as a competitive weapon.

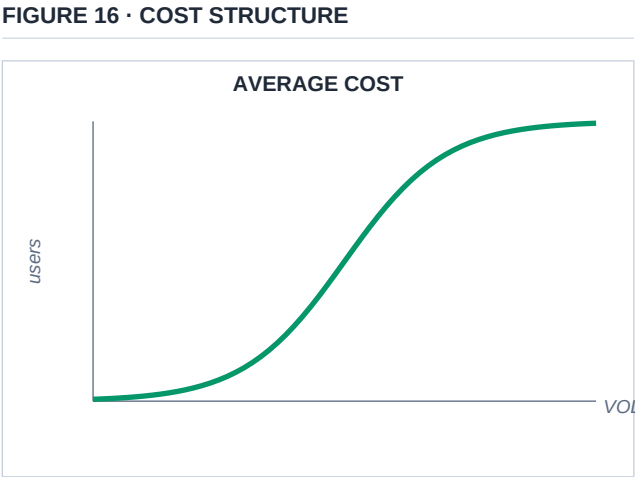


Figure 16. Average cost falls steeply with volume because marginal cost is near zero. Once the fixed cost is amortised, every additional unit is pure margin — the central engine of digital scale economies (Shapiro and Varian, 1999).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Identify which costs are fixed (one-time creation) vs. variable (per-unit production or distribution). Optimise pricing for differentiation (versioning, bundling, discrimination) rather than cost-plus.	Pricing strategy for any digital product, investor diligence on platform firms, and the foundational session of any digital-economics course.	Information-economics scholars, digital-product CFOs, pricing strategy teams, and competition economists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with classical microeconomics — where the marginal-cost curve drives optimal output — the information-goods framing inverts the logic: pricing is set by what users will pay, not by what production costs. Compared with the broader 'fixed-cost' economics, it identifies digital information as a special case where marginal cost approaches zero.

RELATED FRAMEWORKS IN THIS COURSE

- Atoms → Bits · Slibrary 1
- Versioning · Slibrary 8
- Price Discrimination · Slibrary 8
- Long Tail · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

Brynjolfsson, E. and McAfee, A. (2014) The Second Machine Age. New York, NY: W.W. Norton.
Rifkin, J. (2014) The Zero Marginal Cost Society. New York, NY: Palgrave Macmillan.
Shapiro, C. and Varian, H.R. (1999) Information Rules: A Strategic Guide to the Network Economy. Boston, MA: Harvard Business School Press.

Experience Goods

SLIBRARY 08 · SHEET 2 / 5

Quality unknown until used — and the consequences for digital pricing and trust

PROFESSIONAL DEFINITION

An Experience Good is one whose quality cannot be assessed before consumption. Information goods (films, software, articles) are paradigmatic experience goods — the buyer cannot evaluate them without effectively consuming them. This creates fundamental challenges of trust, signalling, sampling, and the role of reviews and reputation (Nelson, 1970; Shapiro and Varian, 1999).

WHO DEVELOPED IT

Phillip Nelson (1970) introduced the search-vs-experience-goods distinction. Shapiro and Varian (1999) applied it to digital information goods. Tirole (1988) provided the industrial-organisation extension to repeat purchase and reputation.

WHY IT WAS DEVELOPED

Experience-good logic explains why digital businesses invest so heavily in reviews, sampling, free trials, and trust signals — these are the mechanisms by which buyers overcome the inherent information asymmetry.

FIGURE 17 · TRUST MECHANISMS

Sampling

free trials, demos

Reputation

reviews, brands

Signalling

warranties, certifications

Figure 17. Three mechanisms used in digital markets to overcome experience-good uncertainty. Each has its own incentive-design challenges — review fraud, brand dilution, certification gaming (Shapiro and Varian, 1999; Tirole, 1988).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Diagnose which experience-good problem the customer faces; design the corresponding trust mechanism; monitor and govern its integrity (e.g. detect fake reviews).	Pricing and go-to-market design, marketplace integrity programmes, content-licensing deals, and competition-policy analyses of review platforms.	Marketplace product managers, brand-management teams, competition economists, and consumer-protection regulators.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the search-cost models of classical economics, the experience-good framing recognises that for many digital products, no amount of pre-purchase search resolves the uncertainty — only consumption (or reputation) does. Compared with naive trust frameworks, it grounds the analysis in formal information economics.

RELATED FRAMEWORKS IN THIS COURSE

- Zero Marginal Cost · Slibrary 8
- Freemium Model · Slibrary 7
- Versioning · Slibrary 8
- Hyper-Personalisation · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Nelson, P. (1970) 'Information and consumer behavior', Journal of Political Economy, 78(2), pp. 311-329.
Shapiro, C. and Varian, H.R. (1999) Information Rules: A Strategic Guide to the Network Economy. Boston, MA: Harvard Business School Press.
Tirole, J. (1988) The Theory of Industrial Organization. Cambridge, MA: MIT Press.

Versioning

SLIBRARY 08 · SHEET 3 / 5

Producing multiple versions of a digital product to capture more consumer surplus

PROFESSIONAL DEFINITION

Versioning is the strategy of producing multiple versions of a digital product — varying in feature set, capacity, speed, or restrictions — at near-zero incremental production cost, then offering them at different prices. It allows the seller to capture more of the heterogeneous willingness-to-pay across the customer base (Shapiro and Varian, 1999; Varian, 1996).

WHO DEVELOPED IT

Hal Varian developed the formal versioning theory (1996, 1999). Shapiro and Varian (1999) popularised the strategic logic in 'Information Rules' with extensive case examples (LexisNexis, NYT, software vendors).

WHY IT WAS DEVELOPED

Versioning is everywhere in digital business — basic/pro/enterprise software tiers, video-streaming quality levels, ad-funded vs paid editions. Mastering it is essential to setting digital prices.

FIGURE 18 · TIER LADDER

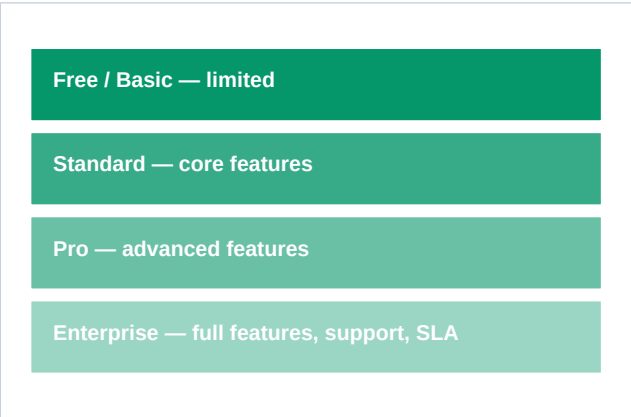


Figure 18. Versioning typically uses three to five tiers. Higher tiers add features with high willingness-to-pay; lower tiers preserve free or low-price acquisition (Shapiro and Varian, 1999).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Identify dimensions on which willingness-to-pay varies (capacity, speed, support, feature breadth). Design versions that segment those dimensions cleanly. Set prices so each tier is rationally chosen by its target segment.	Product-line pricing decisions, SaaS-tier design, B2B price-discrimination strategy, and digital-product portfolio reviews.	Pricing strategists, product-marketing managers, SaaS founders, and information-economics scholars.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-price information goods, versioning multiplies revenue at zero marginal cost — the most elegant exploitation of the information-good cost structure. Compared with simple price discrimination, versioning replaces direct identification of willingness-to-pay with self-selection across versions.

RELATED FRAMEWORKS IN THIS COURSE

Zero Marginal Cost · *Slibrary 8*

Price Discrimination · *Slibrary 8*

Freemium Model · *Slibrary 7*

Subscription Models · *Slibrary 7*

SOURCES (HARVARD REFERENCING STYLE)

Shapiro, C. and Varian, H.R. (1999) Information Rules: A Strategic Guide to the Network Economy. Boston, MA: Harvard Business School Press.
Varian, H.R. (1996) 'Differential pricing and efficiency', First Monday, 1(2).
Varian, H.R. (1999) 'Versioning information goods', Working paper, UC Berkeley.

Price Discrimination

SLIBRARY 08 · SHEET 4 / 5

Charging different prices to different buyers — the granular form digital makes possible

PROFESSIONAL DEFINITION

Price Discrimination charges different prices for the same (or near-same) product to different buyers, based on willingness to pay. Three classical forms exist: first-degree (personalised pricing), second-degree (versioning, bundling), third-degree (group-based, e.g. student discounts). Digital data and AI have moved first-degree from textbook ideal toward practical reality (Pigou, 1920; Varian, 2018; OECD, 2018).

WHO DEVELOPED IT

Arthur Pigou (1920) introduced the three-degree taxonomy. Hal Varian's 2018 NBER paper applied it to AI-driven personalisation. The OECD (2018) 'Personalised Pricing in the Digital Era' documented competition concerns.

WHY IT WAS DEVELOPED

Personalised pricing has become technically feasible at scale — Uber surge, dynamic airline fares, e-commerce A/B-tested offers. Understanding the economics is essential to evaluating fairness, competition, and consumer-protection policy.

FIGURE 19 · THREE DEGREES

Degree	Mechanism	Digital Example
1st	Personalised price	Uber surge
2nd	Versioning, bundling	SaaS tiers
3rd	Group-based	Student plans

Figure 19. The three classical degrees of price discrimination. Digital firms increasingly approach first-degree through real-time personalisation enabled by data and AI (Varian, 2018; OECD, 2018).

HOW IT IS USED

Map a product to the dominant degree of discrimination feasible. Test pricing experiments. Audit for legal and reputational risks (sensitive attributes, anti-discrimination law, EU consumer protection).

WHEN IT IS USED

Pricing strategy reviews, regulatory compliance audits, fairness-in-pricing programmes, and competition-authority investigations of personalised pricing.

BY WHOM IT IS USED

Pricing teams, regulators, consumer-protection ombudsmen, and academic industrial-organisation researchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional price discrimination — limited by the seller's information — data-rich digital firms approach personalised pricing in real time. Compared with the static three-degree taxonomy, modern digital pricing blends all three simultaneously, raising new fairness and competition issues.

RELATED FRAMEWORKS IN THIS COURSE

Versioning · Slibrary 8

Hyper-Personalisation · Slibrary 16

Data as Asset · Slibrary 7

Algorithmic Collusion · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

OECD (2018) Personalised Pricing in the Digital Era. Background note, DAF/COMP(2018)13. Paris: OECD.
Pigou, A.C. (1920) The Economics of Welfare. London: Macmillan.
Varian, H.R. (2018) Artificial intelligence, economics, and industrial organization. NBER Working Paper No. 24839. Cambridge, MA: NBER.

Lock-In & Switching Costs

SLIBRARY 08 · SHEET 5 / 5

Why digital customers stay — and what that means for competition

PROFESSIONAL DEFINITION

Lock-In describes the situation where customers face significant costs to switch from one digital provider to another — costs that may be technical (data migration), contractual, psychological (familiarity), or social (network effects). Switching costs translate directly into pricing power, durability of revenue, and competition-policy concerns (Shapiro and Varian, 1999; Klemperer, 1995).

WHO DEVELOPED IT

Paul Klemperer (1995) provided the formal economic theory in his Review of Economic Studies survey. Shapiro and Varian (1999) made it central to digital strategy. The EU Digital Markets Act (2022) explicitly targets switching-cost reduction in gatekeeper ecosystems.

WHY IT WAS DEVELOPED

Switching costs are the single most important non-network-effect source of platform durability. Diagnosing them is essential for both incumbents (defending) and entrants (attacking) — and for regulators.

FIGURE 20 · COST CATEGORIES

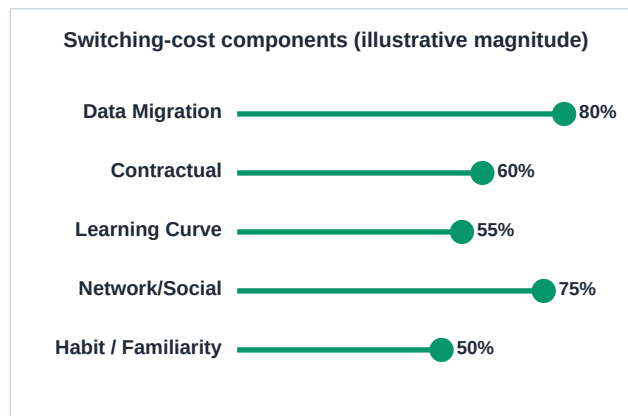


Figure 20. Switching costs decompose into multiple sources, each with its own remediation. Data-portability mandates (e.g. GDPR Art. 20, DMA) target the technical layer; education and trial offers target the learning layer (Shapiro and Varian, 1999).

HOW IT IS USED

Inventory the firm's switching costs by category. For incumbents: invest in those that are legally defensible and customer-positive. For entrants: design migration tools that neutralise the largest layer first.

WHEN IT IS USED

Defensive strategy reviews, competition-authority market-power analyses, DMA data-portability compliance, and SaaS contract negotiations.

BY WHOM IT IS USED

Strategy teams, antitrust economists, procurement leads, and competition-policy researchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the simple 'customer-loyalty' framing, switching-cost analysis reveals the structural — and sometimes regulator-actionable — sources of stickiness. Compared with general retention literature, it ties retention to enterprise value through pricing power and competition-policy exposure.

RELATED FRAMEWORKS IN THIS COURSE

[Subscription Models · Slibrary 7](#)
[Cloud Computing IaaS/PaaS/SaaS · Slibrary 5](#)
[DMA — Digital Markets Act · Slibrary 12](#)
[Platform Governance · Slibrary 11](#)

SOURCES (HARVARD REFERENCING STYLE)

European Union (2022) Regulation (EU) 2022/1925 — Digital Markets Act. Brussels: Official Journal of the European Union.
 Klemperer, P. (1995) 'Competition when consumers have switching costs', Review of Economic Studies, 62(4), pp. 515-539.
 Shapiro, C. and Varian, H.R. (1999) Information Rules: A Strategic Guide to the Network Economy. Boston, MA: Harvard Business School Press.

Winner-Takes-All Markets

SLIBRARY 09 · SHEET 1 / 5

Why digital markets so often produce a single dominant firm

PROFESSIONAL DEFINITION

A Winner-Takes-All (WTA) market is one in which the leading firm captures a disproportionate share of profits — typically because of strong network effects, switching costs, or scale economies in data. Once tipped, runners-up cannot recover even with substantially better products (Frank and Cook, 1995; Eisenmann et al., 2006; OECD, 2020).

WHO DEVELOPED IT

The phrase originates with Frank and Cook (1995) in 'The Winner-Take-All Society'. Eisenmann, Parker and Van Alstyne (2006) translated it for digital markets. The OECD (2020) and the UK CMA (2022) use it in formal market-power assessments of GAFAM.

WHY IT WAS DEVELOPED

WTA dynamics make ordinary 'market share' analysis misleading: small share gaps translate into enormous profit gaps. They also reshape competition policy — intervention becomes more urgent because tipping is irreversible.

FIGURE 21 · TIPPING DYNAMIC

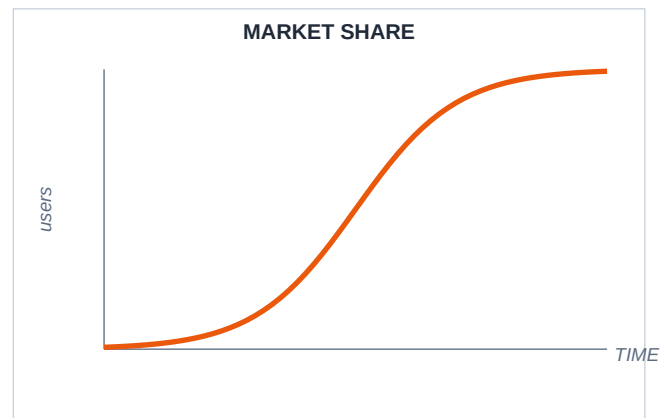


Figure 21. WTA dynamics: once the leader passes a threshold, share concentration accelerates. Search (Google), social (Facebook) and PC OS (Windows) all show this pattern (Eisenmann et al., 2006).

HOW IT IS USED

Diagnose WTA risk via three tests: strength of network effects, height of switching costs, and presence of multi-homing. If all three favour concentration, expect winner-takes-most rather than competitive equilibrium.

WHEN IT IS USED

Strategic market-entry decisions, antitrust market-definition exercises, and DMA gatekeeper-designation analyses.

BY WHOM IT IS USED

Strategy consultants, competition authorities, technology economists, and venture capitalists evaluating market winnability.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional concentration ratios (CR4, HHI), the WTA framing recognises that digital markets often have one true winner and multiple irrelevant survivors. Compared with general 'monopoly' rhetoric, it specifies the structural conditions under which durable dominance arises.

RELATED FRAMEWORKS IN THIS COURSE

[Tipping Point · Slibrary 6](#)
[GAFAM Concentration · Slibrary 9](#)
[Two-Sided Markets · Slibrary 9](#)
[DMA — Digital Markets Act · Slibrary 12](#)

SOURCES (HARVARD REFERENCING STYLE)

Eisenmann, T., Parker, G. and Van Alstyne, M. (2006) 'Strategies for two-sided markets', Harvard Business Review, 84(10), pp. 92-101.
 Frank, R.H. and Cook, P.J. (1995) The Winner-Take-All Society. New York, NY: Free Press.
 OECD (2020) Going Digital integrated policy framework. Paris: OECD Publishing.

Two-Sided Markets — Market Structure View

SLIBRARY 09 · SHEET 2 / 5

Rochet & Tirole's framework as applied to market definition and competition

PROFESSIONAL DEFINITION

Viewed at the market-structure level, a Two-Sided Market is one in which a platform intermediates between two distinct user groups whose value depends on each other. Market boundaries cannot be drawn on one side alone: the relevant antitrust market necessarily spans both sides simultaneously (Rochet and Tirole, 2003; Filistrucchi et al., 2014; OECD, 2018).

WHO DEVELOPED IT

Rochet and Tirole (2003, 2006) established the formal theory; Filistrucchi, Geradin and Van Damme (2014) developed the antitrust application; the US Supreme Court adopted a two-sided market analysis in *Ohio v. American Express* (2018).

WHY IT WAS DEVELOPED

Mis-defining the market on a single side is the most common error in digital antitrust cases. Two-sided market analysis is now mandatory in EU, US and UK competition reviews of platform mergers.

FIGURE 22 · TWO-SIDED MARKET

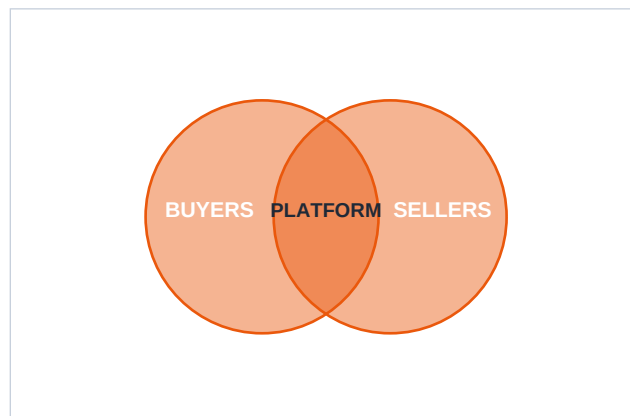


Figure 22. The platform mediates between two interdependent groups. Antitrust must consider both sides simultaneously: a small price increase on one side may be constrained by competition on the other (Filistrucchi et al., 2014).

HOW IT IS USED

When defining the market: (1) identify the two sides; (2) map cross-side externalities; (3) assess substitution patterns on each side; (4) draw market boundaries that span both.

WHEN IT IS USED

Merger reviews of digital platforms, designation of DMA gatekeepers, and academic industrial-organisation case analyses.

BY WHOM IT IS USED

Competition authorities, antitrust economists, M&A lawyers, and industrial-organisation faculty.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with one-sided market definition, two-sided analysis prevents both false-positive and false-negative findings of dominance. Compared with the original Rochet-Tirole pricing model, this market-structure perspective is the form most used by lawyers and regulators.

RELATED FRAMEWORKS IN THIS COURSE

[Two-Sided Markets · Slibrary 6](#)
[Multi-Sided Platforms · Slibrary 9](#)
[DMA — Digital Markets Act · Slibrary 12](#)
[Marketplace vs Platform · Slibrary 9](#)

SOURCES (HARVARD REFERENCING STYLE)

Filistrucchi, L., Geradin, D. and Van Damme, E. (2014) 'Identifying two-sided markets', *World Competition*, 37(1), pp. 33-59.
 OECD (2018) *Rethinking Antitrust Tools for Multi-Sided Platforms*. Paris: OECD Publishing.
 Rochet, J.-C. and Tirole, J. (2003) 'Platform competition in two-sided markets', *Journal of the European Economic Association*, 1(4), pp. 990-1029.

Multi-Sided Platforms

SLIBRARY 09 · SHEET 3 / 5

Beyond two sides — orchestrating ecosystems with three or more interdependent groups

PROFESSIONAL DEFINITION

A Multi-Sided Platform (MSP) connects three or more distinct user groups whose interactions create value. iOS connects users, app developers, and advertisers; Visa connects cardholders, merchants, and issuing banks; Uber connects riders, drivers, and Uber Eats restaurants. Pricing, governance, and competitive dynamics become more complex with each added side (Hagiu and Wright, 2015; Cusumano et al., 2019).

WHO DEVELOPED IT

Andrei Hagiu and Julian Wright (2015) provided the canonical academic taxonomy in Management Science. Cusumano, Gawer and Yoffie (2019) developed the strategy-school treatment in 'The Business of Platforms'.

WHY IT WAS DEVELOPED

Most major digital platforms today are multi-sided. Two-sided thinking under-predicts their dynamics — especially the role of complementor firms whose business models depend on platform decisions.

FIGURE 23 · ECOSYSTEM SIDES

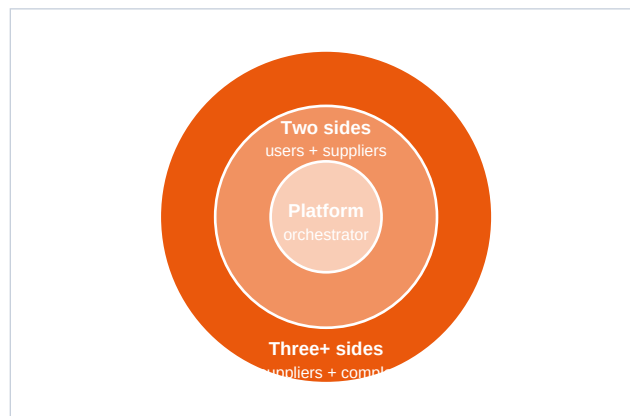


Figure 23. Multi-sided platforms layer additional sides onto the basic two-sided structure. iOS adds the developer side; Visa adds banks; Uber adds restaurants. Each additional side multiplies governance complexity (Hagiu and Wright, 2015).

HOW IT IS USED

Map all sides; identify the dominant cross-side externality pairs; design tiered pricing so high-WTP sides subsidise low-WTP sides; build governance for each side.

WHEN IT IS USED

Platform-business strategy, ecosystem governance design, and antitrust analysis of complex platforms (Apple App Store, Google Play, Amazon Marketplace + AWS).

BY WHOM IT IS USED

Platform CEOs, ecosystem strategists, antitrust regulators, and Stern School / HBS / Insead faculty.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with two-sided market theory, the MSP framing handles the empirical complexity of real-world platforms. Compared with general 'ecosystem' rhetoric, it preserves the formal economics of cross-side externalities while extending to n-sides.

RELATED FRAMEWORKS IN THIS COURSE

[Two-Sided Markets · Slibrary 9](#)
[Platform vs Pipeline · Slibrary 3](#)
[Multi-Sided Platforms · Slibrary 11](#)
[Platform Governance · Slibrary 11](#)

SOURCES (HARVARD REFERENCING STYLE)

Cusumano, M.A., Gawer, A. and Yoffie, D.B. (2019) The Business of Platforms. New York, NY: HarperBusiness.
 Hagiu, A. and Wright, J. (2015) 'Multi-sided platforms', International Journal of Industrial Organization, 43, pp. 162-174.
 Parker, G.G., Van Alstyne, M.W. and Choudary, S.P. (2016) Platform Revolution. New York, NY: W.W. Norton.

Marketplace vs. Platform

SLIBRARY 09 · SHEET 4 / 5

A subtle but consequential distinction in business-model design

PROFESSIONAL DEFINITION

A Marketplace facilitates transactions between buyers and sellers (eBay, Etsy, Amazon Marketplace). A Platform enables a wider range of value-creating activities, including non-transactional ones (iOS, YouTube, Salesforce AppExchange). All marketplaces are platforms, but not all platforms are marketplaces. The distinction matters for strategy, governance, and regulation (Hagiu, 2014; Choudary, 2015).

WHO DEVELOPED IT

Andrei Hagiu (2014) drew the boundary in his Harvard Business Review article 'Strategic decisions for multisided platforms'. Sangeet Choudary (2015) developed the practitioner framing in 'Platform Scale'.

WHY IT WAS DEVELOPED

Confusing the two types leads to misallocated investment: marketplaces optimise for transaction volume; platforms optimise for ecosystem participation. The right metrics and governance differ accordingly.

FIGURE 24 · MARKETPLACE VS PLATFORM

Dimension	Marketplace	Platform
Goal	Match buyers + sellers	Enable broad value creation
Economics	Take rate per transaction	Various revenue streams
Examples	Etsy, eBay	iOS, YouTube

Figure 24. Marketplace vs platform comparison. The marketplace business model is a subset of the platform business model. Strategic choice depends on the type of value being created (Hagiu, 2014; Choudary, 2015).

HOW IT IS USED

Diagnose which model the firm operates. Marketplaces optimise for liquidity, take rate, and trust mechanisms. Platforms optimise for participation breadth and complementor investment.

WHEN IT IS USED

Business-model design, M&A diligence on digital firms, and DMA gatekeeper characterisation.

BY WHOM IT IS USED

Platform-business founders, strategy consultants, regulators, and B-school strategy instructors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the catch-all term 'platform', this distinction preserves analytic precision. Compared with the older 'intermediary vs facilitator' framing, it captures the complementor economics of modern technology platforms.

RELATED FRAMEWORKS IN THIS COURSE

[Multi-Sided Platforms · Slibrary 9](#)
[Platform vs Pipeline · Slibrary 3](#)
[Two-Sided Markets · Slibrary 9](#)
[Platform Governance · Slibrary 11](#)

SOURCES (HARVARD REFERENCING STYLE)

Choudary, S.P. (2015) Platform Scale. Boston, MA: Platform Thinking Labs.
 Hagiu, A. (2014) 'Strategic decisions for multisided platforms', MIT Sloan Management Review, 55(2), pp. 71-80.
 Parker, G.G., Van Alstyne, M.W. and Choudary, S.P. (2016) Platform Revolution. New York, NY: W.W. Norton.

GAFAM Concentration

SLIBRARY 09 · SHEET 5 / 5

Five US firms, one-sixth of global equity market capitalisation

PROFESSIONAL DEFINITION

GAFAM (Google/Alphabet, Apple, Facebook/Meta, Amazon, Microsoft) refers to the five largest US technology firms by market capitalisation. Together they have repeatedly represented around 15-20% of global equity market value, making their performance, spending, and governance decisions of macroeconomic significance (Bloomberg, 2024; Forbes, 2024).

WHO DEVELOPED IT

The acronym GAFAM (originally GAFA, before Microsoft was added) circulated in French and EU policy debates from c.2014. The acronym MAANG/MAMAA is also used in US markets. Bloomberg, S&P, and the European Commission track the companies' aggregate footprint.

WHY IT WAS DEVELOPED

GAFAM's concentration is the empirical foundation of contemporary competition policy in digital markets — the EU DMA, US antitrust suits, and Chinese platform regulation all respond to the structural reality of GAFAM-scale firms.

FIGURE 25 · MARKET CAP

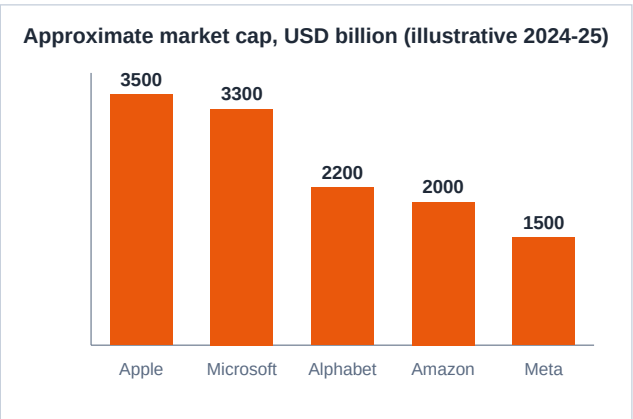


Figure 25. Approximate market capitalisations (USD billions, illustrative 2024-25). Together, GAFAM has repeatedly exceeded the GDP of all but a handful of countries (Bloomberg, 2024).

HOW IT IS USED

Use GAFAM as a benchmark concentration set: how much of any digital sub-market do they own; how much R&D, lobbying, or cloud capacity is GAFAM-dominated; what is the rest-of-economy exposure to GAFAM cloud.

WHEN IT IS USED

Sovereignty reviews, DMA designation, antitrust investigations, supply-chain risk assessments, and macroeconomic strategy work.

BY WHOM IT IS USED

Competition regulators, sovereign-wealth funds, pension fund strategists, and policy researchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional industry concentration metrics (e.g. by sector), GAFAM groups the empirically dominant firms cross-sector. Compared with general 'Big Tech' rhetoric, the acronym names the specific entities and enables consistent measurement.

RELATED FRAMEWORKS IN THIS COURSE

Winner-Takes-All Markets · Slibrary 9

DMA — Digital Markets Act · Slibrary 12

Antitrust in Digital Markets · Slibrary 12

Cloud Computing IaaS/PaaS/SaaS · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

Bloomberg (2024) Global market cap rankings 2024. New York, NY: Bloomberg L.P.
European Commission (2023) Digital Markets Act: Designations of Gatekeepers. Brussels: European Commission.
Forbes (2024) The World's Largest Public Companies 2024. New York, NY: Forbes Media.

Fintech

SLIBRARY 10 · SHEET 1 / 5

Financial services reimagined digitally — payments, lending, wealth, insurance

PROFESSIONAL DEFINITION

Fintech (Financial Technology) is the application of digital technologies to financial services. It spans payments, lending, wealth management, insurance, capital markets, regtech, and crypto. Fintech firms compete with — and increasingly partner with — incumbent banks, with significantly different cost structures and regulatory exposures (Arner, Barberis and Buckley, 2016; KPMG, 2024).

WHO DEVELOPED IT

The modern fintech wave was catalysed by the 2008 financial crisis (eroding trust in incumbents) and smartphone penetration. Arner, Barberis and Buckley (2016) provide the definitive academic taxonomy. KPMG's Pulse of Fintech is the most cited industry source for global investment data.

WHY IT WAS DEVELOPED

Fintech is the largest digital vertical by funding and market impact. Understanding its structure is essential for any digital strategy work touching financial services — which is now most of them.

FIGURE 26 · FINTECH SUB-VERTICALS

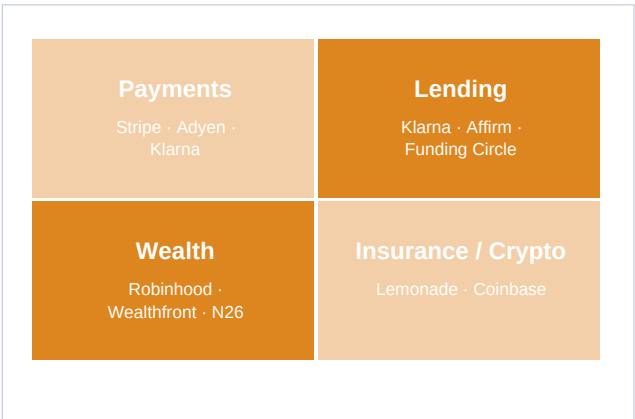


Figure 26. The four main fintech sub-verticals. Each has its own dominant business models and regulatory regimes. Crypto and digital assets are sometimes treated as a fifth sub-vertical (Arner et al., 2016).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Map a fintech firm to its sub-vertical(s); identify the relevant licences, regulator, and incumbents; assess unit economics against incumbent benchmarks (typically lower cost-to-serve, higher acquisition cost).	Investment diligence on fintech start-ups, incumbent-bank digital strategy reviews, and EU PSD2/PSD3 compliance work.	Banking strategy teams, fintech founders, financial-services regulators, and equity-research analysts.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the older e-banking framing, fintech is built API-first, mobile-first, and often regulator-first (using regulatory sandboxes). Compared with treating 'tech in finance' as a single category, the sub-vertical structure enables sharper competitive analysis.

RELATED FRAMEWORKS IN THIS COURSE

- API Economy · Slibrary 5
- Regtech · Slibrary 10
- Insurtech · Slibrary 10
- Subscription Models · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

Arner, D.W., Barberis, J. and Buckley, R.P. (2016) 'The evolution of Fintech: a new post-crisis paradigm?', Georgetown Journal of International Law, 47(4), pp. 1271-1319.
KPMG (2024) Pulse of Fintech H2 2024. Amstelveen: KPMG International.
OECD (2024) The Future of Financial Markets and Fintech. Paris: OECD Publishing.

Legaltech

SLIBRARY 10 · SHEET 2 / 5

AI-powered contract review, legal research, and access to justice

PROFESSIONAL DEFINITION

Legaltech is the application of digital technologies — increasingly large language models — to legal services. It spans contract analysis, legal research, e-discovery, dispute resolution, document automation, and consumer-facing access-to-justice tools. The shift from rule-based systems to LLM-assisted workflows since 2023 has accelerated both investment and disruption (Susskind, 2023; Stanford CodeX, 2024).

WHO DEVELOPED IT

Richard Susskind has chronicled legal-services transformation since 1996; his 2023 edition of 'Tomorrow's Lawyers' addresses the LLM wave directly. Stanford CodeX is the leading academic centre. Major firms include Harvey, Casetext (now Thomson Reuters), and DoNotPay.

WHY IT WAS DEVELOPED

Legaltech is one of the fastest-growing AI-application verticals. Lawyers, law firms, and corporate legal departments must understand the technology to manage risk and capture productivity gains.

FIGURE 27 · LEGALTECH STACK

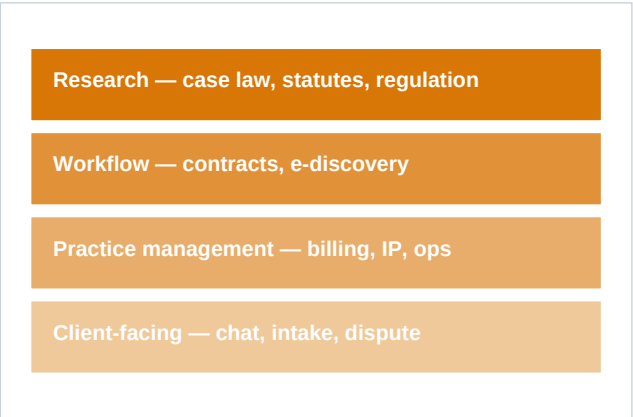


Figure 27. The legaltech stack: each layer is a distinct market. LLM-augmented research and contract layers attract the largest investment in 2024-2025 (Susskind, 2023; Stanford CodeX, 2024).

HOW IT IS USED

For law firms: pilot LLM tools in research and contract layers; benchmark productivity gains; redesign billing models. For consumers: evaluate access-to-justice tools for accuracy and regulatory authorisation.

WHEN IT IS USED

Law-firm strategy reviews, corporate-legal-department procurement, regulatory-sandbox programmes, and bar-association policy work.

BY WHOM IT IS USED

Law-firm managing partners, general counsels, legal-technology vendors, and legal-services regulators.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with earlier rules-based legaltech (template generators, tagged search), LLM-driven tools handle unstructured legal language directly. Compared with general AI applications, legaltech faces uniquely demanding accuracy and confidentiality requirements.

RELATED FRAMEWORKS IN THIS COURSE

- AI Spectrum · Slibrary 15
- Generative AI Economics · Slibrary 15
- Regtech · Slibrary 10
- AI & Copyright · Slibrary 18

SOURCES (HARVARD REFERENCING STYLE)

Stanford CodeX (2024) The CodeX Techindex. Stanford, CA: Stanford Law School.
Susskind, R. (2023) Tomorrow's Lawyers. 3rd edn. Oxford: Oxford University Press.
Thomson Reuters (2024) 2024 State of the Legal Market Report. Eagan, MN: Thomson Reuters.

Insurtech

SLIBRARY 10 · SHEET 3 / 5

Digital insurance — instant quotes, usage-based pricing, AI claims

PROFESSIONAL DEFINITION

Insurtech is the digital transformation of the insurance industry. It includes online distribution (Lemonade, Hippo), usage-based products (telematics auto), AI-driven underwriting, parametric insurance, and claims automation. Insurtech firms compete with traditional insurers on cost-to-serve and customer experience while facing the same prudential regulation (EIOPA, 2024; Capgemini, 2023).

WHO DEVELOPED IT

The term emerged c.2015 in industry analyses by EY and CB Insights. EIOPA (the European Insurance and Occupational Pensions Authority) provides the most rigorous regulatory treatment. Capgemini's World Insurtech Report is a standard industry reference.

WHY IT WAS DEVELOPED

Insurance has been slower to digitalise than banking but is now catching up. Understanding insurtech is essential for incumbent insurers, reinsurers, and the regulators that supervise them.

FIGURE 28 · CLAIMS LOOP

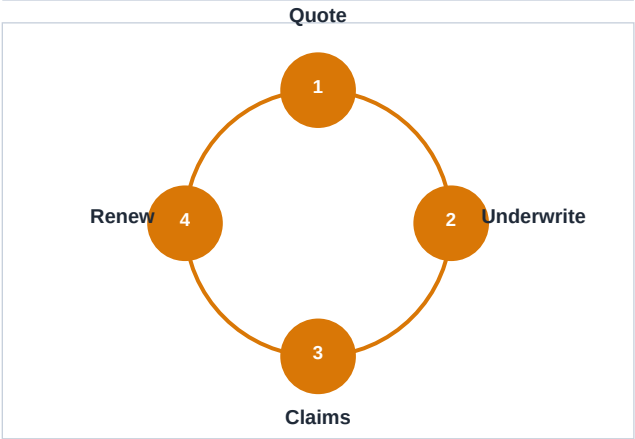


Figure 28. The insurance value loop. Insurtech firms compress each stage with automation: instant quotes, AI underwriting, photo-based claims, and behaviour-based renewal pricing (Capgemini, 2023).

HOW IT IS USED

Decompose the insurance value chain; identify the steps with highest manual cost or customer friction; pilot insurtech partners or in-house automation; track loss ratio vs. expense ratio.

WHEN IT IS USED

Insurance strategy reviews, insurtech-vendor diligence, regulatory engagement, and reinsurance partnership design.

BY WHOM IT IS USED

Insurance executives, insurtech founders, reinsurers, EIOPA-supervised firms, and B-school risk-management courses.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional digital-distribution insurance, insurtech goes further into the underwriting and claims stages. Compared with fintech, insurance technology faces longer-tail risks and stricter solvency requirements, shaping a different innovation trajectory.

RELATED FRAMEWORKS IN THIS COURSE

Fintech · Slibrary 10

AI in Business — General Spectrum · Slibrary 15

Hyper-Personalisation · Slibrary 16

Data as Asset · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

Capgemini (2023) World Insurtech Report 2023. Paris: Capgemini Research Institute.
EIOPA (2024) Financial Stability Report June 2024. Frankfurt am Main: European Insurance and Occupational Pensions Authority.
Eling, M. and Lehmann, M. (2018) 'The impact of digitalization on the insurance value chain', Geneva Papers on Risk and Insurance — Issues and Practice, 43(3), pp. 359-396.

Regtech

SLIBRARY 10 · SHEET 4 / 5

Using technology to manage regulatory compliance at scale

PROFESSIONAL DEFINITION

Regtech (Regulatory Technology) applies digital technologies — automation, AI, distributed ledger — to compliance and regulatory reporting in financial services and other regulated industries. Use cases include AML/KYC monitoring, transaction surveillance, regulatory reporting, fraud detection, and ESG disclosure (FCA, 2018; IIF, 2024).

WHO DEVELOPED IT

The UK Financial Conduct Authority popularised the term and ran the first regulatory sandbox in 2016. The Institute of International Finance and the Bank for International Settlements have developed the global policy frame. Major vendors include NICE Actimize, ComplyAdvantage, and Behavox.

WHY IT WAS DEVELOPED

Compliance costs have risen sharply since 2008. Regtech can reduce them while improving detection rates — but introduces model-risk and explainability challenges.

FIGURE 29 · COMPLIANCE FUNNEL

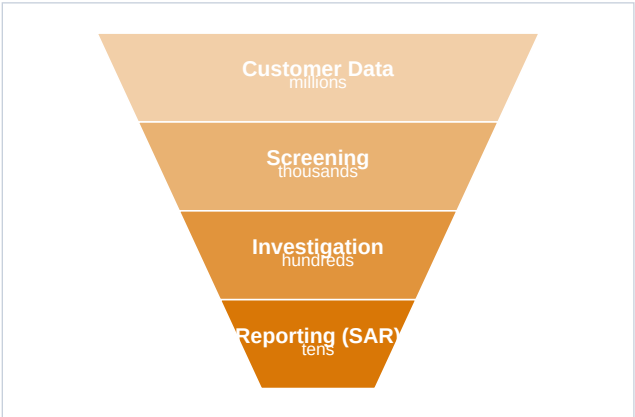


Figure 29. The compliance funnel. Regtech automates the upper stages (data ingestion, rule-based screening) while humans focus on the narrowing investigative stages (FCA, 2018; IIF, 2024).

HOW IT IS USED

Inventory existing manual compliance workflows; identify stages amenable to automation; pilot regtech vendors with measurable false-positive-rate targets; redesign the compliance operating model around automation.

WHEN IT IS USED

AML/KYC programme design, regulatory-reporting transformation, and Basel-III/Solvency-II implementation.

BY WHOM IT IS USED

Chief Compliance Officers, financial-services regulators, regtech vendors, and big-four advisory teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with manual compliance, regtech can scale to handle the volumes and complexity of modern financial flows. Compared with broader fintech, regtech is procurement-driven (B2B/B2G) rather than consumer-facing, with longer sales cycles but stickier customers.

RELATED FRAMEWORKS IN THIS COURSE

- Fintech · Slibrary 10
- API Economy · Slibrary 5
- AI in Business — General Spectrum · Slibrary 15
- DSA — Digital Services Act · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

FCA (2018) Project Innovate and Innovation Hub: Reports. London: Financial Conduct Authority.
IIF (2024) The Future of Regtech: A Global Outlook. Washington, DC: Institute of International Finance.
Arner, D.W., Barberis, J. and Buckley, R.P. (2017) 'FinTech, RegTech, and the reconceptualization of financial regulation', Northwestern Journal of International Law and Business, 37(3), pp. 371-413.

Proptech

SLIBRARY 10 · SHEET 5 / 5

Digital transformation of real estate — search, leasing, smart buildings

PROFESSIONAL DEFINITION

Proptech (Property Technology) is the digital transformation of real estate. It spans consumer-facing search and leasing platforms (Zillow, Rightmove), institutional asset and facility management software (Yardi, MRI), construction technology (BIM, modular construction), and smart-building IoT systems (JLL/CRE Tech Survey, 2024; PWC/ULI, 2024).

WHO DEVELOPED IT

The term emerged c.2014 in industry analyses; major academic treatment came from MIT's Real Estate Innovation Lab and Saïd Business School's Future of Real Estate Initiative. Industry references include the Urban Land Institute and JLL's annual CRE Tech Survey.

WHY IT WAS DEVELOPED

Real estate is the world's largest asset class but historically a digital laggard. Proptech is bringing data-driven decision-making to land use, leasing, and operations — with significant economic, sustainability, and urban-planning implications.

FIGURE 30 · PROPTECH STACK

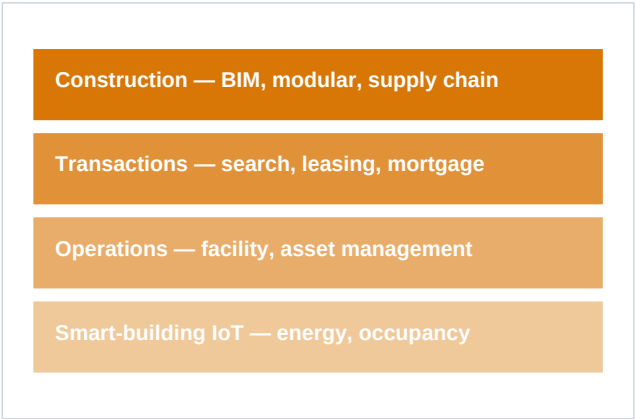


Figure 30. The proptech stack: four layers, each with distinct economics and customer types. Smart-building IoT is most rapidly growing in 2024-2025 driven by ESG-disclosure requirements (PWC/ULI, 2024).

HOW IT IS USED

Map a real-estate firm's tech estate to the four layers; identify gaps; pilot vendors; build the data infrastructure necessary for ESG and EU CSRD/EPBD reporting.

WHEN IT IS USED

Real-estate strategy reviews, ESG-disclosure programme design, smart-city planning, and proptech investment diligence.

BY WHOM IT IS USED

Real-estate executives, urban planners, ESG analysts, proptech founders, and the European Public Real Estate Association (EPRA).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the older 'real-estate IT' framing, proptech encompasses the entire lifecycle from construction to operation. Compared with general digital-transformation approaches, it integrates physical (sensors, building systems) and digital (platforms, data) layers explicitly — central to net-zero and smart-city agendas.

RELATED FRAMEWORKS IN THIS COURSE

Cloud Computing IaaS/PaaS/SaaS · Slibrary 5

Data as Asset · Slibrary 7

Industry Digitization Index · Slibrary 4

Hyper-Personalisation · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

JLL (2024) Global Real Estate Technology Survey 2024. Chicago, IL: JLL.
PWC and ULI (2024) Emerging Trends in Real Estate Europe 2025. London: PWC and Urban Land Institute.
Baum, A. (2017) PropTech 3.0: The future of real estate. Oxford: University of Oxford Saïd Business School.

Chicken-and-Egg Problem

SLIBRARY 11 · SHEET 1 / 5

How platforms ignite when neither side will join without the other

PROFESSIONAL DEFINITION

The Chicken-and-Egg Problem is the launch challenge facing every two-sided platform: buyers will not join without sellers, and sellers will not join without buyers. Platforms must break the deadlock through subsidy, single-side seeding, marquee anchors, or simulating the missing side (Caillaud and Jullien, 2003; Evans and Schmalensee, 2010).

WHO DEVELOPED IT

Caillaud and Jullien (2003) provided the first formal model. Evans and Schmalensee (2010) developed the practitioner taxonomy of solutions. Parker, Van Alstyne and Choudary (2016) popularised it in 'Platform Revolution'.

WHY IT WAS DEVELOPED

Most platform launches fail at the chicken-and-egg stage. Knowing the standard playbooks — and which fits which platform — is the central tactical question of platform launch.

FIGURE 31 · CHICKEN-AND-EGG

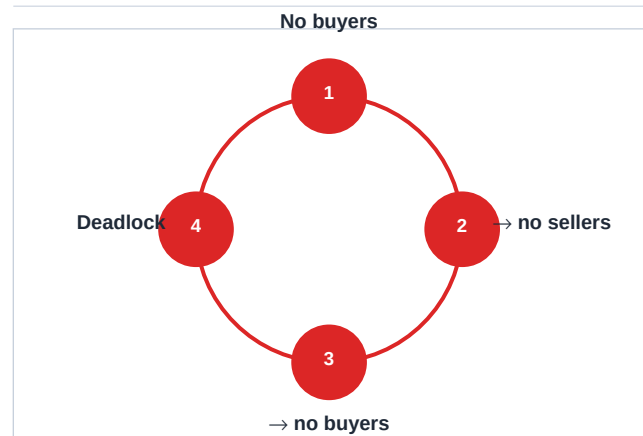


Figure 31. The chicken-and-egg deadlock. Standard solutions: subsidise one side (payment networks pay merchants); seed one side with quality content (YouTube paying creators); use a marquee anchor (Visa as a launch partner) (Evans and Schmalensee, 2010).

HOW IT IS USED

Identify which side is the 'tipping side' (typically the more cost-sensitive). Subsidise it heavily. Or seed it manually. Or simulate it (e.g. fake content). Iterate until the platform tips.

WHEN IT IS USED

Platform-launch planning, two-sided business plans, and post-mortems of failed platform launches.

BY WHOM IT IS USED

Platform founders, growth-marketing leaders, venture-capital partners, and platform-economics scholars.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-sided launch playbooks, the chicken-and-egg framing makes salient the bilateral coordination problem. Compared with generic 'cold start' rhetoric, it specifies which side to ignite first and how.

RELATED FRAMEWORKS IN THIS COURSE

Two-Sided Markets · Slibrary 6

Subsidised Side Pricing · Slibrary 11

Tipping Point · Slibrary 6

Direct & Indirect Network Effects · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

Caillaud, B. and Jullien, B. (2003) 'Chicken and egg: competition among intermediation service providers', RAND Journal of Economics, 34(2), pp. 309-328.
 Evans, D.S. and Schmalensee, R. (2010) 'Failure to launch: critical mass in platform businesses', Review of Network Economics, 9(4).
 Parker, G.G., Van Alstyne, M.W. and Choudary, S.P. (2016) Platform Revolution. New York, NY: W.W. Norton.

Subsidised Side Pricing

SLIBRARY 11 · SHEET 2 / 5

Charge the high-WTP side, subsidise the elastic side — the platform pricing law

PROFESSIONAL DEFINITION

Subsidised Side Pricing is the principle that two-sided platforms set price below cost (often free, sometimes negative) on the side with high price elasticity and large cross-side externalities, recouping margin on the other side. Whether the user or the advertiser pays is a strategic choice, not a feature of the technology (Rochet and Tirole, 2003; Eisenmann et al., 2006).

WHO DEVELOPED IT

Rochet and Tirole (2003, 2006) established the formal economics. Eisenmann, Parker and Van Alstyne (2006) translated it into management practice in Harvard Business Review. Belleflamme and Peitz (2018) provide the textbook treatment.

WHY IT WAS DEVELOPED

Platform pricing differs fundamentally from single-product pricing. Without the subsidised-side principle, services like Google Search, Facebook, and Visa look economically irrational.

FIGURE 32 · WHO PAYS

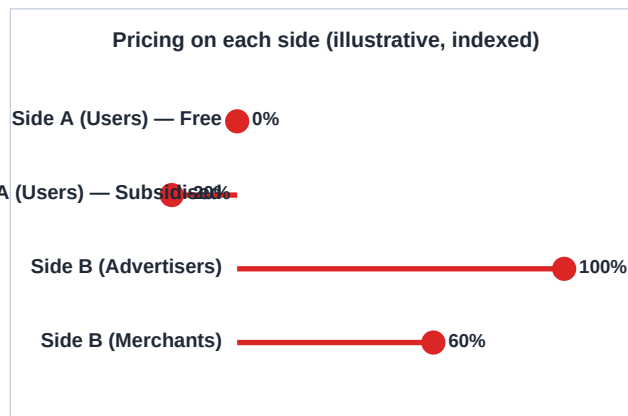


Figure 32. Pricing across the two sides of a platform. The 'free' side is often subsidised below cost; the paying side is priced based on its willingness-to-pay and the value of access to the subsidised side (Eisenmann et al., 2006).

HOW IT IS USED

Identify each side's elasticity, cross-side externality, and outside option. Set price such that the marginal user attracted on the cheap side generates more cross-side value than the subsidy.

WHEN IT IS USED

Platform pricing-strategy reviews, gatekeeper-fee analyses (Apple App Store, Amazon Marketplace), and competition-policy assessments of zero-priced services.

BY WHOM IT IS USED

Platform CFOs and pricing strategists, antitrust economists, and DMA-enforcement teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-side pricing logic, subsidised-side pricing explains why much of the modern digital economy is offered at no monetary price. Compared with general loss-leader strategies, it has formal economic foundations and is supported by Nobel-laureate research.

RELATED FRAMEWORKS IN THIS COURSE

[Two-Sided Markets · Slibrary 9](#)
[Multi-Sided Platforms · Slibrary 9](#)
[Price Discrimination · Slibrary 8](#)
[Platform Governance · Slibrary 11](#)

SOURCES (HARVARD REFERENCING STYLE)

Belleflamme, P. and Peitz, M. (2018) 'Platforms and network effects', in Corchón, L. and Marini, M. (eds.) Handbook of Game Theory and Industrial Organization. Cheltenham: Edward Elgar.
 Eisenmann, T., Parker, G. and Van Alstyne, M. (2006) 'Strategies for two-sided markets', Harvard Business Review, 84(10), pp. 92-101.
 Rochet, J.-C. and Tirole, J. (2003) 'Platform competition in two-sided markets', Journal of the European Economic Association, 1(4), pp. 990-1029.

Platform Governance

SLIBRARY 11 · SHEET 3 / 5

Rules, trust, dispute resolution, and content moderation in platform ecosystems

PROFESSIONAL DEFINITION

Platform Governance is the system of rules, enforcement mechanisms, and dispute-resolution processes through which a platform manages interactions among its users and complementors. Effective governance balances participation breadth, integrity, and platform profitability — and increasingly answers to regulators (Boudreau and Hagiu, 2009; Klonick, 2018; European Commission, 2022 — DSA).

WHO DEVELOPED IT

Boudreau and Hagiu (2009) developed the academic framework in 'Platform Rules'. Kate Klonick (2018) analysed content moderation as 'the new governors' in Harvard Law Review. The EU Digital Services Act (2022) is the most ambitious regulatory codification.

WHY IT WAS DEVELOPED

Governance is now the platform's primary product feature. Bad governance destroys trust, drives away both sides, and triggers regulator intervention. It is also where social and political conflicts increasingly play out.

FIGURE 33 · GOVERNANCE PILLARS

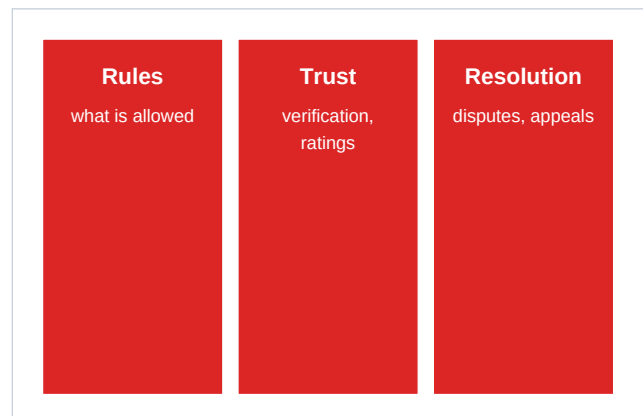


Figure 33. The three pillars of platform governance: rules (terms of service), trust mechanisms (KYC, reviews), and dispute resolution (appeals, arbitration). The DSA now sets minimum standards for each (European Commission, 2022).

HOW IT IS USED

Audit each pillar; benchmark against best practice and regulatory minima; build transparency reporting; design complementor-side participation in rule-setting.

WHEN IT IS USED

Platform-launch planning, post-incident reviews, DSA/DMA compliance programmes, and ecosystem-strategy refresh cycles.

BY WHOM IT IS USED

Trust & Safety teams, platform legal counsel, content-moderation researchers, and EU/UK/US digital regulators.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with platform-as-product framings, platform-as-governance recognises the platform as a quasi-regulatory entity itself. Compared with self-regulation rhetoric, the DSA framework introduces accountability mechanisms that interact with private rules.

RELATED FRAMEWORKS IN THIS COURSE

[Multi-Sided Platforms · Slibrary 9](#)
[Marketplace vs Platform · Slibrary 9](#)
[DSA — Digital Services Act · Slibrary 12](#)
[Algorithmic Bias · Slibrary 20](#)

SOURCES (HARVARD REFERENCING STYLE)

Boudreau, K.J. and Hagiu, A. (2009) 'Platform rules: multi-sided platforms as regulators', in Gawer, A. (ed.) *Platforms, Markets and Innovation*. Cheltenham: Edward Elgar.
 European Union (2022) Regulation (EU) 2022/2065 — Digital Services Act. Brussels: Official Journal of the European Union.
 Klonick, K. (2018) 'The new governors: the people, rules, and processes governing online speech', *Harvard Law Review*, 131(6), pp. 1598-1670.

Switching Costs in Platforms

SLIBRARY 11 · SHEET 4 / 5

The platform's perspective on lock-in — what regulators target and what defends margin

PROFESSIONAL DEFINITION

Switching Costs in platforms are the frictions — technical, contractual, social, psychological — that keep users on a platform even when a better alternative exists. From the platform's perspective they are the foundation of pricing power and customer lifetime value; from the regulator's perspective they are the central target of data-portability and interoperability mandates (Shapiro and Varian, 1999; European Union, 2022 — DMA).

WHO DEVELOPED IT

Klemperer (1995) and Shapiro and Varian (1999) developed the economics. The EU Digital Markets Act (2022) and the GDPR (2018, Art. 20) are the major regulatory responses. The UK CMA's market studies of cloud, mobile ecosystems, and music streaming examine switching costs in detail.

WHY IT WAS DEVELOPED

Switching costs are the durability driver in mature platform markets. Diagnosing them is essential for incumbents (defending margin), entrants (attacking), and regulators (intervening).

FIGURE 34 · COST CATEGORIES

Switching-cost components in platform markets (illustrative)

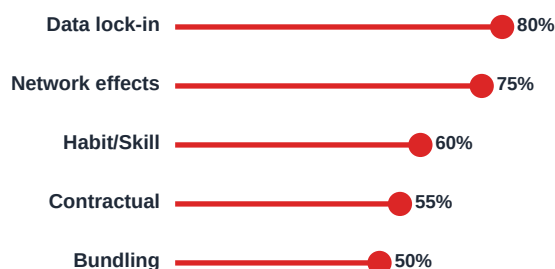


Figure 34. The dominant switching-cost components in platform markets. Data and network costs are typically the largest; the DMA targets the former through portability mandates (European Union, 2022).

HOW IT IS USED

Identify which switching-cost category dominates; assess regulator focus; design defensive (incumbent) or offensive (entrant) plays accordingly.

WHEN IT IS USED

DMA compliance reviews, platform-strategy refreshes, and competition-authority investigations.

BY WHOM IT IS USED

Platform legal and strategy teams, antitrust economists, and DMA-enforcement officials.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the generic switching-cost framing, the platform-specific view emphasises the role of network effects and complementor lock-in. Compared with classical antitrust, the DMA's ex-ante intervention model is novel — directly mandating interoperability and portability rather than waiting for case-by-case enforcement.

RELATED FRAMEWORKS IN THIS COURSE

[Lock-In & Switching Costs · Slibrary 8](#)
[DMA — Digital Markets Act · Slibrary 12](#)
[Data Economies of Scale · Slibrary 11](#)
[Platform Governance · Slibrary 11](#)

SOURCES (HARVARD REFERENCING STYLE)

European Union (2022) Regulation (EU) 2022/1925 — Digital Markets Act. Brussels: Official Journal of the European Union.
 Klemperer, P. (1995) 'Competition when consumers have switching costs', *Review of Economic Studies*, 62(4), pp. 515-539.
 Shapiro, C. and Varian, H.R. (1999) *Information Rules: A Strategic Guide to the Network Economy*. Boston, MA: Harvard Business School Press.

Data Economies of Scale

SLIBRARY 11 · SHEET 5 / 5

More users → more data → better product → more users (the data flywheel as moat)

PROFESSIONAL DEFINITION

Data Economies of Scale describe the self-reinforcing dynamic in which firms with more users generate more data, train better models, deliver better products, and so attract yet more users. Unlike traditional economies of scale (which exhaust at some efficient size), data scale economies in AI-driven products exhibit no obvious upper bound, becoming a primary durability mechanism (Varian, 2018; OECD, 2022; Crémer et al., 2019).

WHO DEVELOPED IT

Varian (2018) provided the formal economics. The European Commission's Crémer-de Montjoye-Schweitzer report (2019) translated it into competition-policy language. OECD (2022) incorporated it into market-power assessments.

WHY IT WAS DEVELOPED

Data scale economies are the most-cited rationale for current digital-competition policy. Whether they are sufficiently strong to justify ex-ante regulation is itself an active empirical debate.

FIGURE 35 · DATA FLYWHEEL

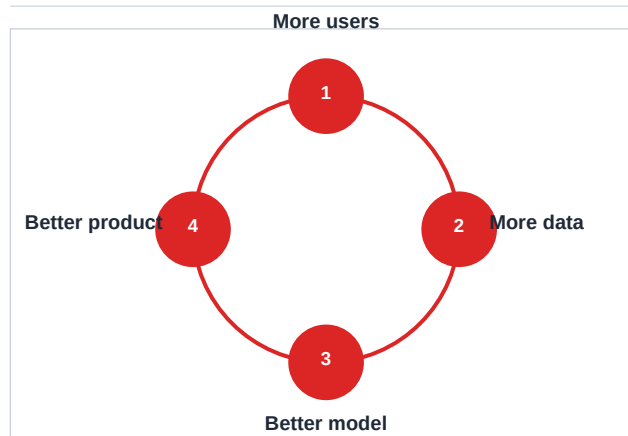


Figure 35. The data flywheel. Each turn improves the product, attracting more users — the dominant theory of why GAFAM's positions are durable (Varian, 2018; Crémer et al., 2019).

HOW IT IS USED

Test the flywheel empirically: do additional users actually improve the product, and by how much? Beyond a saturation point, scale stops mattering — diagnosing that point is critical to both incumbent and entrant strategy.

WHEN IT IS USED

Competition-policy analysis, AI-strategy reviews, M&A diligence on data-rich firms, and gatekeeper-designation logic.

BY WHOM IT IS USED

Antitrust economists, AI-strategy teams, regulators, and platform-economics scholars.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional economies of scale (with a U-shaped average-cost curve), data scale economies appear to compound indefinitely in some applications. Compared with general 'big data' rhetoric, the formal framework specifies the empirical tests and identifies the saturation conditions.

RELATED FRAMEWORKS IN THIS COURSE

[Data as Asset · Slibrary 7](#)
[AI ROI — The \\$252B Question \(MIT\) · Slibrary 15](#)
[DMA — Digital Markets Act · Slibrary 12](#)
[Multi-Sided Platforms · Slibrary 9](#)

SOURCES (HARVARD REFERENCING STYLE)

Crémer, J., de Montjoye, Y.-A. and Schweitzer, H. (2019) Competition Policy for the Digital Era. Brussels: European Commission.
 OECD (2022) The Evolving Concept of Market Power in the Digital Economy. Paris: OECD Publishing.
 Varian, H.R. (2018) Artificial intelligence, economics, and industrial organization. NBER Working Paper No. 24839. Cambridge, MA: NBER.

DMA — Digital Markets Act

SLIBRARY 12 · SHEET 1 / 5

EU 2022 — ex-ante obligations on 'gatekeeper' platforms

PROFESSIONAL DEFINITION

The Digital Markets Act (Regulation (EU) 2022/1925) imposes ex-ante obligations on designated 'gatekeeper' platforms — large core platform services with durable market positions and significant gateway roles. Obligations include data-portability, interoperability, prohibitions on self-preferencing, and bans on combining personal data across services without consent (European Union, 2022; European Commission, 2023).

WHO DEVELOPED IT

Drafted by the European Commission's DG CONNECT under Margrethe Vestager. Adopted by the European Parliament and Council in 2022. The first six gatekeepers — Alphabet, Amazon, Apple, ByteDance, Meta, and Microsoft — were designated in September 2023; Booking.com followed in 2024.

WHY IT WAS DEVELOPED

The DMA represents the world's first comprehensive ex-ante digital-competition regime. Understanding its mechanics is essential for any digital strategy, antitrust analysis, or platform-business plan touching the EU.

FIGURE 36 · DMA OBLIGATIONS

Obligation	Effect
Interop	messaging, side-loading
Portability	real-time data export
No self-preference	ranking neutrality
Consent	no cross-service data combining

Figure 36. Selected DMA obligations and their intended effect. Failure to comply carries fines of up to 10% of global turnover (20% for repeat offences) (European Union, 2022).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
For gatekeepers: implement compliance programme, designate compliance officer, prepare annual report. For complementors and entrants: identify which obligations create new opportunities (e.g. messaging interoperability, portable user data).	Annual compliance cycles, product-launch reviews touching gatekeeper services, competition-litigation, and regulatory-engagement programmes.	Platform compliance teams, the European Commission DG COMP, complementor businesses, and antitrust scholars.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with case-by-case antitrust enforcement, the DMA introduces ex-ante rules — preventing harm rather than remedying it after the fact. Compared with sectoral regulation (telecoms, banking), it targets dominant firms by category rather than industry.

RELATED FRAMEWORKS IN THIS COURSE

- DSA — Digital Services Act · Slibrary 12
- Antitrust in Digital Markets · Slibrary 12
- GAFAM Concentration · Slibrary 9
- Data Economies of Scale · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

European Commission (2023) Designating gatekeepers under the Digital Markets Act. Brussels: European Commission.
European Union (2022) Regulation (EU) 2022/1925 — Digital Markets Act. Brussels: Official Journal of the European Union.
Cabral, L., Haucap, J., Parker, G., Petropoulos, G., Valletti, T. and Van Alstyne, M. (2021) The EU Digital Markets Act: A Report from a Panel of Economic Experts. Luxembourg: Publications Office of the EU.

DSA — Digital Services Act

SLIBRARY 12 · SHEET 2 / 5

EU 2022 — content moderation, transparency, and user rights for online platforms

PROFESSIONAL DEFINITION
The Digital Services Act (Regulation (EU) 2022/2065) sets harmonised obligations on online intermediaries operating in the EU. Obligations scale with size: minimal for all hosts, more for platforms, most for Very Large Online Platforms (VLOPs) and Very Large Online Search Engines (VLOSEs) with 45M+ EU users. Content moderation, transparency, and user redress are central (European Union, 2022; European Commission, 2023).

WHO DEVELOPED IT
Drafted by the European Commission alongside the DMA. Designation of the first 19 VLOPs/VLOSEs (Alibaba AliExpress, Amazon Store, Apple App Store, Booking, Facebook, Google Search, Google Maps, Google Play, Google Shopping, Instagram, LinkedIn, Pinterest, Snapchat, TikTok, Twitter/X, Wikipedia, YouTube, Zalando, Bing) occurred in April 2023.

WHY IT WAS DEVELOPED
The DSA codifies content-moderation norms across the EU's 450M+ population. It is the world's most consequential platform-content regulation and a model for emerging UK (OSA), Brazil, and Canada regimes.

FIGURE 37 · TIERED OBLIGATIONS

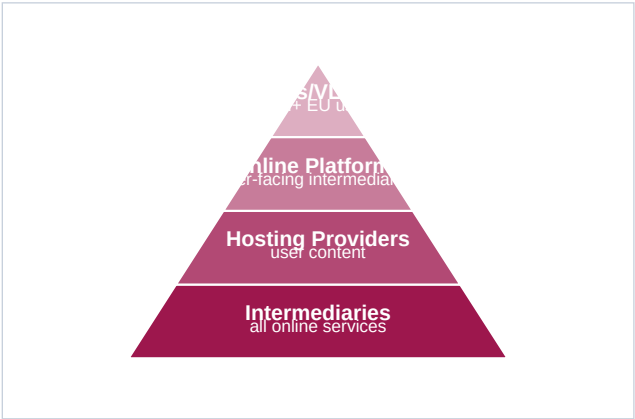


Figure 37. The DSA's tiered obligations. VLOPs/VLOSEs face systemic-risk assessment, audit, and crisis-response duties on top of the general regime (European Union, 2022; European Commission, 2023).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Identify the firm's tier; map the applicable obligations; build content-moderation, transparency-reporting, and user-redress workflows; engage Trusted Flaggers and researchers.	Annual transparency reporting, crisis-response activation, designating compliance function, and product-launch reviews.	Trust & Safety leadership, in-house counsel, the European Commission's DSA enforcement team, and Trusted Flagger civil-society organisations.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL
Compared with previous EU e-commerce directives, the DSA introduces tiered obligations scaling with platform size and risk. Compared with US Section 230, it imposes affirmative duties of care while preserving conditional liability protection.

RELATED FRAMEWORKS IN THIS COURSE

- DMA — Digital Markets Act · Slibrary 12
- Platform Governance · Slibrary 11
- Algorithmic Bias · Slibrary 20
- Antitrust in Digital Markets · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

European Commission (2023) DSA: Designation of VLOPs and VLOSEs. Brussels: European Commission.
European Union (2022) Regulation (EU) 2022/2065 — Digital Services Act. Brussels: Official Journal of the European Union.
Husovec, M. (2024) Principles of the Digital Services Act. Oxford: Oxford University Press.

Antitrust in Digital Markets

SLIBRARY 12 · SHEET 3 / 5

Google · Apple · Amazon · Meta · Microsoft — all under case

PROFESSIONAL DEFINITION

Antitrust in Digital Markets refers to the application of competition law (Article 102 TFEU in the EU; Section 2 of the Sherman Act in the US) to the conduct and structure of digital firms. As of 2025, every GAFAM firm is the subject of at least one major competition case in the US, EU, UK, or other jurisdictions, with a growing tendency toward structural remedies (DOJ, 2024; European Commission, 2024).

WHO DEVELOPED IT

Major active cases include US v. Google (search, ad-tech), US v. Apple (App Store), FTC v. Meta (Instagram, WhatsApp acquisition), and EU Commission cases against Apple, Amazon, Google, and Microsoft. Lina Khan's 2017 Yale Law Journal article 'Amazon's Antitrust Paradox' shifted scholarly opinion.

WHY IT WAS DEVELOPED

These cases will determine the structure of the digital economy for the next decade. Outcomes will shape platform business models, complementor opportunities, and consumer prices.

FIGURE 38 · ACTIVE CASES

Firm	Major Cases (illustrative)
Google	Search · Ad-tech (US, EU)
Apple	App Store · Music (EU, US)
Amazon	Marketplace (EU, FTC)
Meta	Acquisitions (FTC)
Microsoft	Cloud · Activision (UK, EU)

Figure 38. Selected active digital-antitrust cases (illustrative as of 2024-2025). Cases evolve continuously; refer to the European Commission and US DOJ press rooms for current status (DOJ, 2024; European Commission, 2024).

HOW IT IS USED

Track active cases relevant to the firm's strategy. Assess impact on competitive dynamics. Consider remedies (behavioural vs structural) and their likely effect on the firm or its rivals.

WHEN IT IS USED

Strategic planning cycles, M&A diligence, regulatory engagement programmes, and B-school competition-policy modules.

BY WHOM IT IS USED

Antitrust economists, in-house competition counsel, strategy teams, and academic legal scholars.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional antitrust (which focused on price effects), modern digital-antitrust cases emphasise structural foreclosure and innovation harm. Compared with regulator-led enforcement only, the wave of private actions and state-AG suits in the US creates a more pluralistic enforcement environment.

RELATED FRAMEWORKS IN THIS COURSE

DMA — Digital Markets Act · Slibrary 12

GAFAM Concentration · Slibrary 9

Essential Facilities Doctrine · Slibrary 12

Algorithmic Collusion · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

DOJ (2024) Antitrust Division: Major Active Cases. Washington, DC: US Department of Justice.
European Commission (2024) Antitrust enforcement priorities in digital markets. Brussels: European Commission.
Khan, L. (2017) 'Amazon's antitrust paradox', Yale Law Journal, 126(3), pp. 710-805.

Algorithmic Collusion

SLIBRARY 12 · SHEET 4 / 5

When AI pricing algorithms tacitly collude — no human conspiracy needed

PROFESSIONAL DEFINITION

Algorithmic Collusion describes the phenomenon in which independent pricing algorithms — particularly those using reinforcement learning — converge on supra-competitive prices without explicit communication or human direction. Unlike classical cartels, no agreement need exist; the equilibrium emerges from algorithmic interaction (Calvano et al., 2020; OECD, 2017; Ezrachi and Stucke, 2016).

WHO DEVELOPED IT

Ezrachi and Stucke (2016) introduced the concept in 'Virtual Competition'. Calvano, Calzolari, Denicolò and Pastorello (2020) provided the seminal experimental demonstration. The OECD (2017) and EU competition authorities have taken up the policy implications.

WHY IT WAS DEVELOPED

Algorithmic pricing is now standard in airlines, e-commerce, energy, and ride-hailing. If algorithms can tacitly collude, traditional cartel law — which requires evidence of agreement — may be inadequate.

FIGURE 39 · CONVERGENCE

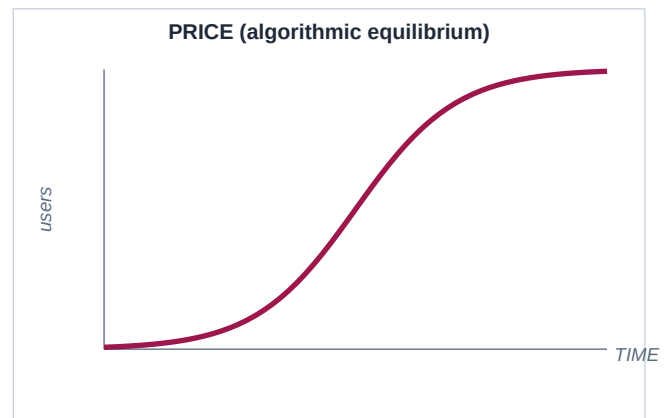


Figure 39. Q-learning algorithms in laboratory experiments converge to supra-competitive prices without communication. The implications for competition policy are profound (Calvano et al., 2020).

HOW IT IS USED

For firms: audit pricing algorithms for collusive emergence, especially in concentrated markets. For regulators: invest in market surveillance tools and consider extending cartel definitions.

WHEN IT IS USED

Pricing-algorithm design reviews, competition-policy formulation, and academic IO research.

BY WHOM IT IS USED

Pricing analytics teams, antitrust regulators (especially DG COMP and FTC), and competition economists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with classical cartels (which require communication and agreement), algorithmic collusion may emerge from market interactions of independently-developed pricing systems. Compared with traditional 'tacit collusion', it can occur in markets with many firms — expanding the scope of competition concern significantly.

RELATED FRAMEWORKS IN THIS COURSE

[Price Discrimination · Slibrary 8](#)
[Antitrust in Digital Markets · Slibrary 12](#)
[AI Spectrum · Slibrary 15](#)
[DMA — Digital Markets Act · Slibrary 12](#)

SOURCES (HARVARD REFERENCING STYLE)

Calvano, E., Calzolari, G., Denicolò, V. and Pastorello, S. (2020) 'Artificial intelligence, algorithmic pricing, and collusion', *American Economic Review*, 110(10), pp. 3267-3297.
 Ezrachi, A. and Stucke, M.E. (2016) *Virtual Competition*. Cambridge, MA: Harvard University Press.
 OECD (2017) *Algorithms and Collusion: Competition Policy in the Digital Age*. Paris: OECD Publishing.

Essential Facilities Doctrine

SLIBRARY 12 · SHEET 5 / 5

Some infrastructure is so vital that competitors must be granted access

PROFESSIONAL DEFINITION

The Essential Facilities Doctrine is the antitrust principle that a dominant firm controlling infrastructure essential to competitors — and not feasibly duplicable — may be required to grant access on fair and non-discriminatory terms. Originating in rail and telecoms, the doctrine has been applied (cautiously) to digital platforms — App Stores, payment systems, cloud, and AI compute (Areeda, 1990; European Commission, 2022).

WHO DEVELOPED IT

The doctrine is associated with *US v. Terminal Railroad* (1912) and articulated by Areeda (1990) in 'Essential Facilities: An Epithet in Need of Limiting Principles'. The EU Bronner case (1998) restricted but preserved it. The DMA partially codifies it for digital gatekeepers.

WHY IT WAS DEVELOPED

Whether App Stores, search results, cloud platforms, or AI compute are 'essential facilities' is the central legal question of the next decade of digital antitrust. The doctrine is the bridge between case-by-case enforcement and ex-ante regulation.

FIGURE 40 · THE TEST

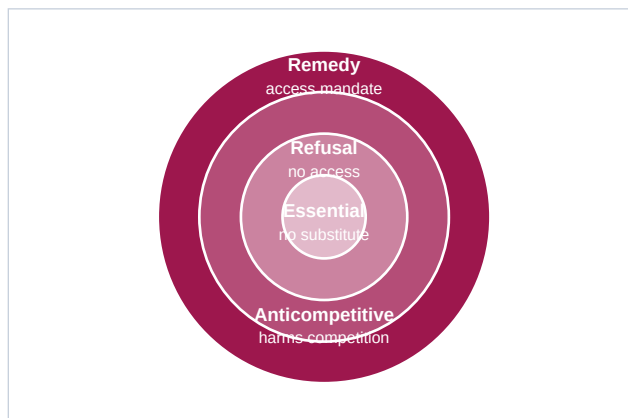


Figure 40. The classical four-part test: (1) facility is essential, (2) access is denied, (3) effect is anticompetitive, (4) granting access is feasible. All four must hold to mandate access (Areeda, 1990; European Commission, 2022).

HOW IT IS USED

Identify the candidate facility; assess substitutability; evaluate refusal effects; consider remedy feasibility. Use the test as a bridge between competition law and ex-ante regulation.

WHEN IT IS USED

Competition-law analyses of refusal-to-deal, DMA gatekeeper proceedings, interoperability rule-making, and academic antitrust seminars.

BY WHOM IT IS USED

Competition lawyers, regulators, antitrust scholars, and platform-strategy advisors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with general antitrust prohibitions (e.g. abuse of dominance), the essential-facilities doctrine specifies precise conditions under which a duty to deal arises. Compared with the DMA's blanket interoperability mandates, it preserves case-by-case scrutiny — although critics argue this is too slow for fast-moving digital markets.

RELATED FRAMEWORKS IN THIS COURSE

Antitrust in Digital Markets · Slibrary 12

DMA — Digital Markets Act · Slibrary 12

Cloud Computing IaaS/PaaS/SaaS · Slibrary 5

API Economy · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

Areeda, P. (1990) 'Essential facilities: an epithet in need of limiting principles', *Antitrust Law Journal*, 58(3), pp. 841-853.
European Commission (2022) Article 102 Enforcement Priorities. Brussels: European Commission.
Court of Justice of the European Union (1998) *Bronner v. Mediaprint*, Case C-7/97. Luxembourg: CJEU.

BCG 10-20-70 Rule

SLIBRARY 13 · SHEET 1 / 5

AI transformation success: 10% algorithms · 20% tech · 70% people & process

PROFESSIONAL DEFINITION

The BCG 10-20-70 Rule is an empirical generalisation by Boston Consulting Group (2022-2024) about where AI-transformation effort and value live. Roughly 10% of the challenge is algorithm selection, 20% is technology integration, and 70% is organisational change — people, process, governance, and culture (BCG, 2023; BCG and MIT SMR, 2024).

WHO DEVELOPED IT

Developed by BCG's Center for Digital Government and Henderson Institute. Drawn from BCG's 'AI at Scale' research with MIT Sloan Management Review (2019-2024). The framing echoes earlier IT-implementation literature (e.g. Markus and Robey, 1988) on socio-technical systems.

WHY IT WAS DEVELOPED

Most AI initiatives fail not on technology but on adoption. The 10-20-70 ratio re-allocates management attention away from model selection and toward the change-management work that determines outcomes.

FIGURE 41 · 10-20-70 SPLIT

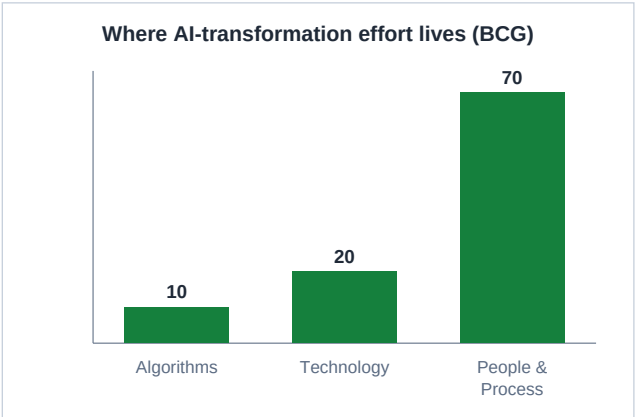


Figure 41. The 10-20-70 split. Most teams spend their attention on the 10% (model choice) while neglecting the 70% (organisational adoption) — the inverse of the empirical pattern that distinguishes successful from failed transformations (BCG and MIT SMR, 2024).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Audit the firm's AI investment by ratio. If technology and algorithm spend dominate, rebalance toward change management, training, and process redesign. Set KPIs against the 70% category, not just model performance.	AI-strategy reviews, transformation programme design, AI Centre-of-Excellence operating-model decisions, and post-mortem of failed AI initiatives.	Chief AI Officers, transformation leads, BCG and competing strategy consultants, and B-school AI-strategy modules.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the technology-first framing of much consulting work, the 10-20-70 rule centres adoption. Compared with classical change-management literature, it provides calibrated proportions — usable in board-level investment cases.

RELATED FRAMEWORKS IN THIS COURSE

McKinsey 7S — Digital Edition · Slibrary 13

Digital Maturity Model · Slibrary 13

Reskilling vs Upskilling · Slibrary 14

Industry Digitization Index · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

BCG (2023) AI at Scale: From Pilots to Profits. Boston, MA: Boston Consulting Group.
BCG and MIT Sloan Management Review (2024) Achieving Individual and Organizational Value with AI. Cambridge, MA: MIT Sloan Management Review.
Markus, M.L. and Robey, D. (1988) 'Information technology and organizational change', Management Science, 34(5), pp. 583-598.

McKinsey 7S — Digital Edition

SLIBRARY 13 · SHEET 2 / 5

Strategy · Structure · Systems · Skills · Style · Staff · Shared Values — re-applied to digital

PROFESSIONAL DEFINITION

The McKinsey 7S framework — originally developed by Peters, Waterman and Phillips (1980) — identifies seven internal elements that must be aligned for an organisation to succeed: strategy, structure, systems, skills, style, staff, and shared values. The Digital Edition asks how each element shifts when an organisation digitises (Peters and Waterman, 1982; McKinsey Digital, 2023).

WHO DEVELOPED IT

Originally Peters, Waterman and Phillips (1980, 'Structure is not Organization'). Updated by McKinsey Digital (2023) for digital transformation. The framework remains central to strategy-execution writing.

WHY IT WAS DEVELOPED

Many digital transformations focus narrowly on systems (technology) and skills (training). The 7S framework forces simultaneous attention to all seven elements — preventing alignment failures that block transformation.

FIGURE 42 · 7S DIAGRAM

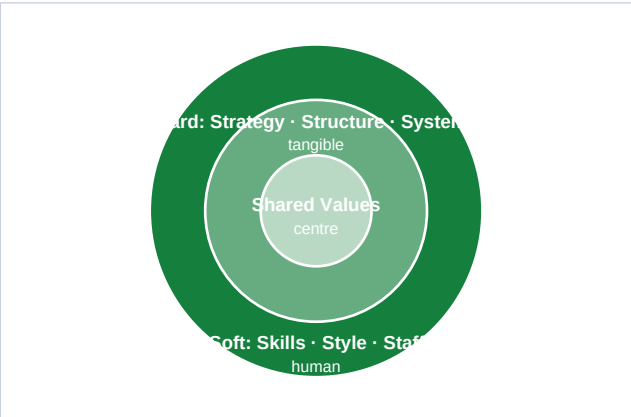


Figure 42. The 7S framework with shared values at the centre. The Digital Edition emphasises that hard elements (systems) cannot move without simultaneous soft-element shifts (style, staff) (McKinsey Digital, 2023).

HOW IT IS USED

Map each element's current state, target state, and gap. Sequence interventions so alignment is achieved at each phase. Avoid the common error of changing systems while leaving structure and style untouched.

WHEN IT IS USED

Transformation programme design, M&A integration, post-acquisition value capture, and B-school strategy-execution courses.

BY WHOM IT IS USED

Transformation directors, McKinsey and competing consultants, HR leadership, and CFOs of transforming firms.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with strategy-only frameworks, 7S incorporates execution. Compared with purely structural diagnostics, it captures the soft elements that determine real behaviour.

RELATED FRAMEWORKS IN THIS COURSE

- BCG 10-20-70 Rule · Slibrary 13
- Digital Maturity Model · Slibrary 13
- Transformation Roadmap · Slibrary 13
- Change Resistance · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

McKinsey Digital (2023) Rewired: The McKinsey Guide to Outcompeting in the Age of Digital and AI. Hoboken, NJ: Wiley.
Peters, T.J. and Waterman, R.H. (1982) In Search of Excellence. New York, NY: Harper & Row.
Waterman, R.H., Peters, T.J. and Phillips, J.R. (1980) 'Structure is not organization', Business Horizons, 23(3), pp. 14-26.

Digital Maturity Model

SLIBRARY 13 · SHEET 3 / 5

Five stages: Experimenting → Emerging → Expanding → Exploiting → Embedded

PROFESSIONAL DEFINITION

A Digital Maturity Model is a staged progression by which an organisation moves from ad-hoc digital experiments to systematically embedded digital capability. The five-stage model used widely in 2024-2025 maps each stage to typical capabilities, governance, investment, and outcomes (Westerman et al., 2014; Deloitte, 2024).

WHO DEVELOPED IT

Westerman, Bonnet and McAfee (2014, 'Leading Digital') developed the foundational academic version with the Capgemini-MIT Center for Digital Business. Deloitte's Digital Maturity Model is a widely-used commercial implementation.

WHY IT WAS DEVELOPED

Without a maturity diagnostic, leaders mis-set priorities — investing in advanced capabilities before foundational ones are in place. The model provides a shared language and a roadmap.

FIGURE 43 · MATURITY LADDER

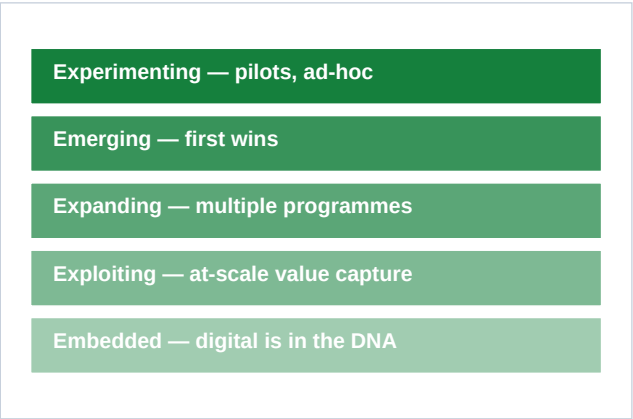


Figure 43. The five-stage maturity ladder. Most large incumbents in 2024-2025 sit at Stage 3 (Expanding); fewer than 15% reach Stage 5 (Embedded) (Westerman et al., 2014; Deloitte, 2024).

HOW IT IS USED

Diagnose the current stage across multiple dimensions (strategy, capability, culture, data). Identify the next-stage priorities. Set expectations: stages take 2-3 years each.

WHEN IT IS USED

Annual transformation reviews, board-level digital strategy refreshes, and benchmarking against industry peers.

BY WHOM IT IS USED

Chief Digital Officers, transformation directors, Deloitte and Capgemini consultants, and B-school transformation modules.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with binary 'digital vs not' framings, the maturity model captures the incremental nature of organisational change. Compared with technology-only maturity models, the version used here weights culture and governance equally.

RELATED FRAMEWORKS IN THIS COURSE

BCG 10-20-70 Rule · Slibrary 13

McKinsey 7S — Digital Edition · Slibrary 13

Industry Digitization Index · Slibrary 4

Transformation Roadmap · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

Deloitte (2024) Digital Maturity Model. London: Deloitte Insights.
Westerman, G., Bonnet, D. and McAfee, A. (2014) Leading Digital: Turning Technology into Business Transformation. Boston, MA: Harvard Business Review Press.
Kane, G.C., Palmer, D., Phillips, A.N., Kiron, D. and Buckley, N. (2017) Achieving Digital Maturity. Cambridge, MA: MIT Sloan Management Review.

Transformation Roadmap

SLIBRARY 13 · SHEET 4 / 5

Diagnose → Design → Deliver → Sustain — staged execution of digital change

PROFESSIONAL DEFINITION

A Transformation Roadmap is a staged execution plan for moving an organisation from a diagnosed current state to a designed target state. The four-phase model (Diagnose-Design-Deliver-Sustain) is widely used by major consultancies and codifies lessons from decades of transformation success and failure (BCG, 2023; McKinsey Digital, 2023; Kotter, 1995).

WHO DEVELOPED IT

John Kotter's eight-step change model (1995) is the academic foundation. The four-phase implementation map is used in nearly identical form by BCG, McKinsey, Bain, and Deloitte. Sirkin, Keenan and Jackson (2005) provided BCG's empirical 'DICE' refinement.

WHY IT WAS DEVELOPED

Roadmap discipline distinguishes successful transformations from indefinite experimentation. It also makes board-level governance possible — each phase has clear deliverables and exit criteria.

FIGURE 44 · FOUR PHASES

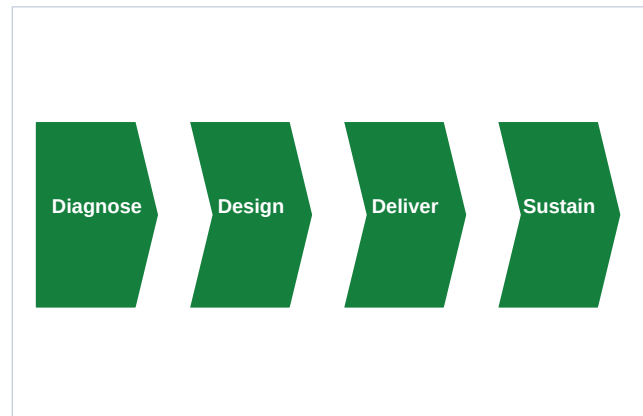


Figure 44. The four-phase roadmap. Each phase has distinct work, governance, and exit criteria. Diagnose-only initiatives stall; Deliver without Diagnose mis-fires (McKinsey Digital, 2023; Kotter, 1995).

HOW IT IS USED

Sequence the phases. Resist the temptation to skip Diagnose; resist the temptation to remain in Design forever. Build Sustain capabilities (CoEs, training, ongoing governance) before declaring victory.

WHEN IT IS USED

Setting up transformation programme governance, board approvals of multi-year transformation budgets, and post-go-live programme transitions.

BY WHOM IT IS USED

Transformation programme directors, programme management offices, and CHROs.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with project-management approaches (which focus on delivery only), the transformation roadmap front-loads diagnosis and back-loads sustainment — recognising that change is a cycle, not a project.

RELATED FRAMEWORKS IN THIS COURSE

[McKinsey 7S — Digital Edition · Slibrary 13](#)
[Change Resistance · Slibrary 13](#)
[BCG 10-20-70 Rule · Slibrary 13](#)
[Digital Maturity Model · Slibrary 13](#)

SOURCES (HARVARD REFERENCING STYLE)

BCG (2023) Transformation: The Imperative to Change. Boston, MA: Boston Consulting Group.
 Kotter, J.P. (1995) 'Leading change: why transformation efforts fail', Harvard Business Review, 73(2), pp. 59-67.
 Sirkin, H.L., Keenan, P. and Jackson, A. (2005) 'The hard side of change management', Harvard Business Review, 83(10), pp. 108-118.

Change Resistance

SLIBRARY 13 · SHEET 5 / 5

Individual · Collective · Structural — knowing which you are fighting

PROFESSIONAL DEFINITION

Change Resistance is the set of forces — individual, collective, and structural — that oppose organisational change. Effective transformation diagnoses the dominant resistance type and applies the matched intervention. Generic 'change-management' programmes tend to address only one type and so fail to move the others (Lewin, 1947; Kotter, 1995; Heifetz et al., 2009).

WHO DEVELOPED IT

Kurt Lewin (1947) introduced the unfreeze-change-refreeze and force-field models. Kotter (1995) named resistance as the principal cause of transformation failure. Heifetz, Grashow and Linsky (2009) developed the adaptive-leadership response.

WHY IT WAS DEVELOPED

Resistance is not the irrational obstacle it is sometimes framed as — it often reflects rational concerns about loss. Diagnosing the type matters because individual concerns respond to communication; structural concerns respond only to redesign.

FIGURE 45 · THREE TYPES

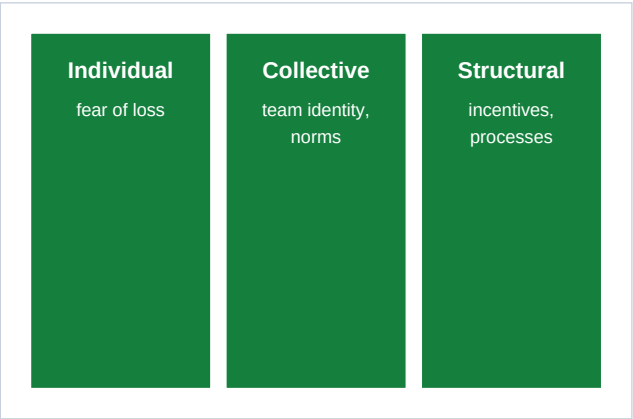


Figure 45. The three resistance types. Each requires a different intervention: individual coaching, collective dialogue, or structural redesign. Misdiagnosis is the principal cause of stalled change (Heifetz et al., 2009).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Map the dominant resistance type by stakeholder group. Apply matched interventions. Track whether resistance is decreasing, displacing, or hardening, and adapt accordingly.	Mid-transformation programme reviews, post-merger integration, and culture-change initiatives.	Change-management leads, organisation-development specialists, executive coaches, and HR leadership.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with 'win hearts and minds' rhetoric, the three-type taxonomy makes the diagnosis precise. Compared with purely psychological framings, it captures structural drivers — often the most consequential.

RELATED FRAMEWORKS IN THIS COURSE

Transformation Roadmap · Slibrary 13

McKinsey 7S — Digital Edition · Slibrary 13

BCG 10-20-70 Rule · Slibrary 13

Reskilling vs Upskilling · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Heifetz, R.A., Grashow, A. and Linsky, M. (2009) The Practice of Adaptive Leadership. Boston, MA: Harvard Business Press.
Kotter, J.P. (1995) 'Leading change: why transformation efforts fail', Harvard Business Review, 73(2), pp. 59-67.
Lewin, K. (1947) 'Frontiers in group dynamics', Human Relations, 1(1), pp. 5-41.

Reskilling vs. Upskilling

SLIBRARY 14 · SHEET 1 / 5

Reskilling = new role · Upskilling = same role, new capabilities

PROFESSIONAL DEFINITION

Reskilling is the acquisition of capabilities required for a fundamentally different role; Upskilling is the acquisition of additional capabilities for the existing role. Confusing the two is the most common error in workforce-transformation planning, leading to under-investment in the harder reskilling programmes (WEF, 2025; OECD, 2024).

WHO DEVELOPED IT

Drawn from OECD and WEF skills literature (2018-2025). The WEF's 'Reskilling Revolution' platform (launched at Davos 2020) is the most-cited reference framework. Brynjolfsson, Mitchell and Rock (2018) provided the academic foundation in their NBER paper.

WHY IT WAS DEVELOPED

Per WEF (2025), 39% of workers' core skills will change by 2030. Workforce planners must size both reskilling and upskilling programmes. Reskilling is dramatically more expensive and time-consuming — often six-to-twelve-months full-time.

FIGURE 46 · TWO PATHWAYS

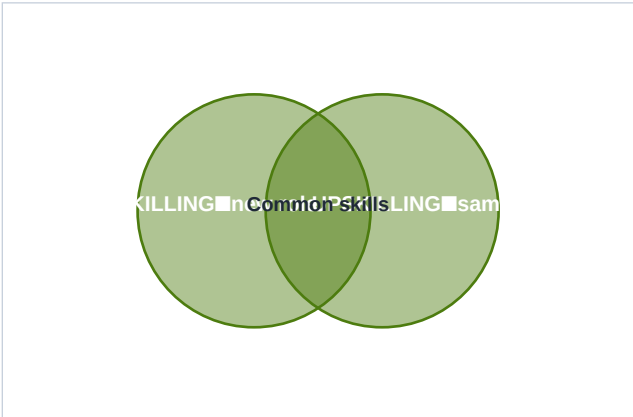


Figure 46. The two pathways. Reskilling moves a worker between job-families (e.g. retail clerk → data analyst); upskilling adds capability within a family (e.g. analyst → AI-augmented analyst) (WEF, 2025; OECD, 2024).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Inventory the workforce by job-family vulnerability. For displaceable roles, design reskilling programmes early. For augmentable roles, design upskilling programmes. Track time-to-productivity, retention, and ROI.	Workforce-strategy reviews, AI-deployment planning, M&A workforce integration, and CHRO-led talent-strategy refreshes.	Chief HR Officers, workforce planners, OECD and WEF analysts, and trade-union strategists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with generic 'training' rhetoric, the reskilling/upskilling distinction differentiates investment intensity by an order of magnitude. Compared with legacy L&D budgets, the framing forces alignment with strategic workforce planning.

RELATED FRAMEWORKS IN THIS COURSE

WEF Future of Jobs · Slibrary 4

BCG 10-20-70 Rule · Slibrary 13

Skills Matrix 2025-2030 · Slibrary 14

Task Automation Potential · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Brynjolfsson, E., Mitchell, T. and Rock, D. (2018) 'What can machines learn, and what does it mean for occupations and the economy?', AEA Papers and Proceedings, 108, pp. 43-47.
OECD (2024) OECD Skills Outlook 2024. Paris: OECD Publishing.
WEF (2025) The Future of Jobs Report 2025. Geneva: World Economic Forum.

Task Automation Potential

SLIBRARY 14 · SHEET 2 / 5

How much of a typical role can current AI plausibly automate?

PROFESSIONAL DEFINITION

Task Automation Potential measures the share of tasks within a role that current AI systems can plausibly automate. Decomposing roles into tasks (rather than treating them as monolithic) reveals that most jobs are partially — not fully — exposed. The Brynjolfsson-Eloundou-Mitchell methodology underpins most contemporary estimates (Eloundou et al., 2023; OECD, 2024).

WHO DEVELOPED IT

Eloundou, Manning, Mishkin and Rock (2023) — 'GPTs are GPTs' — provided the seminal methodology and estimates for LLM exposure. Felten, Raj and Seamans (2023) developed complementary occupation-level estimates. The OECD has integrated similar approaches into its Employment Outlook.

WHY IT WAS DEVELOPED

Automation potential is the most-quoted statistic in workforce-transformation planning. Knowing which tasks (not jobs) are exposed enables targeted investment in augmentation vs displacement strategies.

FIGURE 47 · AUTOMATION SHARE

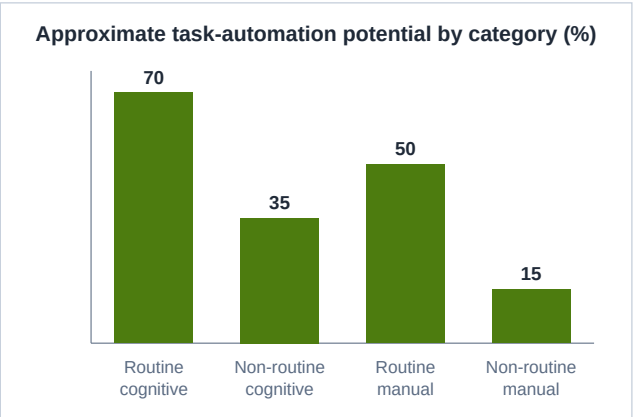


Figure 47. Estimated task-automation potential by category (illustrative, based on Eloundou et al., 2023). Routine cognitive work is the most exposed; non-routine manual work the least.

HOW IT IS USED

Decompose jobs into constituent tasks. Score each task for automation potential. Identify roles where automation potential >50% of tasks (likely redesign candidates) vs roles below (likely augmentation candidates).

WHEN IT IS USED

Workforce-strategy planning, AI-deployment business cases, redesign of roles in AI-rich workflows.

BY WHOM IT IS USED

Workforce planners, OECD and BLS labour economists, consultants advising on AI deployment.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with job-level estimates (which gave alarmist headlines), the task-level approach reveals that most jobs are partially — not fully — exposed. Compared with generic 'AI will take jobs' rhetoric, it gives operational guidance.

RELATED FRAMEWORKS IN THIS COURSE

Reskilling vs Upskilling · Slibrary 14

Step-Function AI Impact (MIT) · Slibrary 14

WEF Future of Jobs · Slibrary 4

AI Spectrum · Slibrary 15

SOURCES (HARVARD REFERENCING STYLE)

Eloundou, T., Manning, S., Mishkin, P. and Rock, D. (2023) GPTs are GPTs: An early look at the labor market impact potential of large language models. arXiv:2303.10130.
Felten, E., Raj, M. and Seamans, R. (2023) 'Occupational, industry, and geographic exposure to artificial intelligence', Strategic Management Journal, 44(11), pp. 2195-2217.
OECD (2024) OECD Employment Outlook 2024. Paris: OECD Publishing.

Step-Function AI Impact (MIT)

SLIBRARY 14 · SHEET 3 / 5

Horton: AI's labour effects jump at capability thresholds, rather than smoothly

PROFESSIONAL DEFINITION

Step-Function AI Impact is the empirical observation — articulated by John Horton (MIT) and the MIT IDE — that AI's effects on labour markets are not gradual but jump discretely at capability thresholds. Once a model crosses a competence threshold for a task class, exposed roles re-organise rapidly rather than incrementally (MIT IDE, 2025; Horton, 2024).

WHO DEVELOPED IT

John Horton (MIT Sloan) developed the empirical case in research at the MIT Initiative on the Digital Economy. The MIT IDE 2025 Annual Report formalises the step-function framing across multiple labour markets (translation, copywriting, support).

WHY IT WAS DEVELOPED

Linear projections of AI labour-market effects systematically miss the step-function nature: nothing happens for years, then a single capability release triggers wholesale redesign. Workforce planners need to provision for sudden adjustment.

FIGURE 48 · STEP FUNCTION

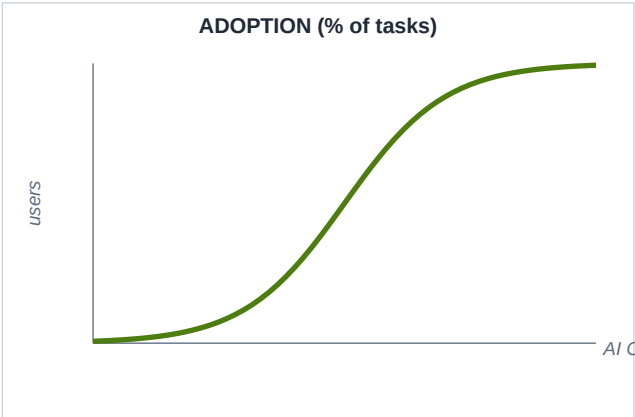


Figure 48. Step-function impact: AI adoption stays low until a capability threshold is crossed, then jumps. Translation 2017, copywriting 2023, software engineering 2024-2025 (MIT IDE, 2025; Horton, 2024).

HOW IT IS USED

Identify which task classes are near a capability threshold. Pre-position adjustment resources (reskilling budgets, transition support) ahead of the jump rather than responding after.

WHEN IT IS USED

Workforce-scenario planning, capital-budget formulation for AI projects, and labour-market policy formulation.

BY WHOM IT IS USED

MIT IDE researchers, ministries of labour, OECD analysts, and HR strategy teams in AI-exposed industries.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with linear automation projections (e.g. Frey and Osborne, 2017), the step-function model better fits the observed pattern of AI adoption. Compared with naïve exponential framings, it identifies the capability thresholds that trigger discrete transitions.

RELATED FRAMEWORKS IN THIS COURSE

- Task Automation Potential · Slibrary 14
- AI Spectrum · Slibrary 15
- AI ROI — The \$252B Question (MIT) · Slibrary 15
- WEF Future of Jobs · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

Horton, J.J. (2024) Large language models as simulated economic agents. NBER Working Paper No. 31122. Cambridge, MA: NBER.
MIT Initiative on the Digital Economy (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Sloan School of Management.
Frey, C.B. and Osborne, M.A. (2017) 'The future of employment: how susceptible are jobs to computerisation?', Technological Forecasting and Social Change, 114, pp. 254-280.

Gig Economy Platforms

SLIBRARY 14 · SHEET 4 / 5

Uber · Deliveroo · Upwork · Fiverr — flexible but precarious

PROFESSIONAL DEFINITION

The Gig Economy is a labour-market segment in which workers earn through short-term, platform-mediated tasks rather than long-term employment. Defining features include: algorithmic management, piecework pricing, contractor classification, and (typically) limited social protection (De Stefano, 2016; ILO, 2021; Aloisi and De Stefano, 2022).

WHO DEVELOPED IT

Conceptual roots in non-standard-employment labour-economics literature. Recent academic treatment by De Stefano (2016) and Aloisi and De Stefano (2022). The ILO (2021) report 'World Employment and Social Outlook' provides global data.

WHY IT WAS DEVELOPED

Gig work has expanded sharply since 2010 and now intersects with major regulatory battles (the EU Platform Work Directive 2024, UK Uber BV v Aslam Supreme Court 2021, California AB5).

FIGURE 49 · GIG VS EMPLOYMENT

Dimension	Employee	Gig Worker
Hours	Set, paid time	Per task
Management	Manager	Algorithm
Protection	Statutory	Limited
Pay structure	Salary/hourly	Piecework

Figure 49. Gig vs. employment comparison. The legal status of gig workers (employee, worker, contractor) is now the central regulatory question across major jurisdictions (De Stefano, 2016; ILO, 2021).

HOW IT IS USED

For platforms: comply with new EU Platform Work Directive (2024) presumption of employment. For workers: assess platform terms, classification, and protection. For policymakers: balance flexibility against social protection.

WHEN IT IS USED

EU Platform Work Directive transposition (2026), labour-market policy formulation, and platform-business legal-risk reviews.

BY WHOM IT IS USED

Platform legal counsel, ministries of labour, trade unions, ILO and OECD analysts, and labour-economics scholars.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with employee-vs-self-employed binary, the gig framing recognises that platform workers face algorithmic management more akin to employment. Compared with general non-standard-employment literature, it focuses on the specific dynamics of digital platform mediation.

RELATED FRAMEWORKS IN THIS COURSE

- Multi-Sided Platforms · Slibrary 9
- Two-Sided Markets · Slibrary 6
- Platform Governance · Slibrary 11
- Reskilling vs Upskilling · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Aloisi, A. and De Stefano, V. (2022) Your Boss is an Algorithm. Oxford: Hart Publishing.
De Stefano, V. (2016) The rise of the just-in-time workforce. Conditions of Work and Employment Series No. 71. Geneva: ILO.
ILO (2021) World Employment and Social Outlook 2021: The role of digital labour platforms. Geneva: International Labour Organization.

Skills Matrix 2025-2030

SLIBRARY 14 · SHEET 5 / 5

Analytical thinking · AI & big data · resilience · leadership — the WEF skills agenda

PROFESSIONAL DEFINITION

The Skills Matrix 2025-2030 is the WEF Future of Jobs Report's projection of which skills will rise and fall in importance over the period. Top rising skills include analytical thinking, AI and big data, resilience and adaptability, technology literacy, and leadership and social influence (WEF, 2025; OECD, 2024).

WHO DEVELOPED IT

Compiled by the World Economic Forum from a survey of >800 leading employers across 55 economies (WEF, 2025). Complementary frameworks from OECD (Skills Outlook 2024) and the European Skills Agenda (2020-2025).

WHY IT WAS DEVELOPED

The Skills Matrix is the most-cited reference in workforce-strategy planning. It anchors training-investment decisions, university curriculum reform, and individual career-development planning.

FIGURE 50 · TOP SKILLS

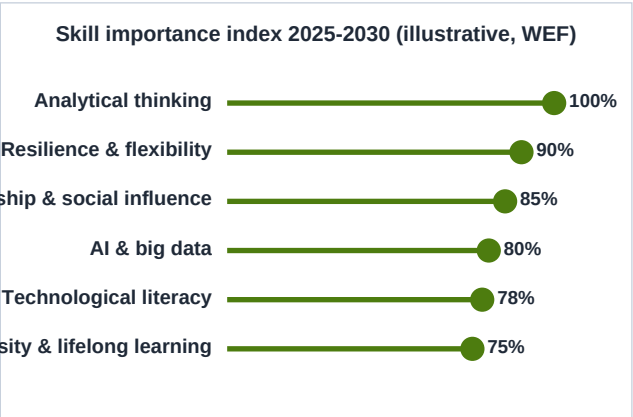


Figure 50. Top rising skills 2025-2030 (illustrative, WEF). The mix of cognitive, self-management, and technology skills reflects an adapting-to-AI labour market rather than a purely technical one (WEF, 2025).

HOW IT IS USED

Compare the firm's skills inventory against the rising-skills set. Prioritise gap-closing in the largest exposures. Update talent-attraction, learning-and-development, and performance-management systems accordingly.

WHEN IT IS USED

Annual workforce planning, university curriculum reviews, and individual career-development conversations.

BY WHOM IT IS USED

Chief HR Officers, university leaders, education ministries, career-development professionals, and workforce-strategy consultants.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with hard-vs-soft-skills dichotomies, the WEF matrix shows the integration of cognitive and self-management skills with technical literacy. Compared with national or sectoral skills frameworks, it provides a globally comparable benchmark.

RELATED FRAMEWORKS IN THIS COURSE

Reskilling vs Upskilling · Slibrary 14

WEF Future of Jobs · Slibrary 4

Task Automation Potential · Slibrary 14

BCG 10-20-70 Rule · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

European Commission (2020) European Skills Agenda for sustainable competitiveness, social fairness and resilience. Brussels: European Commission.
OECD (2024) OECD Skills Outlook 2024. Paris: OECD Publishing.
WEF (2025) The Future of Jobs Report 2025. Geneva: World Economic Forum.

AI Spectrum

SLIBRARY 15 · SHEET 1 / 5

Narrow AI → General AI → Superintelligence — the capability ladder

PROFESSIONAL DEFINITION

The AI Spectrum classifies AI systems by capability scope: Artificial Narrow Intelligence (ANI, performs specific tasks), Artificial General Intelligence (AGI, performs most human cognitive tasks), and Artificial Superintelligence (ASI, exceeds human capability across domains). All deployed AI today is narrow; AGI/ASI remain hypothetical, though the boundary is increasingly debated (Russell and Norvig, 2021; Bostrom, 2014).

WHO DEVELOPED IT

The taxonomy is established in mainstream AI textbooks (Russell and Norvig, 2021). Bostrom (2014) developed the safety-focused ASI literature. Goertzel and Pennachin (2007) coined 'AGI' as a research-programme label.

WHY IT WAS DEVELOPED

Conflating narrow AI with AGI is the most common error in popular and managerial discussion. The spectrum allows precise framing of what current systems can and cannot do — and what investment levels are appropriate.

FIGURE 51 · CAPABILITY LADDER

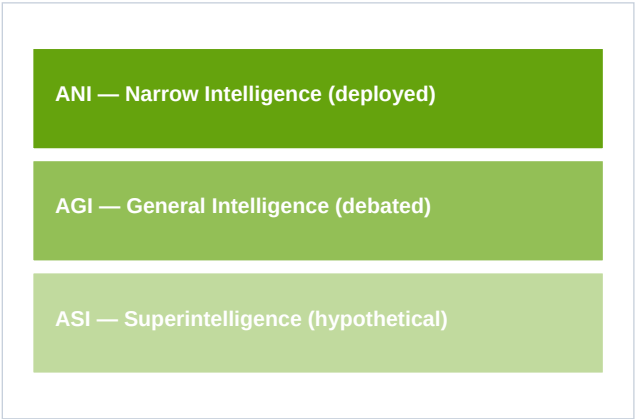


Figure 51. The AI capability ladder. All commercial AI today is narrow — including GPT-class systems, despite their breadth. AGI is contested; ASI is theoretical (Russell and Norvig, 2021).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Classify AI deployments by capability layer. For ANI: focus on integration, governance, ROI. For AGI debates: separate research, policy, and commercial implications.	AI-strategy reviews, board-level AI literacy briefings, regulatory-engagement preparation, and AI-policy formulation.	AI-strategy teams, board directors, AI-policy researchers, and B-school AI-introduction modules.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with binary 'AI vs not' framings, the spectrum allows precise capability discussion. Compared with discipline-specific taxonomies (ML, NLP, CV), it provides a high-level scaffold for cross-domain dialogue.

RELATED FRAMEWORKS IN THIS COURSE

- Generative AI Economics · Slibrary 15
- AI ROI — The \$252B Question (MIT) · Slibrary 15
- EU AI Act · Slibrary 20
- Step-Function AI Impact (MIT) · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Bostrom, N. (2014) Superintelligence: Paths, Dangers, Strategies. Oxford: Oxford University Press.
Goertzel, B. and Pennachin, C. (eds.) (2007) Artificial General Intelligence. Berlin: Springer.
Russell, S. and Norvig, P. (2021) Artificial Intelligence: A Modern Approach. 4th edn. Hoboken, NJ: Pearson.

Generative AI Economics

SLIBRARY 15 · SHEET 2 / 5

Inference cost curves · moats · the 'thin-wrapper' problem

PROFESSIONAL DEFINITION
Generative AI Economics studies the cost structures, revenue models, and competitive dynamics of firms producing or applying generative AI. Key features: very high training fixed costs, falling but non-zero inference costs, contested durability of model moats, and the 'thin-wrapper' challenge — the difficulty of building defensible products on top of foundation models (Capgemini, 2024; OECD, 2024; MIT IDE, 2025).

WHO DEVELOPED IT
Industry analysis primarily by Capgemini Research Institute, Bain, BCG, and McKinsey (2023-2025). Academic treatment by MIT IDE (2025), which integrates capacity, cost, and value-creation dimensions.

WHY IT WAS DEVELOPED
Generative AI is the largest infrastructure investment wave since cloud. Understanding its economics is essential to any AI-strategy or platform-investment decision today.

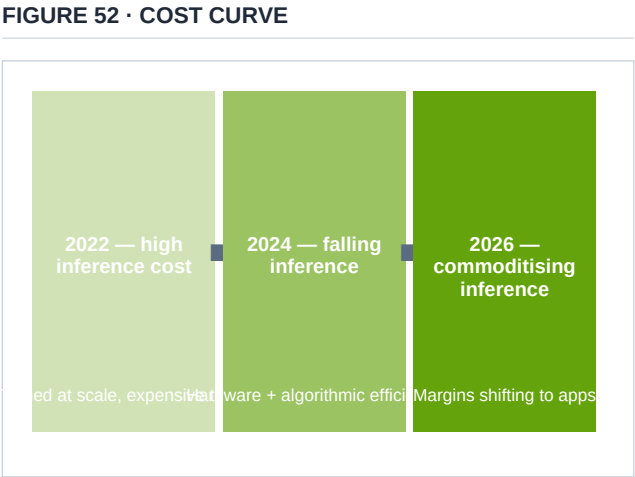


Figure 52. The inference-cost trajectory. As inference commoditises, value migrates to vertical applications — but the 'thin-wrapper' problem warns that simple model wrappers are themselves vulnerable to disintermediation (Capgemini, 2024; MIT IDE, 2025).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Map the AI-product business across three layers (model, infrastructure, application). Identify the layer with durable margin. For thin-wrapper businesses, build differentiation in data, distribution, or workflow integration.	AI-product strategy, generative-AI investment cases, and venture-capital diligence on AI-application start-ups.	AI-product CEOs, foundation-model labs (OpenAI, Anthropic, Google DeepMind), venture capitalists, and corporate-strategy teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL
Compared with the SaaS economics framework, generative-AI economics has dramatically higher fixed costs and a still-unsettled question about model moats. Compared with general 'AI hype' framings, it grounds the analysis in cost curves, capacity, and competitive dynamics.

RELATED FRAMEWORKS IN THIS COURSE

AI Spectrum · Slibrary 15

AI ROI — The \$252B Question (MIT) · Slibrary 15

Cloud Computing IaaS/PaaS/SaaS · Slibrary 5

API Economy · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)
Capgemini Research Institute (2024) Top Tech Trends 2025-2026. Paris: Capgemini.
MIT Initiative on the Digital Economy (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Sloan.
OECD (2024) Initial policy considerations for generative artificial intelligence. Paris: OECD Publishing.

AI ROI — The \$252B Question (MIT)

SLIBRARY 15 · SHEET 3 / 5

Global AI spend grew sharply — but measurable productivity has not

PROFESSIONAL DEFINITION

The AI ROI Gap describes the empirical observation that global enterprise AI spending (estimated \$252 billion+ in 2024) has not yet produced commensurate measured productivity gains in macroeconomic statistics or corporate financials. The MIT IDE characterises this as a 'productivity paradox' analogous to the 1980s computing paradox identified by Solow (MIT IDE, 2025; Brynjolfsson, 1993; Rogowski, 2025).

WHO DEVELOPED IT

MIT Initiative on the Digital Economy (2025) developed the empirical case. Builds on Brynjolfsson's (1993) original IT-productivity-paradox framing. Rob Rogowski's (2025) synthesis 'Exploring AI's Impact: From Hype to ROI' provides the visualisation widely circulated in late 2025.

WHY IT WAS DEVELOPED

The gap shapes board-level conversations about AI investment. If genuine, it implies we are early in a J-curve (productivity comes later); if structural, it implies AI value capture is harder than expected.

FIGURE 53 · SPEND VS PRODUCTIVITY

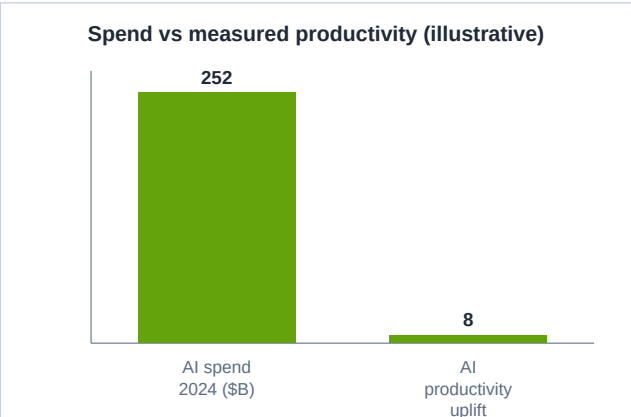


Figure 53. The ROI gap. Investment runs ahead of measured productivity — consistent with the J-curve seen with previous general-purpose technologies (electricity, computing) (MIT IDE, 2025; Brynjolfsson, 1993).

HOW IT IS USED

For boards: stage-gate AI investment with productivity-tracking. For researchers: instrument firms to measure AI-driven productivity rather than relying on self-reports. For policymakers: invest in complementary intangibles (data, training).

WHEN IT IS USED

Annual AI investment reviews, productivity-research design, and macroeconomic policy formulation.

BY WHOM IT IS USED

MIT IDE researchers, corporate CFOs, central-bank economists, and OECD productivity analysts.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the alarmist 'AI bubble' framing, the productivity-paradox framing places current investment in a longer historical context. Compared with naive ROI models, it acknowledges the lag between technology investment and measured productivity.

RELATED FRAMEWORKS IN THIS COURSE

Generative AI Economics · Slibrary 15

BCG 10-20-70 Rule · Slibrary 13

Step-Function AI Impact (MIT) · Slibrary 14

Industry Digitization Index · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

Brynjolfsson, E. (1993) 'The productivity paradox of information technology', Communications of the ACM, 36(12), pp. 66-77.
MIT Initiative on the Digital Economy (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Sloan.
Rogowski, R. (2025) Exploring AI's Impact: From Hype to ROI. Visual synthesis of MIT IDE research. Cambridge, MA.

Human-AI Pairing (MIT)

Aral & Ju: performance depends on personality-AI fit

SLIBRARY 15 · SHEET 4 / 5

PROFESSIONAL DEFINITION

Human-AI Pairing research at MIT IDE (Aral and Ju, 2024) demonstrates that AI-augmented performance depends not only on the AI's capability but also on the matched human's personality, working style, and prior expertise. The same AI tool can produce sharply different productivity outcomes across users — implying that workforce design, not tool selection, often dominates AI-deployment success (MIT IDE, 2025; Aral and Ju, 2024).

WHO DEVELOPED IT

Sinan Aral and Harang Ju at MIT Sloan / MIT IDE conducted the seminal field experiments (2023-2024). The MIT IDE 2025 Annual Report formalises the pairing implications for workforce strategy.

WHY IT WAS DEVELOPED

AI-deployment business cases routinely assume uniform productivity gains across the workforce. The pairing research shows this is empirically wrong, with implications for ROI, training design, and equity.

FIGURE 54 · PAIRING MATRIX

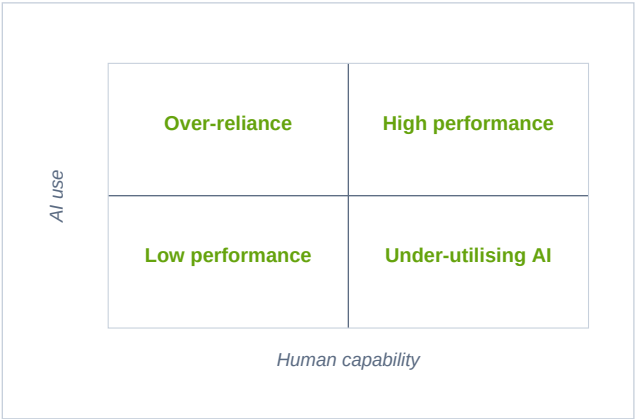


Figure 54. The pairing matrix. The high-performance cell requires both human capability and effective AI use. Over-reliance and under-utilisation are equally important failure modes (Aral and Ju, 2024; MIT IDE, 2025).

HOW IT IS USED

Test AI tools across diverse worker profiles before scaling. Track productivity by user segment. Design training and tool defaults differently for each segment.

WHEN IT IS USED

AI-deployment pilots, workforce-segmentation analyses, and post-deployment productivity assessments.

BY WHOM IT IS USED

Workforce planners, AI-deployment leads, MIT IDE researchers, and HR analytics teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with technology-first AI deployment, pairing research forces a worker-centric approach. Compared with general personality research, it ties the trait-tool match to measurable productivity outcomes.

RELATED FRAMEWORKS IN THIS COURSE

Cognitive Trade-Off (MIT) · Slibrary 15

AI ROI — The \$252B Question (MIT) · Slibrary 15

Reskilling vs Upskilling · Slibrary 14

BCG 10-20-70 Rule · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

Aral, S. and Ju, H. (2024) Human-AI pairing and productivity. Working paper, MIT Sloan School of Management. MIT Initiative on the Digital Economy (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Sloan.
Dell'Acqua, F. et al. (2023) Navigating the jagged technological frontier: Field experimental evidence. HBS Working Paper 24-013. Boston, MA: Harvard Business School.

Cognitive Trade-Off (MIT)

Speed gains from AI trade against critical-thinking atrophy

SLIBRARY 15 · SHEET 5 / 5

PROFESSIONAL DEFINITION

The Cognitive Trade-Off describes the empirical pattern that AI augmentation typically speeds up task completion but, with sustained use, can erode users' independent critical-thinking capability. Mitigation requires deliberate friction in the workflow — tasks where users must engage critically rather than accept AI output uncritically (MIT IDE, 2025; Gerlich, 2025).

WHO DEVELOPED IT

The trade-off was identified by MIT IDE research and contemporaneous studies, notably Gerlich's (2025) Microsoft Research-affiliated work on cognitive offloading. The MIT IDE (2025) Annual Report integrates these findings into design recommendations.

WHY IT WAS DEVELOPED

Productivity-only AI metrics miss the long-run capability cost. Workforce planners and education leaders need to balance immediate output gains against the development of durable human cognitive capacity.

FIGURE 55 · TRADE-OFF

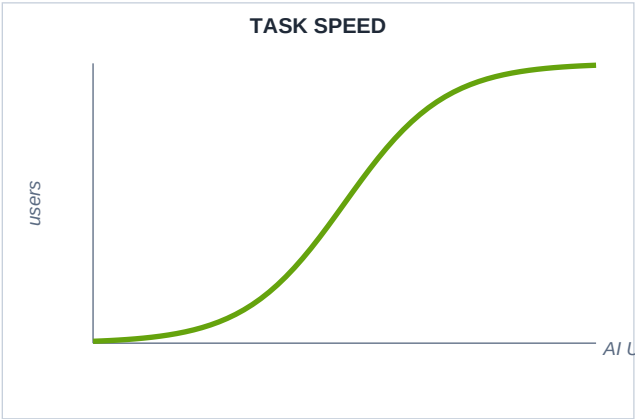


Figure 55. The trade-off curve: speed rises with AI use, but critical-thinking measures decline. Designs that build deliberate friction (review, justify, override) preserve more cognitive engagement (MIT IDE, 2025; Gerlich, 2025).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
In workflow design: build review steps where users must justify or override AI output. In education: distinguish AI-augmented tasks from skill-development tasks. In HR: track both productivity and skill-development metrics.	AI-deployment design reviews, education-policy formulation, and organisational-learning programmes.	AI-product designers, education leaders, learning-and-development leads, and MIT IDE researchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with productivity-only metrics, the trade-off framing surfaces the long-run cost of AI-mediated cognitive offloading. Compared with general 'screen time' framings, it provides specific design and policy levers.

RELATED FRAMEWORKS IN THIS COURSE

- Human-AI Pairing (MIT) · Slibrary 15
- AI Spectrum · Slibrary 15
- Reskilling vs Upskilling · Slibrary 14
- Skills Matrix 2025-2030 · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Gerlich, M. (2025) AI tools and critical thinking: empirical evidence of cognitive offloading. Working paper. MIT Initiative on the Digital Economy (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Sloan.

Risko, E.F. and Gilbert, S.J. (2016) 'Cognitive offloading', Trends in Cognitive Sciences, 20(9), pp. 676-688.

5A Framework (Kotler)

SLIBRARY 16 · SHEET 1 / 5

Aware · Appeal · Ask · Act · Advocate — Kotler's digital-era customer path

PROFESSIONAL DEFINITION

The 5A Framework, developed by Philip Kotler and colleagues (2017, 2021), reframes the consumer journey for the connectivity era as a five-stage path: Aware, Appeal, Ask, Act, Advocate. It supersedes the linear funnel and the 4A model, recognising that advocacy now precedes purchase as often as it follows it (Kotler, Kartajaya and Setiawan, 2017, 2021).

WHO DEVELOPED IT

Philip Kotler, Hermawan Kartajaya, and Iwan Setiawan articulated the framework in 'Marketing 4.0' (2017) and 'Marketing 5.0' (2021). Updates in 'Marketing 6.0' (2023) integrate generative AI and the metaverse.

WHY IT WAS DEVELOPED

Linear funnel models miss the social, non-linear nature of digital consumer journeys. 5A captures advocacy as both an output and an input — the same customer can advocate for one brand while still in the 'Ask' stage for another.

FIGURE 56 · THE 5As

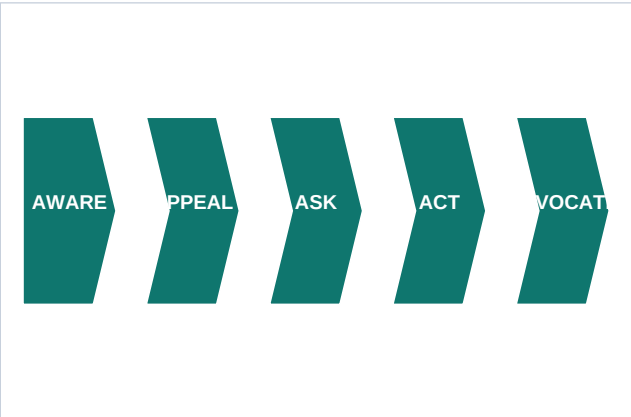


Figure 56. Kotler's 5A path. Unlike the funnel, it loops: advocates feed awareness for the next cohort. Marketing investment must allocate across all five stages (Kotler et al., 2017, 2021).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Map current investment by stage. Most firms over-invest in Aware/Appeal and under-invest in Ask/Advocate. Rebalance toward content, community, and advocacy programmes.	Marketing-strategy reviews, customer-journey redesign, and digital-marketing-channel optimisation.	CMOs, digital-marketing leaders, customer-experience designers, and B-school marketing courses.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with AIDA and the linear funnel, 5A is non-linear and recognises advocacy. Compared with generic 'customer journey' framings, it provides specific stages with matched marketing instruments.

RELATED FRAMEWORKS IN THIS COURSE

- Customer Journey Loop · Slibrary 16
- ZMOT — Zero Moment of Truth · Slibrary 16
- Hyper-Personalisation · Slibrary 16
- Experience Economy · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Kotler, P., Kartajaya, H. and Setiawan, I. (2017) Marketing 4.0: Moving from Traditional to Digital. Hoboken, NJ: Wiley.
Kotler, P., Kartajaya, H. and Setiawan, I. (2021) Marketing 5.0: Technology for Humanity. Hoboken, NJ: Wiley.
Kotler, P., Kartajaya, H. and Setiawan, I. (2023) Marketing 6.0: The Future is Immersive. Hoboken, NJ: Wiley.

ZMOT — Zero Moment of Truth

SLIBRARY 16 · SHEET 2 / 5

Lecinski/Google: consumers decide before they enter the store

PROFESSIONAL DEFINITION

Zero Moment of Truth (ZMOT) is the moment, after stimulus and before purchase, when a consumer searches for and processes online information that shapes their decision. ZMOT is the digital pre-purchase phase that makes or breaks intent before the consumer ever encounters the product physically (Lecinski, 2011; Google, 2011).

WHO DEVELOPED IT

Jim Lecinski coined the term in his 2011 Google book 'ZMOT: Winning the Zero Moment of Truth'. The framing builds on P&G's classic 'First Moment of Truth' (shelf encounter) and 'Second Moment of Truth' (use experience) — adding ZMOT as the digital prequel.

WHY IT WAS DEVELOPED

If the buying decision is made at ZMOT, brand investment must follow. Over 70% of buyers research online before purchase, even for in-store transactions — making ZMOT the highest-leverage marketing investment for many categories.

FIGURE 57 · MOMENTS OF TRUTH

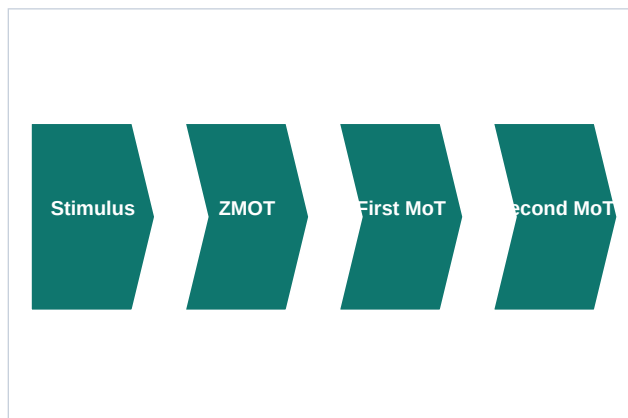


Figure 57. The four moments of truth in the digital era. The stimulus triggers a search; the search resolves at ZMOT; the FMoT (shelf) and SMoT (use) follow. Most research now concentrates investment at ZMOT (Lecinski, 2011).

HOW IT IS USED

Audit the brand's ZMOT presence: search rankings, reviews, comparison sites. Allocate investment to ensure the brand wins the ZMOT search for relevant queries.

WHEN IT IS USED

Digital-marketing planning, search-and-content strategy, and e-commerce conversion optimisation.

BY WHOM IT IS USED

CMOs, search-engine-marketing teams, e-commerce leads, and consumer-products brand managers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with linear funnel models, ZMOT identifies the specific online moment that shapes purchase. Compared with general digital-marketing rhetoric, it links investment directly to a measurable intent moment.

RELATED FRAMEWORKS IN THIS COURSE

5A Framework (Kotler) · Slibrary 16

Hyper-Personalisation · Slibrary 16

Customer Journey Loop · Slibrary 16

Experience Goods · Slibrary 8

SOURCES (HARVARD REFERENCING STYLE)

Google (2011) Winning the Zero Moment of Truth. Mountain View, CA: Google Inc.
Lecinski, J. (2011) ZMOT: Winning the Zero Moment of Truth. Mountain View, CA: Google.
Court, D., Elzinga, D., Mulder, S. and Vetvik, O.J. (2009) 'The consumer decision journey', McKinsey Quarterly, June.

Hyper-Personalisation

SLIBRARY 16 · SHEET 3 / 5

AI-driven one-to-one targeting at scale

PROFESSIONAL DEFINITION

Hyper-Personalisation is the use of AI and real-time data to deliver individually tailored content, products, and experiences at scale. Unlike segmentation (group-level targeting), hyper-personalisation aims at the individual — and increasingly at the individual moment. It is now the dominant marketing paradigm in digital-native firms (Capgemini, 2024; Deloitte Digital, 2026).

WHO DEVELOPED IT

Industry framing developed by Capgemini, Deloitte Digital, and McKinsey since 2020. Academic foundations in the recommender-systems literature (Resnick and Varian, 1997) and Anderson's long-tail work.

WHY IT WAS DEVELOPED

Hyper-personalisation typically lifts conversion 10-30% in well-instrumented categories — while raising privacy, fairness, and dependence concerns. It is now table-stakes for consumer-facing digital businesses.

FIGURE 58 · TARGETING LADDER

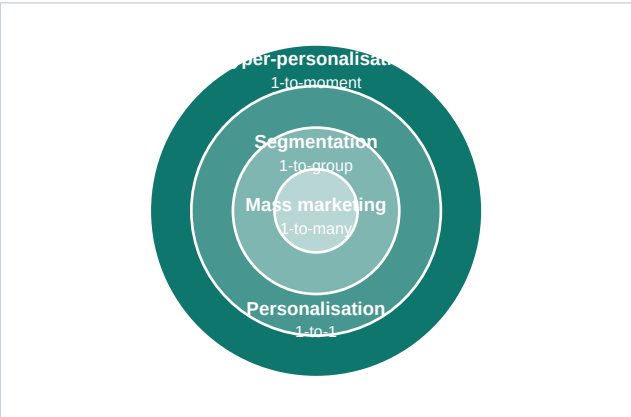


Figure 58. The targeting ladder. Each step requires more data, more compute, and more governance. Hyper-personalisation also raises GDPR and EU AI Act compliance considerations (Capgemini, 2024; Deloitte Digital, 2026).

HOW IT IS USED

Map current targeting capability against the ladder. Identify gaps. Build the data, ML, and governance infrastructure for the next rung. Audit for fairness and consent.

WHEN IT IS USED

Customer-experience strategy reviews, marketing-technology investments, and AI-deployment programmes in customer-facing teams.

BY WHOM IT IS USED

CMOs, customer-data-platform leads, AI-product managers, and consumer-protection regulators.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with classical segmentation, hyper-personalisation operates at individual and momentary granularity. Compared with naive 'use AI for marketing' rhetoric, it specifies what data, models, and consent are required.

RELATED FRAMEWORKS IN THIS COURSE

- Data as Asset · Slibrary 7
- Price Discrimination · Slibrary 8
- 5A Framework (Kotler) · Slibrary 16
- GDPR Principles · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

Capgemini Research Institute (2024) Top Tech Trends 2025-2026. Paris: Capgemini.
Deloitte Digital (2026) The Future of Service: The Age of Intelligent Experience. London: Deloitte.
Resnick, P. and Varian, H.R. (1997) 'Recommender systems', Communications of the ACM, 40(3), pp. 56-58.

Customer Journey Loop

SLIBRARY 16 · SHEET 4 / 5

Not a funnel — a continuous loop of acquire → engage → retain → expand

PROFESSIONAL DEFINITION

The Customer Journey Loop replaces the linear funnel with a continuous cycle: acquire-engage-retain-expand-advocate, with each stage feeding the next acquisition wave. It reflects the subscription-era and platform-era reality that the customer relationship continues — and intensifies — after the first transaction (Pulizzi, 2014; Forrester, 2024).

WHO DEVELOPED IT

Joe Pulizzi (2014) developed the content-driven version. Forrester and Gartner have developed the data-instrumented enterprise versions. The model integrates with Reichheld's (2003) Net Promoter Score work on advocacy.

WHY IT WAS DEVELOPED

Funnel-only thinking under-invests in retention and expansion — the most profitable segments of digital business. The loop reframes the work to continuous engagement.

FIGURE 59 · JOURNEY LOOP

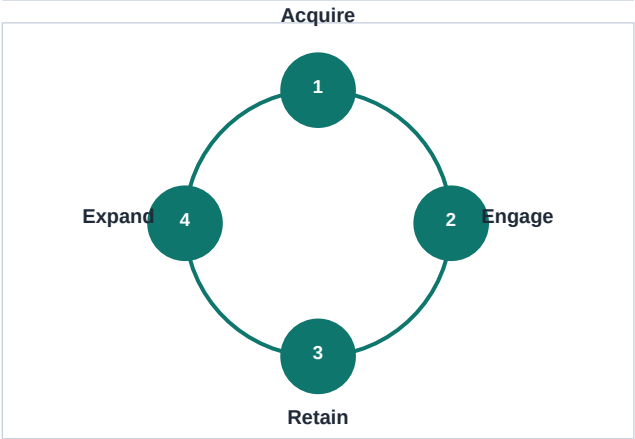


Figure 59. The journey loop. Acquisition is the smallest of the four stages — most value lies in retention and expansion. Advocacy fuels the next acquisition cycle (Pulizzi, 2014; Forrester, 2024).

HOW IT IS USED

Re-allocate marketing investment from acquisition-only to all four stages. Build instrumentation (NPS, CES, retention cohorts) for each stage. Tie compensation to loop-level metrics, not just acquisition.

WHEN IT IS USED

Marketing-strategy and CRM reviews, subscription-business design, and customer-experience investment cases.

BY WHOM IT IS USED

CMOs, customer-success leaders, CFOs of subscription businesses, and customer-experience designers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the marketing-funnel mental model, the journey loop captures the post-purchase phases that dominate subscription economics. Compared with isolated retention metrics, it integrates retention into a unified marketing framework.

RELATED FRAMEWORKS IN THIS COURSE

- Subscription Models · Slibrary 7
- 5A Framework (Kotler) · Slibrary 16
- Experience Economy · Slibrary 16
- Hyper-Personalisation · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Forrester (2024) The Customer Journey Maturity Model. Cambridge, MA: Forrester Research.
Pulizzi, J. (2014) Epic Content Marketing. New York, NY: McGraw-Hill.
Reichheld, F.F. (2003) 'The one number you need to grow', Harvard Business Review, 81(12), pp. 46-54.

Experience Economy

SLIBRARY 16 · SHEET 5 / 5

Pine & Gilmore: consumers pay for memorable experiences, not products

PROFESSIONAL DEFINITION

The Experience Economy framework, developed by Pine and Gilmore (1998, 1999), argues that economic value progresses from commodities → goods → services → experiences → transformations. Each stage commands higher willingness-to-pay because each is more differentiated and personally meaningful. Digital tools have intensified the shift toward experience- and transformation-level offerings (Pine and Gilmore, 1998, 1999, 2019).

WHO DEVELOPED IT

B. Joseph Pine II and James H. Gilmore developed the framework in their seminal 1998 Harvard Business Review article and 1999 book. The 2019 anniversary edition extends the framework to digital and AI-enabled experiences.

WHY IT WAS DEVELOPED

In commoditising digital markets, experience design is the surviving margin. Understanding the framework is essential for any consumer-facing strategy in a market where the product itself is undifferentiated.

FIGURE 60 · ECONOMIC PROGRESSION

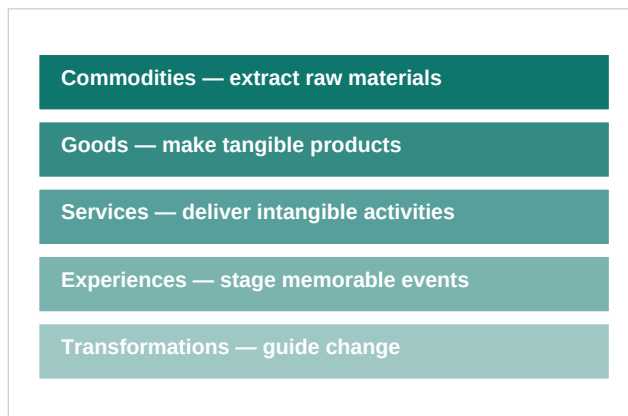


Figure 60. The economic-value ladder. Each step commands more margin than the one below — provided the offering is genuinely differentiated. Digital firms increasingly operate at the top two layers (Pine and Gilmore, 1998, 2019).

HOW IT IS USED

Identify the firm's current value level. Audit competitors at higher levels. Design experience and transformation offerings that customers will pay for as such, rather than as wrappers on commoditised goods.

WHEN IT IS USED

Brand and category strategy reviews, premium-positioning design, and the development of subscription experience offerings.

BY WHOM IT IS USED

Brand strategists, customer-experience designers, premium-segment marketing leads, and B-school strategy and marketing modules.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with goods-services dichotomies, the experience-economy ladder captures the premium tier of value creation. Compared with general 'experience design' rhetoric, it links experience to economic-value progression.

RELATED FRAMEWORKS IN THIS COURSE

[5A Framework \(Kotler\) · Slibrary 16](#)
[Hyper-Personalisation · Slibrary 16](#)
[Customer Journey Loop · Slibrary 16](#)
[Experience Goods · Slibrary 8](#)

SOURCES (HARVARD REFERENCING STYLE)

Pine, B.J. and Gilmore, J.H. (1998) 'Welcome to the experience economy', Harvard Business Review, 76(4), pp. 97-105.
 Pine, B.J. and Gilmore, J.H. (1999) The Experience Economy. Boston, MA: Harvard Business School Press.
 Pine, B.J. and Gilmore, J.H. (2019) The Experience Economy. With a New Preface by the Authors. Boston, MA: Harvard Business Review Press.

GDPR Principles

LIBRARY 17 · SHEET 1 / 5

Lawfulness · fairness · transparency · purpose · minimisation · accuracy · storage · integrity · accountability

PROFESSIONAL DEFINITION

The General Data Protection Regulation (Regulation (EU) 2016/679) sets out seven core principles for processing personal data: lawfulness/fairness/transparency, purpose limitation, data minimisation, accuracy, storage limitation, integrity and confidentiality, and accountability. Together they constitute the world's most influential data-protection regime, with extraterritorial reach (European Union, 2016; EDPB, 2024).

WHO DEVELOPED IT

Drafted by the European Commission and Parliament; in force from 25 May 2018. The European Data Protection Board (EDPB) provides authoritative interpretive guidance. National Data Protection Authorities enforce within member states.

WHY IT WAS DEVELOPED

GDPR has set the global de facto baseline. Even US, UK, and APAC firms structure their data programmes to its requirements. Maximum fines reach €20 million or 4% of global turnover, whichever is higher.

FIGURE 61 · SEVEN PRINCIPLES



Figure 61. The seven GDPR principles. Each must be demonstrably satisfied. 'Accountability' is the meta-principle — controllers must be able to evidence their compliance with the others (European Union, 2016; EDPB, 2024).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Map data-processing activities to the seven principles. Identify gaps. Build documentation, controls, and DPIAs to evidence each principle. Train staff.	Data-protection programme design, DPIAs for new processing, M&A diligence, and regulator engagement.	Data Protection Officers, in-house counsel, EDPB and national DPAs, and corporate boards reviewing GDPR compliance.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with sectoral or notice-and-consent regimes, GDPR is comprehensive and rights-based. Compared with US privacy law (largely sectoral), it provides a unified framework that has been imitated globally — UK, Brazil (LGPD), California (CCPA), Korea, India.

RELATED FRAMEWORKS IN THIS COURSE

- Personal Data as Property · Slibrary 17
- Consent & Legitimate Interest · Slibrary 17
- Data Minimisation · Slibrary 17
- Right to be Forgotten · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

EDPB (2024) Guidelines on the application of GDPR. Brussels: European Data Protection Board.
European Union (2016) Regulation (EU) 2016/679 — General Data Protection Regulation. Brussels: Official Journal of the EU.
Voigt, P. and Von dem Bussche, A. (2017) The EU General Data Protection Regulation (GDPR): A Practical Guide. Cham: Springer.

Personal Data as Property

SLIBRARY 17 · SHEET 2 / 5

Is data the individual's — or the firm's — or both?

PROFESSIONAL DEFINITION

Personal Data as Property is a contested theoretical question: should personal data be treated as property (individuals own it, can sell or licence it), as a fundamental right (inalienable, non-tradeable), or as a hybrid? GDPR adopts the rights model; some US-state and emerging-economy proposals lean toward the property model. The framing has profound consequences for consent, markets, and individual agency (Lessig, 2002; Acquisti et al., 2016; Posner and Weyl, 2018).

WHO DEVELOPED IT

Lawrence Lessig (2002) developed the property critique. Acquisti, Brandimarte and Loewenstein (2016) provided the empirical analysis of consumer choices in 'Privacy and human behavior'. Posner and Weyl (2018) advocated 'data as labour' in 'Radical Markets'.

WHY IT WAS DEVELOPED

The framing determines whether individuals can be paid for their data, whether platforms can claim ownership of derived data, and whether dignity-based or market-based remedies apply. It is the foundational question underlying all digital-rights debate.

FIGURE 62 · TWO FRAMINGS

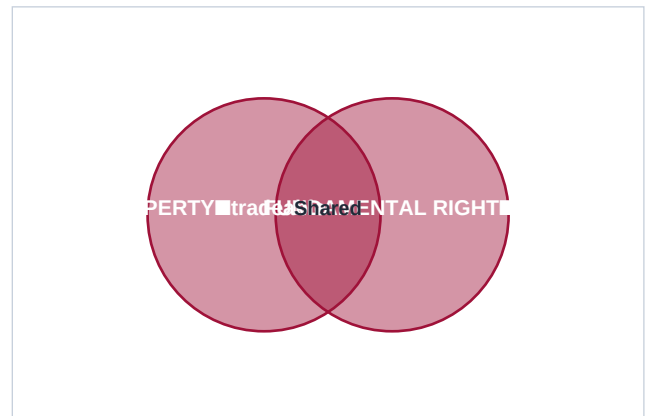


Figure 62. The two competing framings. Each has policy consequences: property allows data markets but risks coercion; rights protect dignity but limit individual choice (Acquisti et al., 2016; Posner and Weyl, 2018).

HOW IT IS USED

When designing a data programme: identify which framing the relevant law adopts; align consent and disclosure accordingly. When advocating policy: be explicit about the framing chosen and its consequences.

WHEN IT IS USED

Data-protection law-reform proposals, DPIAs for novel processing, and academic ethics-and-law discussions.

BY WHOM IT IS USED

Privacy lawyers, academic ethicists, data-rights NGOs, and data-policy researchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with implicit framings in much industry discourse, the explicit choice between property and rights makes consequences visible. Compared with one-size-fits-all approaches, it acknowledges that different jurisdictions have made different choices.

RELATED FRAMEWORKS IN THIS COURSE

[GDPR Principles · Slibrary 17](#)
[Privacy Equation \(MIT\) · Slibrary 19](#)
[Surveillance Capitalism · Slibrary 19](#)
[Behavioural Futures Markets · Slibrary 19](#)

SOURCES (HARVARD REFERENCING STYLE)

Acquisti, A., Brandimarte, L. and Loewenstein, G. (2016) 'Privacy and human behavior in the age of information', *Science*, 347(6221), pp. 509-514.
 Lessig, L. (2002) 'Privacy as property', *Social Research*, 69(1), pp. 247-269.
 Posner, E.A. and Weyl, E.G. (2018) *Radical Markets*. Princeton, NJ: Princeton University Press.

Consent & Legitimate Interest

SLIBRARY 17 · SHEET 3 / 5

GDPR's six lawful bases — consent is only one of them

PROFESSIONAL DEFINITION

GDPR Article 6 lists six lawful bases for processing personal data: consent, contract, legal obligation, vital interests, public task, and legitimate interests. Consent must be freely given, specific, informed, and unambiguous; legitimate interest requires a balancing test against the data subject's rights. Picking the wrong basis is a common and costly compliance failure (European Union, 2016; ICO, 2024).

WHO DEVELOPED IT

GDPR Article 6 sets the six bases. The European Data Protection Board, the UK ICO, and France's CNIL provide the leading interpretive guidance. Major fine cases (Meta, Google) have turned on incorrect basis selection.

WHY IT WAS DEVELOPED

Many firms default to 'consent' without realising this is the most fragile basis (can be withdrawn anytime). Legitimate interest, properly balanced, is often more appropriate and durable.

FIGURE 63 · SIX BASES

Basis	Typical Use
Consent	Marketing, optional features
Contract	Account creation, billing
Legal obligation	Tax, AML reporting
Vital interests	Emergency / life-safety
Public task	Government, public bodies
Legitimate interest	Security, analytics

Figure 63. The six GDPR Article 6 bases. Each has its own conditions and implications. Selecting and documenting the correct basis is foundational to compliance (European Union, 2016; ICO, 2024).

HOW IT IS USED

For each processing activity: document the basis, the test (balancing for legitimate interest), and the alternative considered. Update privacy notices to match. Revisit regularly.

WHEN IT IS USED

DPIAs, processing-inventory exercises, privacy-notice updates, and regulator-investigation responses.

BY WHOM IT IS USED

Data Protection Officers, in-house counsel, EDPB and national DPAs, and B-school data-law modules.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with consent-only regimes (which lead to consent fatigue), the GDPR's six-basis approach allows non-consent processing with appropriate safeguards. Compared with US notice-and-consent, it provides a more nuanced toolkit.

RELATED FRAMEWORKS IN THIS COURSE

- GDPR Principles · Slibrary 17
- Data Minimisation · Slibrary 17
- Right to be Forgotten · Slibrary 17
- Personal Data as Property · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

European Union (2016) Regulation (EU) 2016/679 — General Data Protection Regulation, Article 6. Brussels: Official Journal of the EU.
ICO (2024) Guide to the UK GDPR — Lawful basis for processing. Wilmslow: Information Commissioner's Office.
EDPB (2020) Guidelines 05/2020 on consent under Regulation 2016/679. Brussels: European Data Protection Board.

Data Minimisation

SLIBRARY 17 · SHEET 4 / 5

Collect only what is necessary — not what might be useful

PROFESSIONAL DEFINITION

Data Minimisation is the GDPR Article 5(1)(c) principle that personal data must be adequate, relevant, and limited to what is necessary for the purposes for which it is processed. It runs counter to the 'collect everything in case' instinct of much modern data engineering and is one of the most often-violated principles in practice (European Union, 2016; EDPB, 2024).

WHO DEVELOPED IT

Codified in GDPR Article 5(1)(c). Antecedents in the OECD Guidelines (1980) on the Protection of Privacy and the original 1995 EU Data Protection Directive. Major fine cases (Amazon, Meta) have included minimisation findings.

WHY IT WAS DEVELOPED

Minimisation reduces data-breach exposure, narrows regulator concerns, and aligns with the broader privacy-by-design movement. It is also, somewhat counter-intuitively, often a precondition for unlocking commercial value (cleaner, narrower datasets perform better in many ML applications).

FIGURE 64 · MINIMISATION FUNNEL

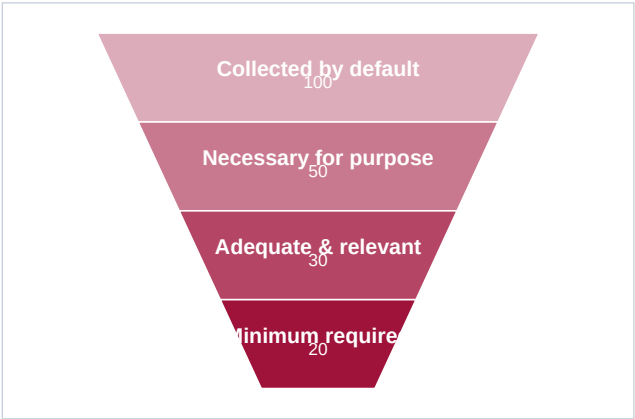


Figure 64. The minimisation funnel. The default collection is typically much larger than the minimum required. Each filter requires explicit justification (European Union, 2016; EDPB, 2024).

HOW IT IS USED

Audit collected fields against documented purposes. Eliminate fields without a specific purpose. Reduce frequency or granularity where possible. Implement privacy-by-design defaults for new systems.

WHEN IT IS USED

DPIAs, system-design reviews, ML-feature-engineering decisions, and regulator engagement.

BY WHOM IT IS USED

DPOs, data engineers, ML engineers, in-house counsel, and EDPB / national DPAs.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with 'collect everything' default of much data engineering, minimisation imposes purposeful design. Compared with case-by-case privacy review, it sets a default of restraint that is more durable as systems evolve.

RELATED FRAMEWORKS IN THIS COURSE

GDPR Principles · Slibrary 17

Consent & Legitimate Interest · Slibrary 17

Right to be Forgotten · Slibrary 17

Data Extraction Imperative · Slibrary 19

SOURCES (HARVARD REFERENCING STYLE)

EDPB (2024) Guidelines on the application of the data minimisation principle. Brussels: European Data Protection Board.
European Union (2016) Regulation (EU) 2016/679 — GDPR, Article 5. Brussels: Official Journal of the EU.
OECD (1980) Guidelines on the Protection of Privacy and Transborder Flows of Personal Data. Paris: OECD.

Right to be Forgotten

SLIBRARY 17 · SHEET 5 / 5

GDPR Article 17 — erasure under specific conditions

PROFESSIONAL DEFINITION

The Right to Erasure (popularly 'Right to be Forgotten') under GDPR Article 17 entitles data subjects to demand erasure of their personal data when one of six conditions is met (e.g. data is no longer necessary, consent is withdrawn, processing is unlawful). Established for search engines by the CJEU's 2014 Google Spain ruling (European Union, 2016; CJEU, 2014).

WHO DEVELOPED IT

Originated in the CJEU's 2014 Google Spain decision (C-131/12). Codified in GDPR Article 17 (2018). Subsequent rulings have addressed scope (territorial limits, exemptions for journalism). The right is exercised in the millions of requests per year.

WHY IT WAS DEVELOPED

The right embodies a distinctive European data-protection philosophy: even lawfully-obtained data does not stay lawful indefinitely. It conflicts with US First Amendment doctrine and global archive interests, raising challenging cross-border issues.

FIGURE 65 · ERASURE PROCESS

Figure 65. The erasure process. Controllers must respond within one month (extendable by two for complex cases). Refusal must cite a specific GDPR exemption (European Union, 2016; EDPB, 2024).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Build a process to receive, verify, evaluate, and execute erasure requests within GDPR timelines. Maintain audit logs of decisions. Train staff. Communicate to data subjects.	Privacy programme design, regulator engagement, and high-profile cases involving public-figure data.	Data Protection Officers, search-engine compliance teams, and national DPAs handling complaints.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with US-style 'speech wins' framings, the right embodies the European balancing of expression and dignity. Compared with sectoral remedies (defamation, privacy torts), it provides a unified, ex-ante framework.

RELATED FRAMEWORKS IN THIS COURSE

- GDPR Principles · Slibrary 17
- Consent & Legitimate Interest · Slibrary 17
- Data Minimisation · Slibrary 17
- Copyright in the Digital Age · Slibrary 18

SOURCES (HARVARD REFERENCING STYLE)

Court of Justice of the European Union (2014) Google Spain SL v AEPD, C-131/12. Luxembourg: CJEU.
EDPB (2024). Guidelines on the right to erasure. Brussels: European Data Protection Board.
European Union (2016) Regulation (EU) 2016/679 — GDPR, Article 17. Brussels: Official Journal of the EU.

Copyright in the Digital Age

SLIBRARY 18 · SHEET 1 / 5

Designed for scarce copies — strained by zero-marginal-cost reproduction

PROFESSIONAL DEFINITION

Copyright is the legal regime granting authors exclusive rights to copy, distribute, and adapt original works. Designed for the print era — when copying required physical production — it is strained by digital reproduction at zero marginal cost. Doctrines like fair use, exhaustion, and TDM exceptions attempt to adapt copyright to digital realities (Lessig, 2004; Patry, 2009; EU Directive 2019/790).

WHO DEVELOPED IT

International framework set by the Berne Convention (1886) and TRIPS (1994). Major academic critiques by Lawrence Lessig (2004), William Patry (2009), and the Creative Commons movement. EU updates: Information Society Directive (2001), DSM Directive (2019).

WHY IT WAS DEVELOPED

Copyright is the single most contested IP regime in the digital era. Music, news, publishing, software, and now AI training all depend on its precise scope and exceptions.

FIGURE 66 · COPYRIGHT STACK

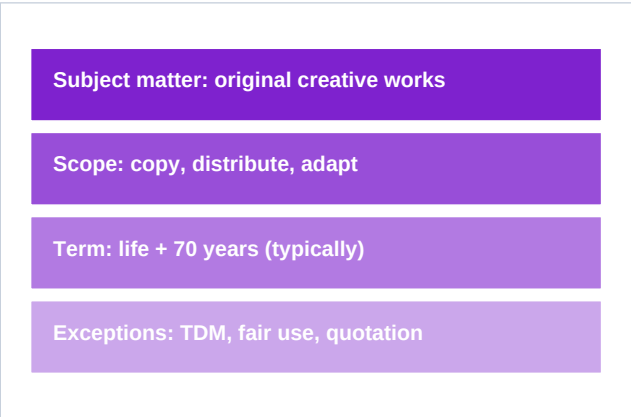


Figure 66. The copyright stack. Each layer has been pressured by digital technology — scope by streaming, term by lobbying, exceptions by AI training (Lessig, 2004; EU Directive 2019/790).

HOW IT IS USED

For creators: choose licensing (CC, traditional). For platforms: build notice-and-takedown and licensing infrastructure. For AI developers: assess TDM exception applicability and licensing alternatives.

WHEN IT IS USED

Content-licensing strategy, platform compliance design, and AI-training data acquisition.

BY WHOM IT IS USED

Content lawyers, platform compliance teams, AI legal counsel, and creator-rights advocacy groups.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with print-era copyright, digital copyright must accommodate massive cheap copying, automated processing, and global reach. Compared with abolitionist alternatives, the current regime preserves the incentive structure while adding exceptions tailored to digital uses.

RELATED FRAMEWORKS IN THIS COURSE

- Creative Commons · Slibrary 18
- AI & Copyright · Slibrary 18
- Trade Secrets vs Patents · Slibrary 18
- Zero Marginal Cost · Slibrary 8

SOURCES (HARVARD REFERENCING STYLE)

European Union (2019) Directive (EU) 2019/790 on copyright and related rights in the Digital Single Market. Brussels: Official Journal of the EU.
Lessig, L. (2004) Free Culture. New York, NY: Penguin Press.
Patry, W. (2009) Moral Panics and the Copyright Wars. Oxford: Oxford University Press.

Creative Commons

SLIBRARY 18 · SHEET 2 / 5

Standardised licences let creators share on their own terms

PROFESSIONAL DEFINITION

Creative Commons (CC) is a non-profit organisation that provides standardised, machine-readable copyright licences allowing creators to share work on graduated terms — from full attribution-only (CC BY) through non-commercial restrictions (CC BY-NC) to share-alike requirements (CC BY-SA). CC underpins Wikipedia, much open-access scholarship, and large datasets used in AI training (Creative Commons, 2024; Lessig, 2004).

WHO DEVELOPED IT

Founded by Lawrence Lessig and others in 2001. Now used on >2 billion works globally. Major adopters include Wikipedia, Flickr, Khan Academy, and most open-access academic publishers.

WHY IT WAS DEVELOPED

CC turns copyright from a default of 'all rights reserved' into a configurable spectrum, enabling sharing while protecting the rights creators care about. It is central to the open-content ecosystem.

FIGURE 67 · LICENCE LADDER

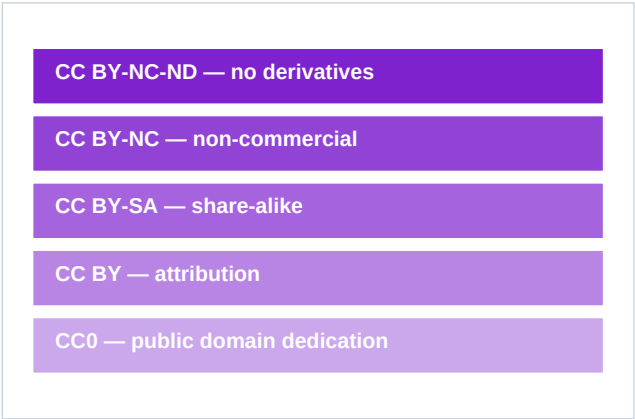


Figure 67. The CC licence ladder, from most permissive (CC0) to most restrictive (CC BY-NC-ND). All-rights-reserved copyright sits beyond the bottom rung (Creative Commons, 2024).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Creators: select the licence reflecting intended use. Users: respect attribution, share-alike, and non-commercial conditions. Platforms: surface licences in metadata.	Content-creation strategy, open-access publishing, and AI-training-data sourcing.	Academic researchers, photographers, educators, AI dataset curators, and content-platform legal teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with all-rights-reserved copyright, CC enables granular sharing. Compared with bespoke open-source licences, CC's standardisation enables efficient discovery and reuse — including for large-scale ML training.

RELATED FRAMEWORKS IN THIS COURSE

- Copyright in the Digital Age · Slibrary 18
- AI & Copyright · Slibrary 18
- Long Tail · Slibrary 7
- Trade Secrets vs Patents · Slibrary 18

SOURCES (HARVARD REFERENCING STYLE)

Creative Commons (2024) State of the Commons 2024. Mountain View, CA: Creative Commons.
Lessig, L. (2004) Free Culture. New York, NY: Penguin Press.
Boyle, J. (2008) The Public Domain: Enclosing the Commons of the Mind. New Haven, CT: Yale University Press.

Patent Trolls

SLIBRARY 18 · SHEET 3 / 5

Non-practising entities buy patents and sue — no products, just litigation

PROFESSIONAL DEFINITION

A Patent Troll (or Non-Practising Entity, NPE) is a firm whose business model is acquiring patents and asserting them through litigation rather than producing goods or services. Trolls exploit the asymmetric cost of patent litigation to extract settlements, particularly in the US software-patent regime. Reform efforts continue (FTC, 2016; Bessen and Meurer, 2014).

WHO DEVELOPED IT

Term coined by Peter Detkin (Intel) in 2001. Major academic analyses by James Bessen and Michael Meurer (2014) of Boston University. The US Federal Trade Commission's 2016 study 'Patent Assertion Entity Activity' provided definitive empirical evidence.

WHY IT WAS DEVELOPED

NPE litigation imposed an estimated \$29 billion+ direct cost on US firms in peak years. The phenomenon has shaped patent reform debates, the Supreme Court's Alice Corp. (2014) ruling on software patents, and corporate patent-defence strategy.

FIGURE 68 · TROLL LOOP

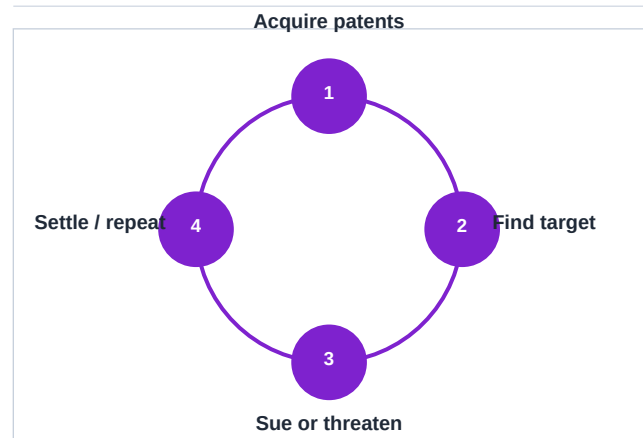


Figure 68. The troll business model. Settlements typically run \$100K-\$10M — below the cost of litigating to a defence verdict. The asymmetry is the engine of the model (FTC, 2016; Bessen and Meurer, 2014).

HOW IT IS USED

For target firms: build defensive patent portfolios, join defensive patent pools (LOT Network, RPX), early-stage patent clearance. For policymakers: reform fee-shifting, patent quality, and venue rules.

WHEN IT IS USED

Patent-portfolio strategy reviews, IP litigation defence planning, and patent-policy reform debates.

BY WHOM IT IS USED

In-house IP counsel, patent-litigation lawyers, antitrust scholars, and corporate IP-strategy teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with productive innovation rivalry, the troll model exploits patent-system asymmetries without producing inventions. Compared with foreign-trade tariff metaphors, the troll critique is internal — a market failure of the patent system itself.

RELATED FRAMEWORKS IN THIS COURSE

[Trade Secrets vs Patents · Slibrary 18](#)
[Copyright in the Digital Age · Slibrary 18](#)
[Antitrust in Digital Markets · Slibrary 12](#)
[AI & Copyright · Slibrary 18](#)

SOURCES (HARVARD REFERENCING STYLE)

Bessen, J. and Meurer, M.J. (2014) 'The direct costs from NPE disputes', *Cornell Law Review*, 99(2), pp. 387-424.
 FTC (2016) *Patent Assertion Entity Activity: An FTC Study*, Washington, DC: Federal Trade Commission.
 Supreme Court of the United States (2014) *Alice Corp. v. CLS Bank International*, 573 U.S. 208.

AI & Copyright

SLIBRARY 18 · SHEET 4 / 5

Training data · AI outputs · fair use — all contested

PROFESSIONAL DEFINITION

AI & Copyright concerns three interlocking questions: (1) Is training a model on copyrighted works infringement? (2) Are AI outputs copyrightable? (3) Are AI outputs themselves infringing? Each question is unsettled across major jurisdictions, with active US litigation (NYT v. OpenAI), EU Directive 2019/790 TDM exceptions, and UK consultations on AI training (Sag, 2023; UK IPO, 2025; Lemley and Casey, 2020).

WHO DEVELOPED IT

Major academic analyses by Matthew Sag (2023) and Mark Lemley (Stanford). Major active cases: NYT v. OpenAI/Microsoft (S.D.N.Y. 2023-), Getty v. Stability AI (UK 2023-), Andersen v. Stability AI (N.D. Cal. 2023-).

WHY IT WAS DEVELOPED

These cases will determine the legality of large-scale AI training, affecting OpenAI, Google, Microsoft, Meta, Anthropic, Stability AI and their customers. The economic stakes are in the hundreds of billions.

FIGURE 69 · THREE QUESTIONS

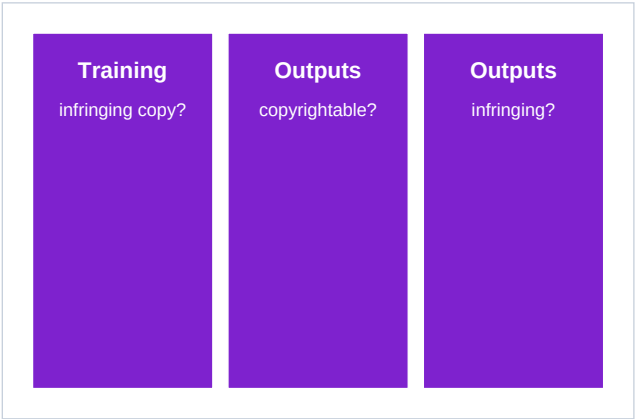


Figure 69. The three open questions. Each is being litigated separately; outcomes may diverge across jurisdictions and use-cases (Sag, 2023; Lemley and Casey, 2020; UK IPO, 2025).

HOW IT IS USED

For AI labs: assess training-data licensing or TDM exception applicability. For creators: track active cases; consider opt-out signals (robots.txt, ai.txt). For users: note that AI-generated outputs may not be copyrightable in the US.

WHEN IT IS USED

AI-product launch reviews, training-data acquisition decisions, and IP-policy formulation.

BY WHOM IT IS USED

AI lab legal teams, content publisher counsel, creator-rights advocates, and IP-policy scholars.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional copyright analysis, AI cases require re-examining each doctrinal element (copy, derivative work, fair use). Compared with regulatory-only responses, the litigation pathway is rapidly producing precedents.

RELATED FRAMEWORKS IN THIS COURSE

- Copyright in the Digital Age · Slibrary 18
- Creative Commons · Slibrary 18
- Trade Secrets vs Patents · Slibrary 18
- EU AI Act · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

Lemley, M.A. and Casey, B. (2020) 'Fair learning', Texas Law Review, 99, pp. 743-785.
Sag, M. (2023) 'Copyright safety for generative AI', Houston Law Review, 61(2), pp. 295-376.
UK IPO (2025) Copyright and AI: Government Response and Consultation. London: UK Intellectual Property Office.

Trade Secrets vs Patents

SLIBRARY 18 · SHEET 5 / 5

Secret = no disclosure, no term limit · Patent = disclosure for 20 years

PROFESSIONAL DEFINITION

Trade Secrets and Patents are the two principal regimes for protecting industrial innovation. Trade secrets give indefinite protection in exchange for keeping the innovation confidential; patents give 20-year exclusive rights in exchange for public disclosure. Each suits different innovation types and competitive contexts (WIPO, 2024; EU Directive 2016/943; US Defend Trade Secrets Act 2016).

WHO DEVELOPED IT

International framework set by TRIPS (1994) and WIPO. EU Trade Secrets Directive (2016/943); US Defend Trade Secrets Act (2016). Coca-Cola's formula is the canonical trade-secret example; pharmaceuticals dominate patent use.

WHY IT WAS DEVELOPED

The choice shapes innovation strategy, M&A diligence, and competitive intelligence exposure. With AI models, the trade-off has new salience: weights are typically trade-secret protected; model architectures are increasingly patented.

FIGURE 70 · TWO REGIMES

Dimension	Trade Secret	Patent
Disclosure	Confidential	Public
Term	Indefinite	20 years
Cost	Internal security	Filing & maintenance
Risk	Reverse engineering	Design-around

Figure 70. The trade-off between trade secrets and patents. The right choice depends on reverse-engineerability, term needs, and disclosure costs (WIPO, 2024).

HOW IT IS USED

For each invention: assess reverse-engineerability, useful term, and competitive intelligence exposure. Build trade-secret protection programmes (NDAs, access controls, audits); file patents for innovations that disclose easily.

WHEN IT IS USED

IP-strategy reviews, M&A diligence, employee-mobility risk reviews, and AI-model IP design.

BY WHOM IT IS USED

Chief IP Officers, R&D leaders, M&A counsel, and competitive-intelligence teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-regime IP strategies, the dual approach optimises across the innovation portfolio. Compared with copyright, both trade secrets and patents protect the underlying innovation, not just expression.

RELATED FRAMEWORKS IN THIS COURSE

- Patent Trolls · Slibrary 18
- Copyright in the Digital Age · Slibrary 18
- AI & Copyright · Slibrary 18
- Data as Asset · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

European Union (2016) Directive (EU) 2016/943 on the protection of trade secrets. Brussels: Official Journal of the EU.
United States (2016) Defend Trade Secrets Act of 2016, Pub. L. No. 114-153. Washington, DC: 114th Congress.
WIPO (2024) WIPO IP Statistics. Geneva: World Intellectual Property Organization.

Zuboff's Definition

SLIBRARY 19 · SHEET 1 / 5

A market logic that claims human experience as free raw material

PROFESSIONAL DEFINITION

Surveillance Capitalism, as defined by Shoshana Zuboff (2019), is a new economic logic that unilaterally claims human experience as free raw material for translation into behavioural data. Some of this data is fed back to improve products; the rest is fed into 'prediction products' sold in 'behavioural futures markets' to firms wanting to predict (and modify) future behaviour (Zuboff, 2015, 2019).

WHO DEVELOPED IT

Shoshana Zuboff (Harvard Business School) coined and developed the framework in her 2015 Journal of Information Technology paper and 2019 book 'The Age of Surveillance Capitalism'. The framework has been adopted by EU and UK regulators in non-doctrinal framing of digital-economy policy.

WHY IT WAS DEVELOPED

Zuboff's framework shifted public debate from 'privacy' (a static property right) to 'surveillance capitalism' (a dynamic economic logic). It is now the dominant critical frame in EU policy circles.

FIGURE 71 · THE LOGIC



Figure 71. Zuboff's chain of value extraction. Human experience is rendered into data, which becomes prediction products sold to advertisers, insurers, employers — anyone wanting to anticipate and shape behaviour (Zuboff, 2019).

HOW IT IS USED

Use the framework to diagnose business models. Distinguish 'data improves product' logics (often acceptable) from 'data sold for prediction' logics (often the surveillance-capitalism core). Identify which data flows fall into which.

WHEN IT IS USED

Critical analysis of digital-economy policy, ethical-strategy reviews, and B-school ethics-and-policy modules.

BY WHOM IT IS USED

Digital-rights NGOs, EU and UK regulators, ethics committees, and academic critical theorists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with privacy framings (which focus on individual rights), surveillance capitalism is structural — it characterises an economic system. Compared with consumer-protection framings, it identifies unilateral data appropriation as the core harm — distinct from misuse of legitimately-obtained data.

RELATED FRAMEWORKS IN THIS COURSE

[Behavioural Futures Markets · Slibrary 19](#)
[Data Extraction Imperative · Slibrary 19](#)
[Instrumentarian Power · Slibrary 19](#)
[Privacy Equation \(MIT\) · Slibrary 19](#)

SOURCES (HARVARD REFERENCING STYLE)

Zuboff, S. (2015) 'Big other: surveillance capitalism and the prospects of an information civilization', Journal of Information Technology, 30(1), pp. 75-89.
 Zuboff, S. (2019) The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power. New York, NY: PublicAffairs.
 Acquisti, A., Brandimarte, L. and Loewenstein, G. (2016) 'Privacy and human behavior in the age of information', Science, 347(6221), pp. 509-514.

Behavioural Futures Markets

SLIBRARY 19 · SHEET 2 / 5

Predictions about you are the product — not you

PROFESSIONAL DEFINITION

Behavioural Futures Markets are markets in which predictions about future human behaviour are bought and sold. The data subject is not the seller; the seller is the data-collecting platform. Buyers include advertisers, insurers, employers, lenders, and political campaigns. Zuboff identifies these markets as the central commercial feature of surveillance capitalism (Zuboff, 2019).

WHO DEVELOPED IT

Zuboff (2019) developed the analytical framework. Empirical studies by Christl and Spiekermann (2016) document the operations of the corporate-surveillance industry; the Wall Street Journal's 'What They Know' series provided the journalistic foundation.

WHY IT WAS DEVELOPED

The framing shifts attention from individual privacy preferences to the market structure in which behavioural predictions are commodified. It changes the diagnosis — and therefore the policy response.

FIGURE 72 · WHO TRANSACTS

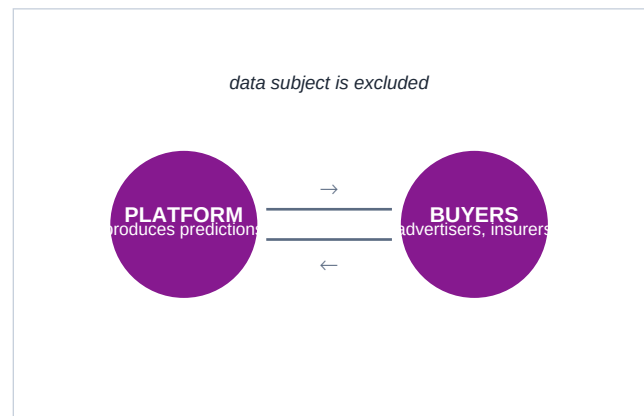


Figure 72. The market structure. The data subject is the source of value but not a party to the transaction — a structural feature, not a contingent failure of consent (Zuboff, 2019).

HOW IT IS USED

Map the prediction-product flows for each digital service. Identify buyers, prices, and feedback into product. Use this map for ethical review and for policy advocacy.

WHEN IT IS USED

Critical-theory analyses of digital business models, regulator-engagement preparation, and ethics-committee reviews.

BY WHOM IT IS USED

Critical theorists, EU and UK competition and consumer regulators, and digital-rights advocacy groups.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with general data-economy framings, the behavioural-futures concept makes the market structure precise. Compared with privacy-as-secrecy framings, it locates the harm in the market rather than the data flow.

RELATED FRAMEWORKS IN THIS COURSE

Surveillance Capitalism · Slibrary 19

Data Extraction Imperative · Slibrary 19

Hyper-Personalisation · Slibrary 16

Personal Data as Property · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

Christl, W. and Spiekermann, S. (2016) Networks of Control: A Report on Corporate Surveillance. Vienna: Facultas.
Zuboff, S. (2019) The Age of Surveillance Capitalism. New York, NY: PublicAffairs.

Couldry, N. and Mejias, U.A. (2019) The Costs of Connection. Stanford, CA: Stanford University Press.

Data Extraction Imperative

SLIBRARY 19 · SHEET 3 / 5

The business model compels ever-deeper data capture

PROFESSIONAL DEFINITION

The Data Extraction Imperative is Zuboff's term for the structural pressure within surveillance-capitalist firms to capture more data, of more kinds, more frequently, and at greater depth. Better predictions require richer data; competitive pressure requires ever-better predictions. The imperative is structural — not the result of individual managerial decisions (Zuboff, 2019).

WHO DEVELOPED IT

Zuboff (2019) named and developed the imperative. Earlier theoretical antecedents in Foucault's analyses of disciplinary power and Bentham's panopticon; empirical documentation in journalism and academic media studies.

WHY IT WAS DEVELOPED

If the imperative is structural, voluntary corporate restraint is unlikely. Effective governance must constrain the market structure rather than rely on firm-level choices.

FIGURE 73 · THE LOOP

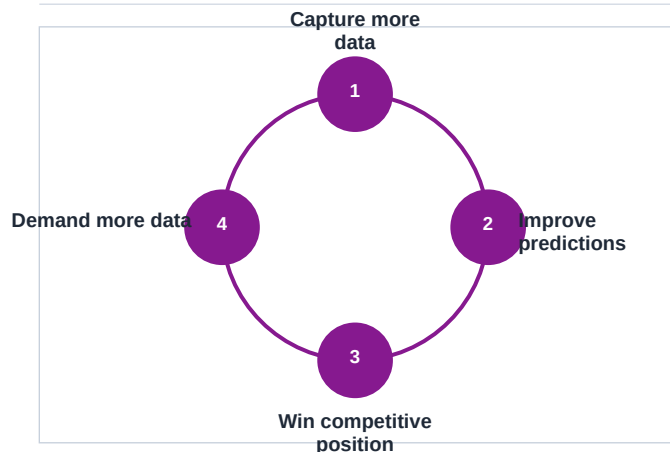


Figure 73. The data-extraction loop. Each turn deepens the surveillance footprint. Without external constraints, the loop tends toward maximal extraction (Zuboff, 2019).

HOW IT IS USED

For policymakers: design ex-ante restrictions (DMA, GDPR minimisation, data caps) that bind the imperative. For firms: recognise the strategic risk of regulatory intervention as the imperative reaches limits.

WHEN IT IS USED

Policy formulation, critical strategy reviews, and academic political-economy analyses.

BY WHOM IT IS USED

Digital regulators, NGO researchers, critical-theory scholars, and ethics-policy specialists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with individual-choice framings, the structural-imperative framing identifies the system-level driver. Compared with general 'data is the new oil' rhetoric, it specifies the competitive dynamic that compels extraction.

RELATED FRAMEWORKS IN THIS COURSE

Surveillance Capitalism · Slibrary 19

Behavioural Futures Markets · Slibrary 19

Instrumentarian Power · Slibrary 19

Data Minimisation · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

Foucault, M. (1977) Discipline and Punish: The Birth of the Prison. New York, NY: Pantheon.
 Zuboff, S. (2019) The Age of Surveillance Capitalism. New York, NY: PublicAffairs.
 Couldry, N. and Mejias, U.A. (2019) The Costs of Connection. Stanford, CA: Stanford University Press.

Instrumentarian Power

SLIBRARY 19 · SHEET 4 / 5

Not totalitarianism — 'we know you, we shape you'

PROFESSIONAL DEFINITION

Instrumentarian Power is Zuboff's term for the form of power exercised by surveillance-capitalist firms: not the brute force of totalitarianism, but the instrumentation of behaviour through ubiquitous data, prediction, and nudges. It operates through knowledge and shaping rather than coercion — 'we know you, and we shape you' (Zuboff, 2019).

WHO DEVELOPED IT

Zuboff (2019) developed the concept, drawing on B.F. Skinner's behaviourism and Hannah Arendt's analyses of totalitarianism. Subsequent application by Couldry and Mejias (2019) in 'The Costs of Connection'.

WHY IT WAS DEVELOPED

The framing answers a key political-theoretic question: surveillance capitalism is concerning not because it resembles 1984, but because it represents a different kind of power — softer, more pervasive, and not addressed by classical anti-totalitarian tools.

FIGURE 74 · TWO POWERS

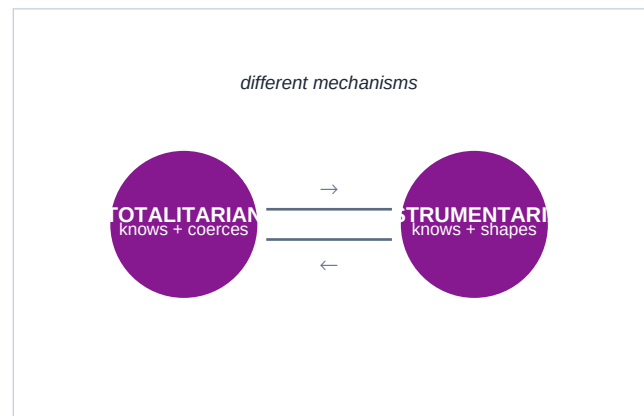


Figure 74. Totalitarian power coerces; instrumentarian power shapes. The latter is compatible with consumer satisfaction and democratic forms — yet undermines autonomous choice in subtler ways (Zuboff, 2019).

HOW IT IS USED

Use the concept to diagnose digital governance failures: classical anti-totalitarian remedies (free speech, due process) may not address instrumentarian harms. New governance models — autonomy rights, algorithmic transparency — are required.

WHEN IT IS USED

Political-theory analyses of digital power, AI-governance frameworks, and critical-theory ethics modules.

BY WHOM IT IS USED

Political theorists, digital-rights advocates, AI-ethics scholars, and policymakers designing governance for AI-driven persuasion.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the surveillance-state framing, instrumentarian power locates the novel feature of digital power in shaping rather than coercion. Compared with classical media-effects theory, it specifies the data-driven mechanism through which shaping now operates.

RELATED FRAMEWORKS IN THIS COURSE

Surveillance Capitalism · Slibrary 19

Behavioural Futures Markets · Slibrary 19

Hyper-Personalisation · Slibrary 16

Algorithmic Bias · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

Couldry, N. and Mejias, U.A. (2019) The Costs of Connection. Stanford, CA: Stanford University Press.
 Zuboff, S. (2019) The Age of Surveillance Capitalism. New York, NY: PublicAffairs.
 Arendt, H. (1951) The Origins of Totalitarianism. New York, NY: Schocken Books.

Privacy Equation (MIT)

SLIBRARY 19 · SHEET 5 / 5

Acquisti: consumers trade personal data for utility — but undervalue it

PROFESSIONAL DEFINITION

The Privacy Equation, developed at MIT and Carnegie Mellon by Alessandro Acquisti, Laura Brandimarte, and George Loewenstein (2015, 2016), formalises consumers' privacy decisions as a trade-off between disclosure utility and disclosure cost. Empirically, consumers systematically undervalue privacy — exhibiting hyperbolic discounting, uncertainty-driven inaction, and context-dependent preferences (Acquisti et al., 2015, 2016).

WHO DEVELOPED IT

Alessandro Acquisti and colleagues at MIT IDE, CMU, and partner institutions. The 'Privacy and human behavior' Science article (2015) is the canonical synthesis. MIT IDE 2025 Annual Report integrates the framework into AI-era recommendations.

WHY IT WAS DEVELOPED

If consumers are systematically biased against their own privacy interests, classical notice-and-consent regimes are inadequate. Effective protection requires structural intervention — defaults, prohibitions, friction — rather than relying on individual choice.

FIGURE 75 · DECISION QUADRANT

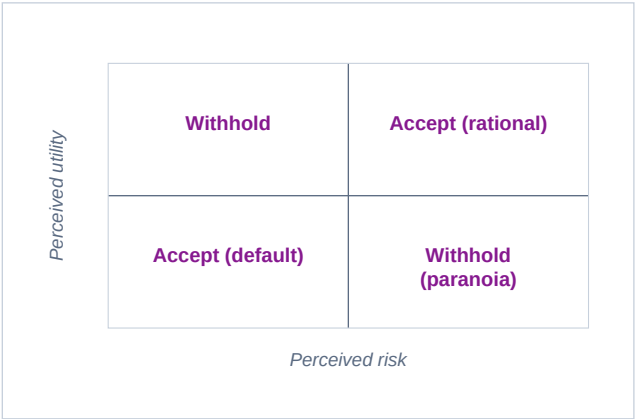


Figure 75. The privacy decision quadrant. Most consumer behaviour falls into the bottom-left 'accept by default' cell — driven by hyperbolic discounting and uncertainty rather than considered choice (Acquisti et al., 2015, 2016).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
For policymakers: shift from notice-and-consent to structural defaults. For firms: design for genuine consent through friction and clarity. For users: recognise the biases and use privacy-preserving defaults.	Privacy-policy formulation, app-design reviews, and behavioural-economics ethics modules.	Privacy researchers (MIT, CMU), regulators designing default rules, behavioural-economics scholars, and ethical-design specialists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with rational-choice privacy models, the Privacy Equation incorporates behavioural-economic biases. Compared with binary privacy-vs-utility framings, it shows how context, defaults, and uncertainty shape decisions.

RELATED FRAMEWORKS IN THIS COURSE

- GDPR Principles · Slibrary 17
- Personal Data as Property · Slibrary 17
- Surveillance Capitalism · Slibrary 19
- Hyper-Personalisation · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Acquisti, A., Brandimarte, L. and Loewenstein, G. (2015) 'Privacy and human behavior in the age of information', Science, 347(6221), pp. 509-514.
Acquisti, A., Taylor, C. and Wagman, L. (2016) 'The economics of privacy', Journal of Economic Literature, 54(2), pp. 442-492.
MIT Initiative on the Digital Economy (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Sloan.

EU AI Act

SLIBRARY 20 · SHEET 1 / 5

Risk-based regulation: unacceptable · high · limited · minimal

PROFESSIONAL DEFINITION

The EU AI Act (Regulation (EU) 2024/1689) is the world's first comprehensive AI regulation. It classifies AI systems by risk level: unacceptable (banned, e.g. social scoring), high-risk (heavy obligations: critical infrastructure, education, employment), limited risk (transparency obligations), and minimal risk (largely unregulated). Enforcement begins phased over 2025-2027 (European Union, 2024; European AI Office, 2024).

WHO DEVELOPED IT

Drafted by the European Commission's DG CNECT under Thierry Breton. Adopted by the European Parliament and Council in March/May 2024; in force 1 August 2024. The European AI Office (within DG CNECT) is the central enforcer; national authorities supervise within member states.

WHY IT WAS DEVELOPED

Like GDPR before it, the AI Act sets the global de facto baseline. Maximum fines reach €35 million or 7% of global turnover. Even non-EU firms must comply if their AI systems are used in the EU.

FIGURE 76 · RISK PYRAMID

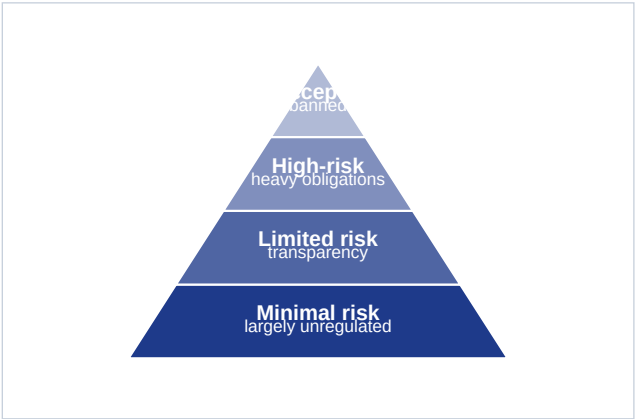


Figure 76. The four-tier risk pyramid. Most AI systems fall into the bottom two tiers; the regulatory weight falls primarily on high-risk systems. General-purpose AI models are governed under a separate, additional regime (European Union, 2024).

HOW IT IS USED

Inventory deployed AI systems; classify by risk tier; build compliance for high-risk systems (data governance, transparency, human oversight, conformity assessment, post-market monitoring).

WHEN IT IS USED

AI-governance programme design, product launches involving AI, and supplier diligence on AI components.

BY WHOM IT IS USED

AI governance leads, in-house counsel, the European AI Office, and B-school AI-policy modules.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with sectoral or self-regulatory approaches, the AI Act is comprehensive and binding. Compared with the GDPR, it adopts a risk-based rather than rights-based structure — recognising that not all AI systems require equal scrutiny.

RELATED FRAMEWORKS IN THIS COURSE

- Algorithmic Bias · Slibrary 20
- Responsibility Gap · Slibrary 20
- AI Risk & Mitigation Matrix (MIT) · Slibrary 20
- Human-Centered AI Blueprint (MIT) · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

European AI Office (2024) Implementing the AI Act: Guidelines and Codes of Practice. Brussels: European Commission.
European Union (2024) Regulation (EU) 2024/1689 — Artificial Intelligence Act. Brussels: Official Journal of the EU.
Veale, M. and Zuiderveen Borgesius, F. (2021) 'Demystifying the Draft EU AI Act', Computer Law Review International, 22(4), pp. 97-112.

Algorithmic Bias

SLIBRARY 20 · SHEET 2 / 5

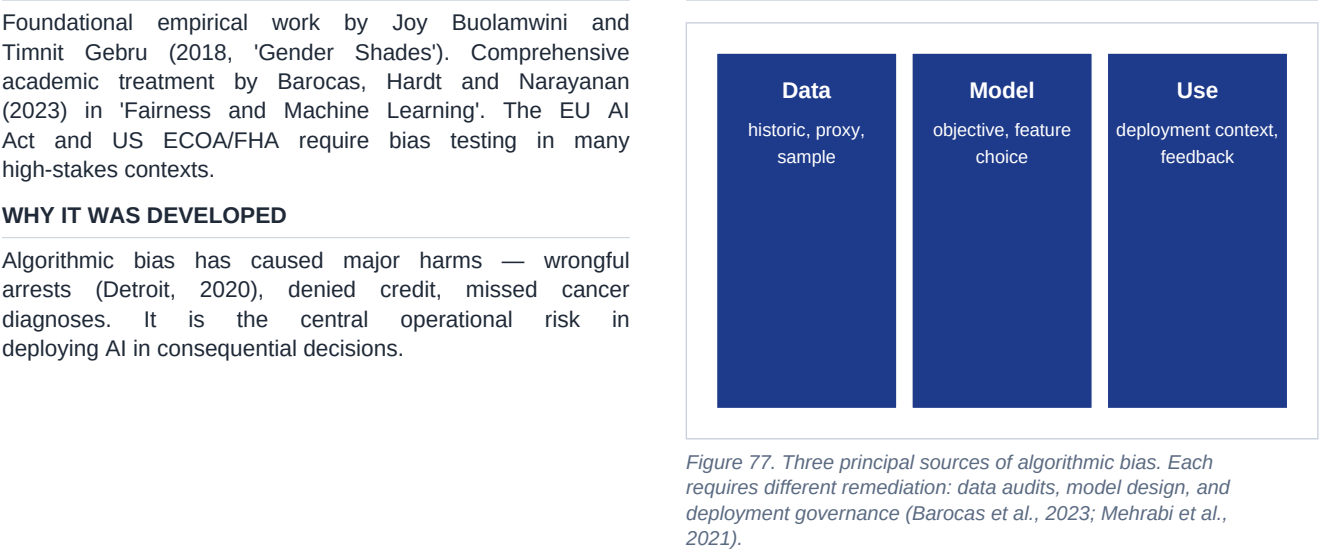
Systematic errors disadvantaging specific groups

PROFESSIONAL DEFINITION

Algorithmic Bias is the systematic error in algorithmic decisions that disadvantages specific groups, typically along protected characteristics (race, gender, age, disability). Sources include biased training data, proxy variables, mis-specified objectives, and feedback loops. Bias has been documented in hiring, lending, criminal justice, and healthcare AI (Mehrabi et al., 2021; Barocas, Hardt and Narayanan, 2023).

WHO DEVELOPED IT

Foundational empirical work by Joy Buolamwini and Timnit Gebru (2018, 'Gender Shades'). Comprehensive academic treatment by Barocas, Hardt and Narayanan (2023) in 'Fairness and Machine Learning'. The EU AI Act and US ECOA/FHA require bias testing in many high-stakes contexts.



HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
Audit deployed systems against fairness metrics (demographic parity, equalised odds, calibration). Document choices and trade-offs. Embed mitigation in MLOps pipelines, not just in launch reviews.	AI-system design and audit, regulatory compliance (EU AI Act, US Fair Lending), and ethics-committee reviews.	ML fairness researchers, AI-governance teams, civil-rights regulators (EEOC, CRE, EU FRA), and B-school AI-ethics modules.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with classical statistical bias, algorithmic bias addresses the systematic and consequential nature of automated decisions. Compared with general fairness philosophy, it provides operational metrics and tooling.

RELATED FRAMEWORKS IN THIS COURSE

- EU AI Act · Slibrary 20
- Responsibility Gap · Slibrary 20
- AI Risk & Mitigation Matrix (MIT) · Slibrary 20
- Human-Centered AI Blueprint (MIT) · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

Barocas, S., Hardt, M. and Narayanan, A. (2023) Fairness and Machine Learning. Cambridge, MA: MIT Press.

Buolamwini, J. and Gebru, T. (2018) 'Gender shades: intersectional accuracy disparities in commercial gender classification', PMLR, 81, pp. 77-91.

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K. and Galstyan, A. (2021) 'A survey on bias and fairness in machine learning', ACM Computing Surveys, 54(6), pp. 1-35.

Responsibility Gap

SLIBRARY 20 · SHEET 3 / 5

Who is accountable when an autonomous system harms?

PROFESSIONAL DEFINITION

The Responsibility Gap is the gap between the harm caused by an autonomous AI system and the persons who can plausibly be held morally and legally responsible. As AI systems make consequential decisions with limited human oversight, traditional responsibility doctrines — designed around human agents — strain or fail (Matthias, 2004; Floridi et al., 2018; Coeckelbergh, 2020).

WHO DEVELOPED IT

Andreas Matthias (2004) named and articulated the gap. Floridi and colleagues (2018, AI4People) developed the policy implications. Coeckelbergh (2020) provides the philosophical synthesis.

WHY IT WAS DEVELOPED

Legal and moral systems require identifiable responsible parties. Without them, victims have no remedy and governance has no leverage. The gap is the conceptual obstacle to coherent AI governance.

FIGURE 78 · CANDIDATES

Candidate	Argument	Limit
Developer	designed system	may not control deployment
Deployer	chose to use	may not understand system
User	operates system	may have no override
AI itself	makes decision	no legal personhood

Figure 78. The four candidates for responsibility in autonomous AI harms. Each has real limits, leaving gaps that current law has not fully closed (Matthias, 2004; Coeckelbergh, 2020).

HOW IT IS USED	WHEN IT IS USED	BY WHOM IT IS USED
For governance: mandate identifiable responsible parties (the EU AI Act's 'provider' and 'deployer' roles do this). For firms: clarify and document allocation of responsibility across the supply chain. For policy: consider strict-liability or insurance-based	AI-governance design, M&A AI-supply-chain reviews, and ethics-and-policy formulation.	AI ethicists, legal-policy researchers, EU AI Office and national AI authorities, and AI-governance leads in firms.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with treating AI as a product (with simple liability), the responsibility-gap framing addresses the unique challenges of autonomous systems. Compared with abolitionist 'don't deploy' approaches, it works toward governance solutions for deployed systems.

RELATED FRAMEWORKS IN THIS COURSE

EU AI Act · Slibrary 20

Algorithmic Bias · Slibrary 20

AI Risk & Mitigation Matrix (MIT) · Slibrary 20

Human-Centered AI Blueprint (MIT) · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

Coeckelbergh, M. (2020) 'Artificial intelligence, responsibility attribution, and a relational justification of explainability', Science and Engineering Ethics, 26, pp. 2051-2068.
Floridi, L. et al. (2018) 'AI4People — an ethical framework for a good AI society', Minds and Machines, 28(4), pp. 689-707.
Matthias, A. (2004) 'The responsibility gap: ascribing responsibility for the actions of learning automata', Ethics and Information Technology, 6(3), pp. 175-183.

AI Risk & Mitigation Matrix (MIT)

SLIBRARY 20 · SHEET 4 / 5

Four risks · matched MIT-backed mitigations

PROFESSIONAL DEFINITION

The AI Risk & Mitigation Matrix synthesises MIT IDE research on AI deployment risks into a four-by-four mapping: four canonical risks (bias, hallucination, autonomy, concentration) matched with empirically-validated mitigations (audit, retrieval, human-in-loop, antitrust). The framework supports operational AI governance (MIT IDE, 2025; Ng et al., 2024).

WHO DEVELOPED IT

Compiled and synthesised by the MIT Initiative on the Digital Economy (2025 Annual Report) drawing on multiple researcher contributions (Aral, Acemoglu, Brynjolfsson, Horton, Acquisti). Rogowski's (2025) visual synthesis circulated widely as a managerial reference.

WHY IT WAS DEVELOPED

Many AI-governance frameworks list risks without matched mitigations or vice versa. The matrix structure supports operational programme design and resource allocation.

FIGURE 79 · RISK-MITIGATION GRID

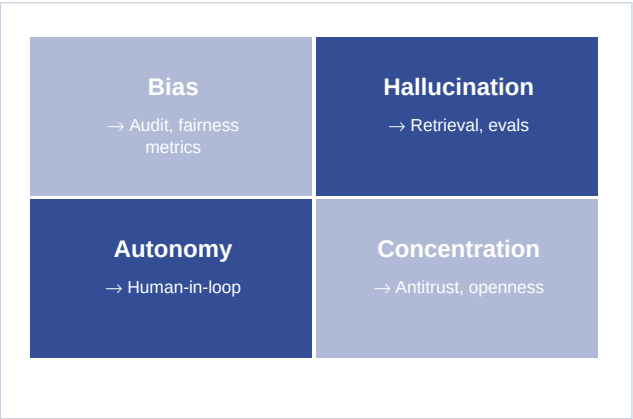


Figure 79. The MIT IDE risk-mitigation grid. Each risk is matched with empirically-validated mitigation. Effective programmes implement all four (MIT IDE, 2025; Rogowski, 2025).

HOW IT IS USED

Inventory deployed AI by risk profile. Implement all four mitigations as appropriate. Track residual risk over time. Update as both AI capability and threat landscape evolve.

WHEN IT IS USED

AI-governance programme design, board-level AI-risk reviews, and supplier-diligence on AI components.

BY WHOM IT IS USED

AI-governance leads, MIT IDE researchers, and large-firm Chief AI Officers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with risk-only frameworks, the matrix matches risks to mitigations. Compared with mitigation-only frameworks, it ensures coverage of all canonical risks. The MIT-backed empirical foundation differentiates it from purely conceptual schemes.

RELATED FRAMEWORKS IN THIS COURSE

- EU AI Act · Slibrary 20
- Algorithmic Bias · Slibrary 20
- Responsibility Gap · Slibrary 20
- Human-Centered AI Blueprint (MIT) · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

MIT Initiative on the Digital Economy (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Sloan.
Ng, A. et al. (2024) AI risk management framework. Stanford, CA: Stanford HAI.
Rogowski, R. (2025) Exploring AI's Impact: From Hype to ROI. Visual synthesis of MIT IDE research.

Human-Centered AI Blueprint (MIT)

SLIBRARY 20 · SHEET 5 / 5

Measure rigorously · design for humans · prepare for autonomy

PROFESSIONAL DEFINITION

The Human-Centered AI (HAI) Blueprint is the MIT IDE's three-pillar framework for deploying AI in a way that augments rather than displaces human capability: (1) measure rigorously (productivity AND wellbeing AND skill development), (2) design for humans (workflows that engage rather than offload), (3) prepare for autonomy (governance for increasingly autonomous systems) (MIT IDE, 2025; Shneiderman, 2022).

WHO DEVELOPED IT

MIT IDE 2025 Annual Report compiled the blueprint from multiple research streams. Drawing on Ben Shneiderman's (2022) 'Human-Centered AI' and Stanford HAI's parallel work. The framework integrates with the AI Risk & Mitigation Matrix as the positive counterpart to risk management.

WHY IT WAS DEVELOPED

Risk-mitigation alone leaves the question of what to build. The HAI blueprint provides an affirmative design philosophy that complements regulatory compliance with constructive design principles.

FIGURE 80 · THREE PILLARS

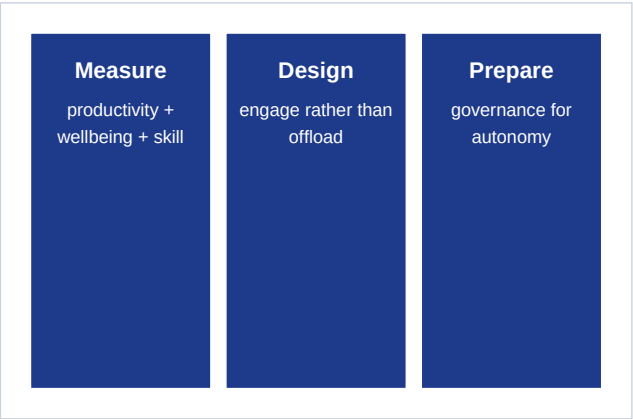


Figure 80. The three pillars of human-centered AI. Each pillar requires both research-driven understanding and design-driven application — bridging MIT IDE's two communities (MIT IDE, 2025; Shneiderman, 2022).

HOW IT IS USED

For programme design: build evaluation across all three measurement dimensions. For product design: apply human-engagement principles (review, justify, override). For governance: prepare ahead for autonomy thresholds rather than reacting after.

WHEN IT IS USED

AI-product design, organisational AI-strategy formulation, and B-school AI-and-society modules.

BY WHOM IT IS USED

AI-product designers, MIT IDE researchers, organisational HR/AI leaders, and corporate-strategy teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with risk-only frameworks, HAI provides the affirmative design counterpart. Compared with general 'human-centered design', it focuses specifically on the AI deployment context where measurement, design, and governance interact.

RELATED FRAMEWORKS IN THIS COURSE

AI Risk & Mitigation Matrix (MIT) · Slibrary 20

Human-AI Pairing (MIT) · Slibrary 15

Cognitive Trade-Off (MIT) · Slibrary 15

EU AI Act · Slibrary 20

SOURCES (HARVARD REFERENCING STYLE)

MIT Initiative on the Digital Economy (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Sloan.
Shneiderman, B. (2022) Human-Centered AI. Oxford: Oxford University Press.
Stanford HAI (2024) AI Index Report 2024. Stanford, CA: Stanford Institute for Human-Centered AI.